

Capnostream™35 Portable Respiratory Monitor PM35MN, with Microstream™ EtCO₂ and Nellcor™ SPO₂ Technologies

Operator's Manual

PN: PT00082177



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1. Introduction

1.1. Introduction

Carefully read this operator's manual and the Directions for Use that accompany Microstream™ etCO₂ sampling lines (FilterLine™ sampling lines, henceforth referred to as sampling lines) and the SpO₂ sensors, in order to use the Capnostream™35 portable respiratory monitor PM35MN, with Microstream™ EtCO₂ and Nellcor™ SPO₂ technologies (henceforth referred to as the Capnostream™35 monitor or the monitor) correctly and safely. Use of the monitor requires full understanding and strict observance of these instructions, the precautionary information, and the specifications.

1.2. Safety Information

1.2.1. General



WARNING:

If uncertain about the accuracy of any measurement, first check the patient's vital signs by alternate means, and then make sure the monitor is functioning correctly.



WARNING:

The monitor should not be used as an apnea monitor.



WARNING:

The monitor should be considered an early warning device. As a trend towards patient deoxygenation is indicated, blood samples should be analyzed by a laboratory co-oximeter to completely understand the patient's condition.



WARNING:

To ensure patient safety, do not place the monitor in any position that might cause it to fall on the patient. If the monitor is mounted, ensure that is securely mounted.



WARNING:

Carefully route patient cabling (SpO₂ sensor and sampling line) to reduce the possibility of patient entanglement or strangulation.



WARNING:

Do not lift the monitor by the SpO₂ sensor cable or sampling line, as they could disconnect from the monitor, causing the monitor to fall on the patient.



WARNING:

Do not obstruct the monitor so that the visual alarm indicators cannot be seen or the audio alarm indicators cannot be heard.



WARNING:

The monitor should not be used adjacent to or stacked with other equipment; if adjacent or stacked use is necessary, the monitor shall be observed to verify normal operation in the configuration in which it will be used.



WARNING:

The use of accessories, transducers, sensors and cables other than those specified may result in increased emission and/or decreased immunity of the equipment and/or system.



WARNING:

Re-use of single-use accessories could pose a cross-contamination risk to the patient or damage the functioning of the monitor.



WARNING:

CO₂ readings, respiratory rate, pulse oximetry readings, and pulse signals can be affected by sensor application errors, certain ambient environmental conditions, and certain patient conditions.



WARNING:

The monitor is a prescription device and is to be operated by qualified healthcare personnel only.



WARNING:

No modification of this equipment is allowed.



WARNING:

If calibration does not take place as instructed in the relevant service manual, the monitor may be out of calibration. A monitor that is out of calibration may provide inaccurate results.



WARNING:

Do not use any monitoring system, sensor, cable, connector, or screen that appears damaged. Remove any damaged equipment from service for inspection by a qualified service technician.



WARNING:

Do not perform service or maintenance on the device while it is in use.



WARNING:

Do not transport damaged or defective lithium cells and batteries via air transport.



Caution:

Storage or transport of the monitor under environmental conditions beyond those mentioned in the specification will affect monitor performance and damage the monitor.



Note:

Temperature sensors will turn the monitor off when it exceeds the permitted temperature.

Devices connected to the monitor must be medical grade only.

The accurate display of the following parameters is required in order to fill the essential performance of the monitor: Carbon dioxide levels in expired breath (CO₂) when monitoring with capnography, and arterial oxygen saturation of blood (SpO₂) and pulse rate when monitoring with pulse oximetry. If the patient is being monitored with both functions, all of these parameters will be displayed. If the patient is being monitored with either or both functions, alarm volume is included in the monitor's essential performance.

1.2.2. MRI Scanning



WARNING:

Do not use oximetry sensors during magnetic resonance imaging (MRI) scanning. Conducted current could cause

burns. The sensors may affect the MRI image, and the MRI unit may affect the accuracy of oximetry measurements.



WARNING:

Do not use the FilterLine™ H Set Infant/Neonatal set or the VitaLine™ H Set Infant/Neonatal set during magnetic resonance imaging (MRI) scanning. Using the FilterLine™ H Set Infant/Neonatal set during MRI scanning could harm the patient.



WARNING:

During MRI scanning, the monitor must be placed outside the MRI suite. When the monitor is used outside the MRI suite, etCO₂ monitoring can be implemented using the FilterLine™ XL sampling line. Reference [4.10.5 Monitoring CO₂ during MRI Scanning](#) on page 90.



Caution:

Use of a CO₂ sampling line with H in its name (indicating that it is for use in humidified environments) during MRI scanning may cause artefacts on the MRI image. The use of non-H sampling lines is advised. For a list of sampling lines, reference [8.1 Microstream™ EtCO₂ Consumables](#) on page 225.

1.2.3. Alarms



WARNING:

Do not silence the audible alarm if patient safety may be compromised.



WARNING:

Always respond immediately to a device alarm since the patient may not be monitored during certain technical/caution alarm conditions.



WARNING:

Before each use, verify that the alarm limits are appropriate for the patient being monitored.



WARNING:

Do not preset different or inappropriate alarm limits for the same or similar equipment in any single area, since this may compromise patient safety.

1.2.4. Fire Hazard



WARNING:

When using the monitor with anesthetics, nitrous oxide or high concentrations of oxygen, connect the gas outlet to a scavenger system.



WARNING:

The monitor is not suitable for use in the presence of flammable anesthetic mixture with air, oxygen or nitrous oxide.



WARNING:

The sampling line may ignite in the presence of O₂ when directly exposed to laser, ESU devices, or high heat. When performing head and neck procedures involving laser, electrosurgical devices or high heat, use these devices with caution to prevent flammability of the sampling line or surrounding surgical drapes.

1.2.5. Electrical



WARNING:

To protect against electric shock hazard, the monitor's cover

is to be removed only by qualified service personnel. There are no user-serviceable parts inside.



WARNING:

To ensure patient electrical isolation, connect only to other equipment with circuits that are electrically isolated.



WARNING:

Do not connect the monitor to a power supply other than that supplied with the monitor.



WARNING:

To avoid the risk of electric shock, the monitor must be connected only to a supply mains with protective earth.



WARNING:

Connect the monitor only to a hospital grade receptacle. The plug must be inserted into a properly wired receptacle; if a properly wired receptacle is not available, a qualified electrician must install one in accordance with the governing electrical code. Do not use extension cords or adapters of any type. The power cord and plug must be intact and undamaged.



WARNING:

Do not connect to an electrical outlet controlled by a wall switch or a dimmer.



WARNING:

Measure the monitor's leakage current whenever an external device is connected to the serial port. Leakage current must not exceed 100 microamperes.



Caution:

Any USB device (excluding a flash memory drive) or PC connected to the monitor must be running on a battery, or a

IEC 60601-1 compliant power supply (Clause 16, ME Systems), or a IEC 60601-1 compliant isolation transformer (Clause 16, ME Systems).



Caution:

All signal input and output (I/O) connectors are intended for connection of devices complying with Clause 16, ME Systems, of IEC 60601-1 only. Connecting additional devices to the monitor may increase chassis or patient leakage currents. To maintain operator and patient safety, consider the requirements of Clause 16, ME Systems, of IEC 60601-1. Measure the leakage currents to confirm that no electric shock hazard exists.



Caution:

Electrical installation of the room or the building in which the monitor is to be used must comply with regulations specified by the country in which the equipment is to be used.



Caution:

Keep power cord, plug and socket clear in case an urgent power supply disconnection is required.

1.2.6. Electro-magnetic Interference

This monitor has been tested and found to comply with the requirements for medical devices according to the standard IEC 60601-1-2. This standard is designed to provide reasonable protection against harmful interference in a typical medical installation.

However, because of the proliferation of radio-frequency transmitting equipment and other sources of electrical noise in healthcare environments (for example: cellular phones, mobile two-way radios, electrical appliances), it is possible that high levels of such interference due to close proximity or strength of a source may result in disruption of performance of this device.



WARNING:

Operating high frequency electrosurgical equipment in the vicinity of the monitor can produce interference in the monitor and cause incorrect measurements. Do not use the monitor with nuclear spin tomography (MRT, NMR, NMT) as the function of the monitor may be disturbed.

1.3. Definitions



Note:

A Note is inserted to point out procedures or conditions which may otherwise be misinterpreted or overlooked and to clarify apparently contradictory or confusing situations.



Caution:

A Caution is inserted to call attention to a procedure which, if not followed exactly, can lead to damage or destruction of the equipment.



WARNING:

A Warning is inserted to call attention to dangerous or hazardous conditions inherent to the operation, cleaning, and maintenance of the equipment which may result in personal injury or death of the operator or patient.

1.4. Contacting Technical Support

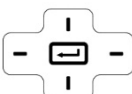
For any technical issue involving the monitor, contact your local Service Center or Covidien Technical Support, as follows:

North America: Tel: 1-888-ORIDION (674-3466), Fax: (781) 453-2722;
 Outside North America: Tel: + (972) 2-589-9104, Fax: + (972) 2-582-
 8868; E-mail: Capnographytechnicalsupport@medtronic.com

1.5. Symbols

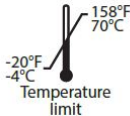
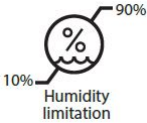
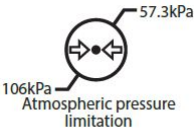
The following symbols appear on the body of the monitor.

Table 1. Symbols that Appear on the Monitor and its Packaging

Symbol	Description
	Monitor ON/OFF button
	Battery/power indicator
	Home button
	Back button
	Temporarily silence audible alarms
	Enter button with directional arrows
	Type BF Defibrillator Proof Protection
	Gas inlet
	Gas outlet

CO₂	CO ₂ port indicator
micro SD	Micro SD card port
 12VDC	12VDC
NELLCOR  SpO₂	Nellcor™ SpO ₂ connection
	USB flash memory connection port
	Monitor connection
FC	Federal Communications Commission
Rx ONLY	For prescription use only
	Caution, consult accompanying documents
	Follow Instructions for Use (in blue)
	Directive on waste from electrical and electronic equipment
	Non-ionizing electromagnetic radiation

Symbols

	Wireless regulation (Australia)
IP54	IPX rating
	ETL Mark
	CE Mark
	Serial number
	Catalog number
	Date of manufacture
	Manufacturer
	Shipping symbols
	Temperature limit for storage and transport
	Humidity limit for storage and transport
	Atmospheric pressure limit for storage and transport

1.6. Who Should Read This Manual

The following persons should read this manual:

- Health Care Professionals who will be using the monitor.
- Equipment managers responsible for ensuring that equipment conforms to institutional policies.
- Researchers or laboratory personnel who will be downloading patient data.
- Technical experts who will be connecting the monitor to a computer.



WARNING:

In the United States, federal law restricts this device to sale by or on the order of a physician.

1.7. Warranty and Disclaimers

1.7.1. Warranty

Oridion Medical 1987 Ltd. ("Oridion Medical") - Warranty for Oridion Monitors:

THIS LIMITED WARRANTY applies to any patient monitor manufactured by Oridion Medical 1987 Ltd. ("Oridion"), ("Products"). Subject to the limitations herein, Oridion warrants that Products, when delivered by Oridion or its authorized distributor, for two (2) years following the delivery date, but no more than 27 months following the date of production, will be free from defects in material and workmanship and will substantially conform to published Oridion specifications for the respective Products and in effect at the time of manufacture. This limited warranty excludes (i) Products

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2. Product Overview

The Capnostream™35 monitor operational check sheet is provided at the end of this chapter to simplify the installation, setup, and getting started processes. Photocopy the Check Sheet from the manual and check off the steps on the Check Sheet as you set up the monitor.

2.1. Overview

This manual provides directions for setting up and operating the monitor.

The Capnostream™35 portable respiratory monitor PM35MN, with Microstream™ EtCO₂ and Nellcor™ SPO₂ technologies, is a portable bedside monitor that continuously monitors a patient's:

- End tidal carbon dioxide (etCO₂) - level of carbon dioxide in exhaled breath
- Respiratory rate (RR), derived from capnography or pulse oximetry
- Oxygen saturation (SpO₂)
- Pulse rate (PR)

The monitor also provides an Integrated Pulmonary Index™ algorithm (henceforth referred to as IPI) value, which is a numerical value that integrates four major parameters measured by the monitor in order to provide a simple indication of the patient's ventilatory status. The integrated parameters are etCO₂, RR, SpO₂, and PR. Only these four parameters are used to calculate IPI; other parameters are not taken into account.

In addition, the monitor provides Apneas per Hour (A/hr) (also known as ASA, Apnea Saturation Alert) and an Oxygen Desaturation Index (ODI), used to help in the identification and quantification of apnea and oxygen desaturation events for patients over age 22, as follows:

A/hr: a count of the number of pauses in breathing (of at least 10 seconds) which the patient experienced, either over the past hour (on the Home screen) or average pauses per hour over a period of time (on the Apnea and O₂ Desaturation screen).

ODI: the number of times that the SpO₂ value dropped 4% or more from baseline and returned to baseline in 240 seconds or less, either in the last hour (on the home screen) or average drops per hour over a period of time (on the Apnea and O₂ Desaturation screen).

The device also provides the OxiMax™ SPD (henceforth referred to as SPD) and Nellcor™ respiration rate features, described in [2.2 Intended Use](#), below.

The Nellcor™ respiration rate and OxiMax™ SPD features are not available on all devices. For more information, contact Capnographytechnicalsupport@medtronic.com.

2.2. Intended Use

The Capnostream™35 monitor is a portable capnograph/pulse oximeter, intended to provide professionally trained health care providers with continuous non-invasive monitoring of carbon dioxide concentration of the expired and inspired breath, respiration rate, arterial oxygen saturation (SpO₂) and pulse rate of adult, pediatric, and neonatal patients. The pulse oximeter is intended for use during both no motion and motion conditions and for patients who are well or poorly perfused.

The Capnostream™35 monitor also provides the clinician with Integrated Pulmonary Index™ (IPI) algorithm, apnea per hour (A/hr) and oxygen desaturation index (ODI) values. The Integrated

Pulmonary Index™ (PI) algorithm is intended for pediatric and adult patients only. A/hr and ODI are intended for age 22 and up.

The device is intended for use in hospitals, hospital-type facilities, during intra-hospital transport, and out-of-hospital Emergency Medical Service applications that include ground and air transport, unless otherwise specified.

The OxiMax™ SPD alert (SPD) feature derived from the pulse oximeter sensor is intended only for facility-use care of adults to detect patterns of desaturation indicative of repetitive reductions in airflow through the upper airway and into the lungs.

The Respiration Rate parameter derived from the pulse oximeter, when used with Nellcor™ adult respiratory sensor, is intended only for the continuous, non-invasive monitoring of respiration rate in adult patients in hospitals and hospital-type facilities.

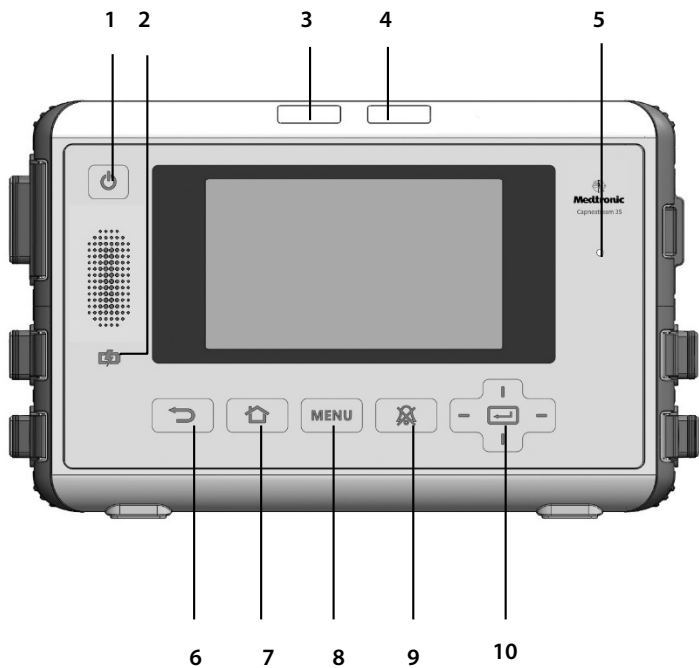
2.3. Buttons, Indicators and Connections

Following are the front, rear, and side views of the monitor showing the display, controls, and external connection points.

2.3.1. Monitor Front Panel

The monitor's front panel controls are described below. Reference [Figure 1. Monitor Front Panel](#), below.

Figure 1. Monitor Front Panel

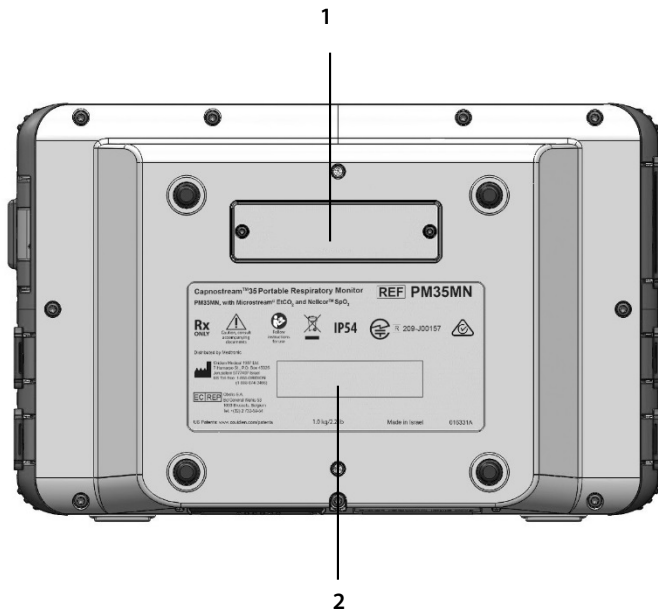


Label	Name	Description	Label	Name	Description
1	On/Off button	Turns the monitor on or off	6	Back button	Returns the user to the previous screen
2	Battery/power indicator	Indicates batteries and power status	7	Home button	Returns the user to the Home screen
3	Medium Priority Alarm LED	Indicates current medium priority	8	Menu button	Opens the Menu screen so a menu

	indicator	alarm status			option can be chosen
4	High Priority Alarm LED indicator	Indicates current high priority alarm status	9	Audio Alarm Paused button	Silences the audible alarms for two minutes
5	Ambient light sensor	Senses level of ambient light and adjusts screen brightness	10	Enter button and directional arrows	Used for navigation and menu selection

2.3.2. Monitor Back Panel

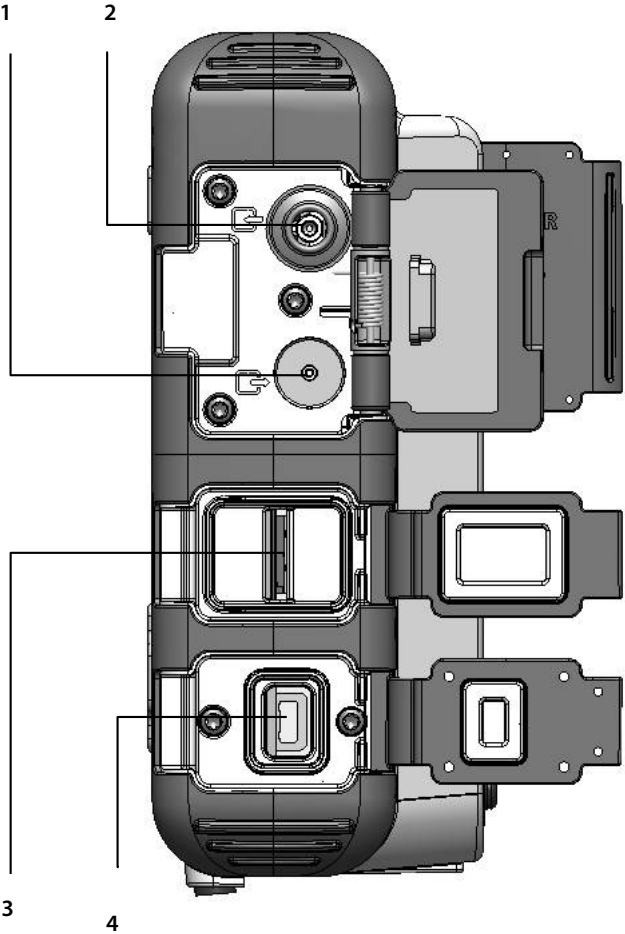
Figure 2. Monitor Back Panel



Label	Function	Description
1	Connector	Connects the monitor with optional accessories
2	Product label	N/A

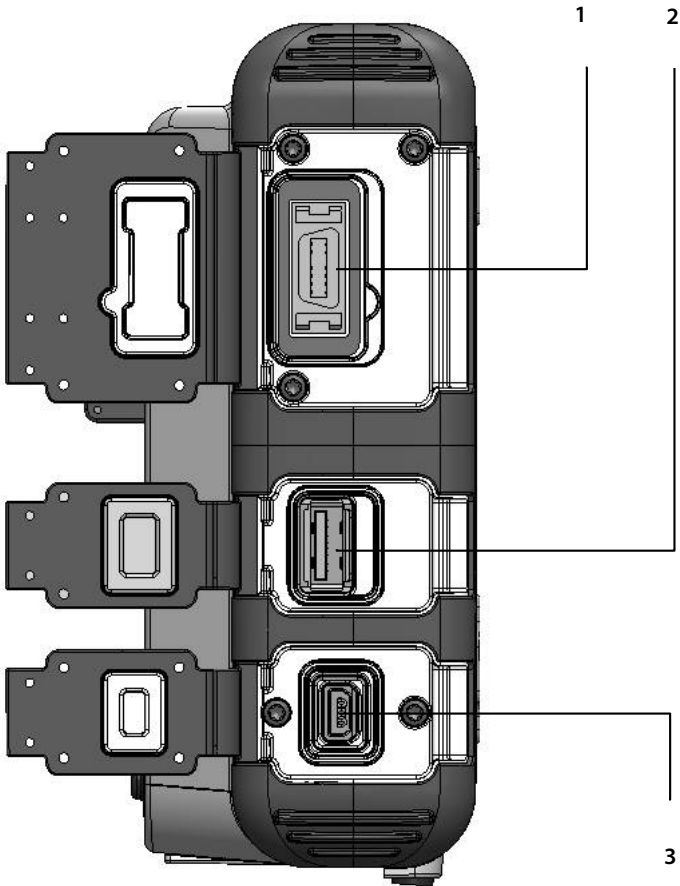
2.3.3. Monitor Right and Left Panels

Figure 3. Monitor Right View



Label	Function	Description
1	Gas outlet	To connect to a scavenger system when the monitor is used in the presence of anesthetic gases. The gas output is a barbed style connector intended for 3/32 inch ID tubing.
2	CO ₂ input connector port	Port for connecting Microstream™ CO ₂ sampling line
3	Micro SD port	Port for micro SD card
4	12VDC connector port	To connect to AC adapter

Figure 4. Monitor Left View



Label	Function	Description
1	Nellcor™ SpO ₂ sensor port	Port for connecting Nellcor™ SpO ₂ sensor

2	USB connector port	Port for connecting USB flash drive
3	Mini USB port	Mini USB port; for technical support use

2.3.4. Display Screen Options

The monitor provides a number of options for the display screen, in order to meet needs of different users and institutions.

2.3.5. Monitoring Display Screen Options

The monitor provides a number of monitoring display screen options, as listed below. Default Home Screen Display #1 will appear when the device is turned on for the first time. By default, Home Screen Displays 1, 2, and 3 will be available when the device is turned on for the first time. To toggle between the screen options, press the right or left directional arrows on the Enter button.

A different Home Screen Display option may be selected as a default using the Institutional Defaults option; this is described in [11.1.7 Home Screen Configuration](#) on page 269.

The Default Home Screen Display #1 will appear when the Home button on the front of the monitor is pressed; likewise, if a different Home Screen Display option is selected as the default, that is the screen that will appear when the Home button is pressed. That default home screen is marked with a home icon () next to the text **HOME** on the screen.

If SPD is enabled, it will appear in the SpO₂ section of all screens.

1. Display Mode Default #1: EtCO₂, RR, SpO₂, and PR values with CO₂ and SpO₂ waveforms (this is the factory default for Default Home Screen)

2. Display Mode Default #2: Numeric Home screen without IPI
3. Display Mode Default #3: EtCO₂, RR, SpO₂, and PR values with IPI trend graph
4. Display Mode Default #4: EtCO₂ and RR values with CO₂ waveform
5. Display Mode Default #5: EtCO₂, RR, SpO₂, and PR values with CO₂ waveform
6. Display Mode Default #6: EtCO₂, RR, SpO₂, and PR values with SpO₂ waveform
7. Display Mode Default #7: EtCO₂, RR, SpO₂, PR, and IPI values with SpO₂ waveform
8. Display Mode Default #8: EtCO₂, RR, SpO₂, PR, and IPI values with CO₂ waveform
9. Display Mode Default #9: EtCO₂, SpO₂, ODI, RR, A/hr, and PR values with IPI trend graph
10. Display Mode Default #10: SpO₂, Nellcor™ RR, and PR values along with SatSeconds™ alarm management parameter and OxiMax™ SPD alarm management parameter, with SpO₂ waveform.

Examples are shown in the screens below.

Note that examples of a standard home screen and a home screen with Nellcor™ RR and OxiMax™ SPD are seen for the Home Screen #1. For all other home screens, one type or the other is shown. For more information about these features, see [4.11 Pulse Oximetry with the monitor](#) on page 92.

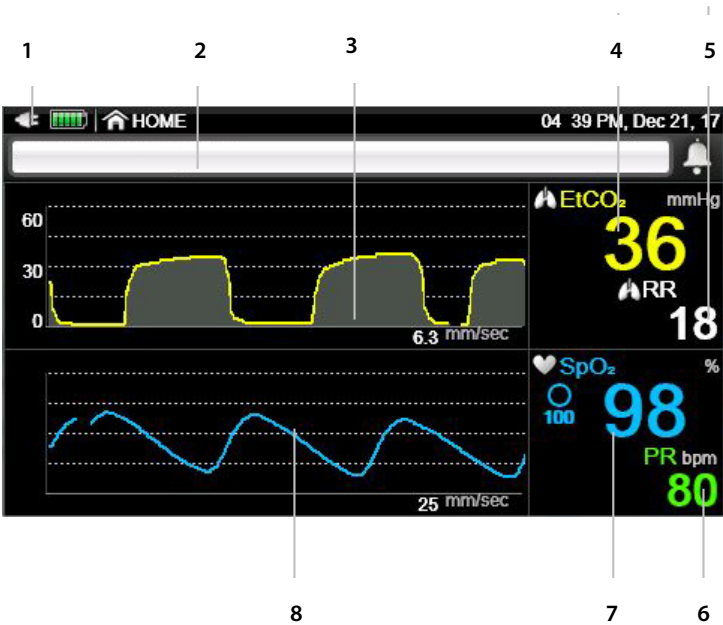


Note:

Note the lung icon that appears next to the EtCO₂ and RR texts, indicating that this data is derived from

capnography, which measures patient respiration, and the heart icon that appears next to the SpO2 data, which indicates that this data is derived from pulse oximetry, which measures data using the patient's pulse. If the device is showing respiration rate data derived from pulse oximetry, a heart icon will appear next to the RR text. For more information, reference [4.11.3 Nellcor™ Respiration Rate](#) on page 93.

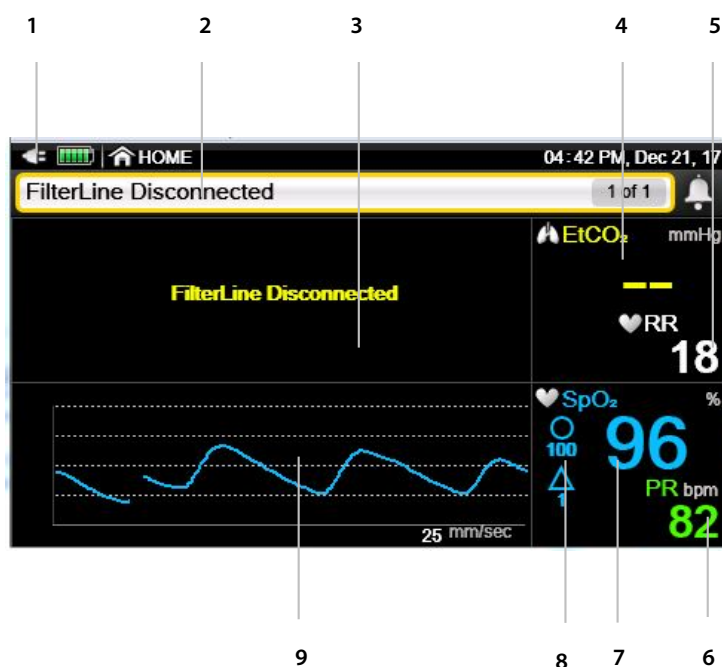
Figure 5. Home Screen Display #1, Standard



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	Message area	Area in which alarm and other messages appear
3	CO ₂ waveform	Display of patient CO ₂ values in waveform format

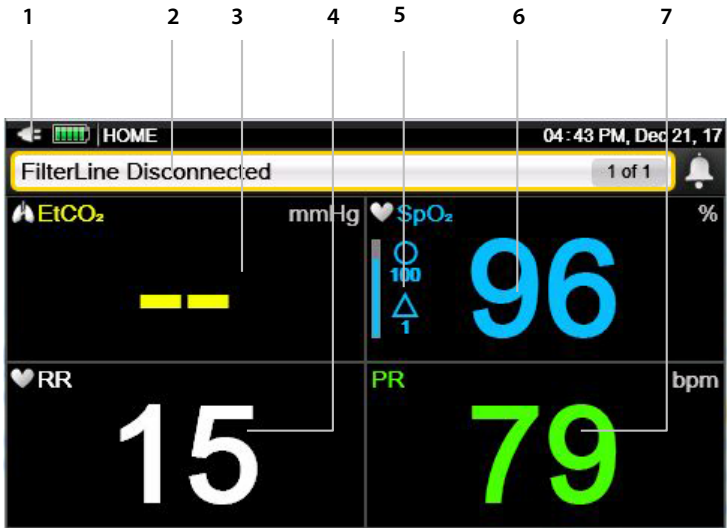
4	EtCO ₂ patient data	Current patient etCO ₂ patient data
5	RR patient data	Current patient RR patient data
6	PR patient data	Current patient PR patient data
7	SpO ₂ patient data	Current patient SpO ₂ patient data
9	SpO ₂ (plethysmograph) waveform	Display of patient SpO ₂ values in waveform format

Figure 6. Home Screen Display #1, with Nellcor™ RR and OxiMax™ SPD Alert



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	Message area	Area in which alarm and other messages appear
3	CO ₂ waveform section	No display of patient CO ₂ values in waveform format is shown, since no CO ₂ sampling line is connected
4	EtCO ₂ patient data section	No current patient etCO ₂ patient data is shown, since no CO ₂ sampling line is connected
5	Nellcor™ RR patient data	Current patient RR patient data based on pulse oximetry; note heart icon
6	PR patient data	Current patient PR patient data
7	SpO ₂ patient data	Current patient SpO ₂ patient data
8	SatSec and OxiMax™ SPD alert icons	SatSec and OxiMax™ SPD alert icons, providing current patient data
9	SpO ₂ (plethysmograph) waveform	Display of patient SpO ₂ values in waveform format

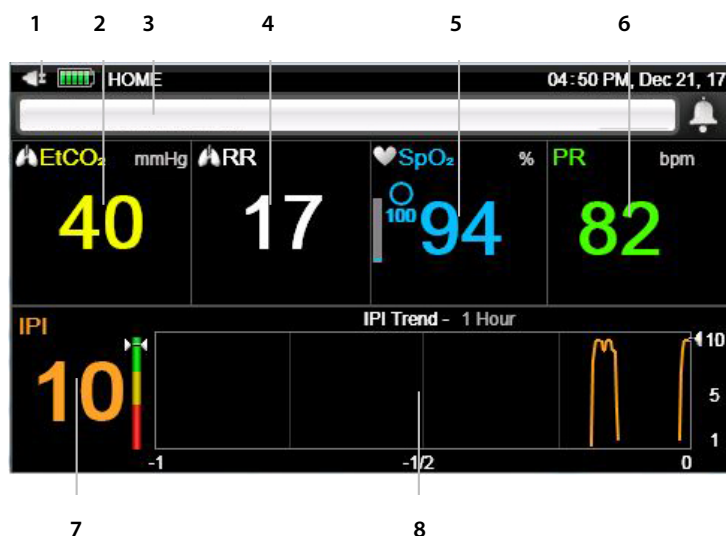
Figure 7. Home Screen Display #2



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	Message area	Area in which alarm and other messages appear
3	EtCO ₂ patient data section	No current patient etCO ₂ patient data is shown, since no CO ₂ sampling line is connected
4	RR patient data	Current patient RR patient data; note the heart icon, indicating that this data is derived from pulse oximetry
5	SatSec and OxiMax™ SPD alert icons	SatSec and OxiMax™ SPD alert icons, providing current patient data

6	SpO ₂ patient data	Current patient SpO ₂ patient data
7	PR patient data	Current patient PR patient data

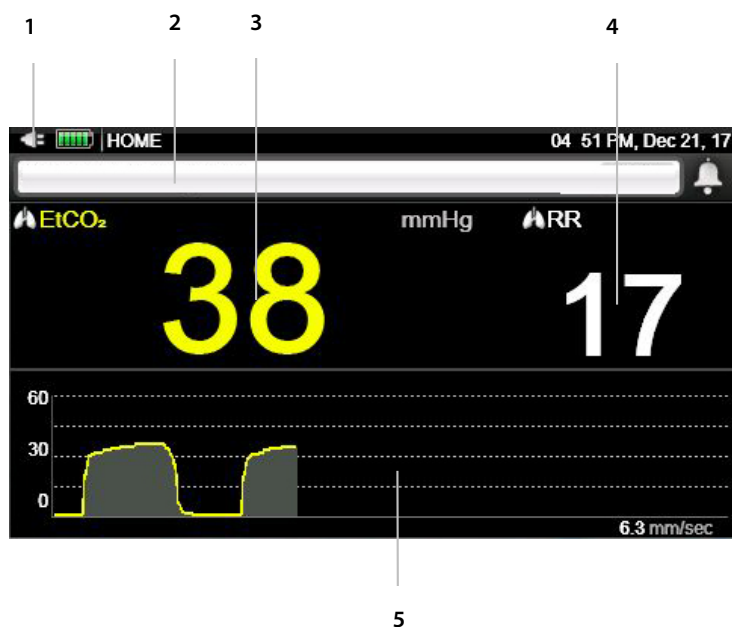
Figure 8. Home Screen Display #3



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	EtCO ₂ patient data	Current patient etCO ₂ patient data
3	Message area	Area in which alarm and other messages appear
4	RR patient data	Current patient RR patient data
5	SpO ₂ patient data	Current patient SpO ₂ patient data

6	PR patient data	Current patient PR patient data
7	IPI value	Current patient Integrated Pulmonary Index™ (IPI) algorithm value
8	IPI trend waveform	Integrated Pulmonary Index™ (IPI) algorithm for current patient as a trend in waveform format

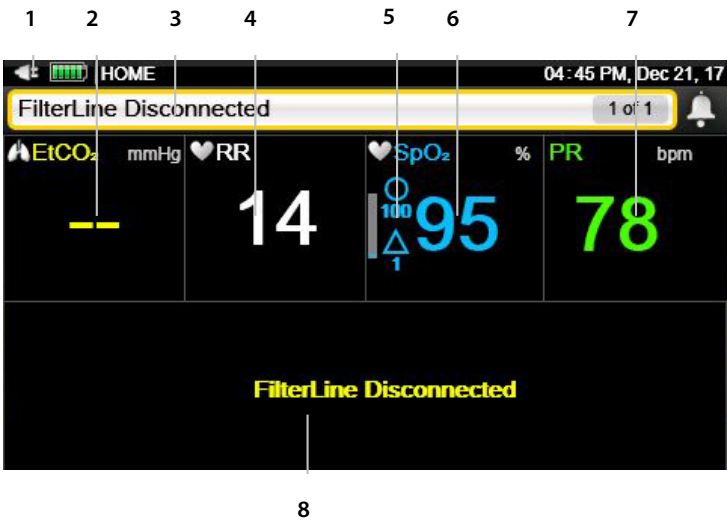
Figure 9. Home Screen Display #4



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connection status of power cord
2	Message area	Area in which alarm and other messages

		appear
3	EtCO ₂ patient data	Current patient etCO ₂ patient data
4	RR patient data	Current patient RR patient data
5	CO ₂ waveform	Display of patient CO ₂ values in waveform format

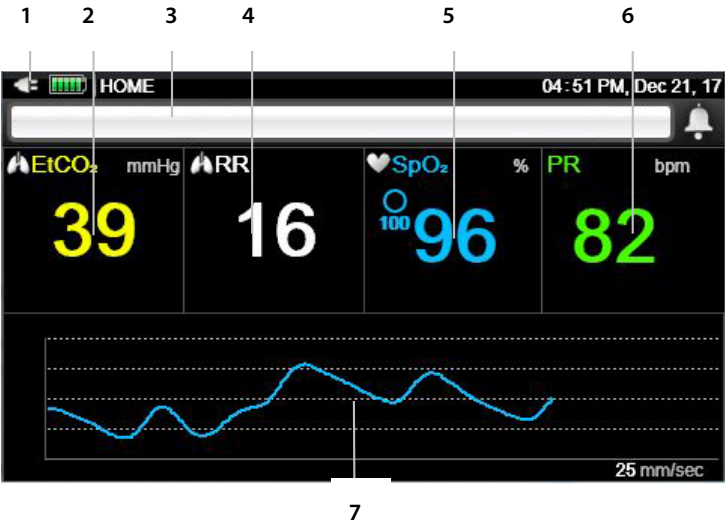
Figure 10. Home Screen Display #5



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	EtCO ₂ patient data section	No current patient etCO ₂ patient data, is shown, since no CO ₂ sampling line is connected

3	Message area	Area in which alarm and other messages appear
4	RR patient data	Current patient RR patient data; note the heart icon, indicating that this data is derived from pulse oximetry
5	SatSec and OxiMax™ SPD alert icons	SatSec and OxiMax™ SPD alert icons, providing current patient data
6	SpO ₂ patient data	Current patient SpO ₂ patient data
7	PR patient data	Current patient PR patient data
8	CO ₂ waveform	Display of patient CO ₂ values in waveform format

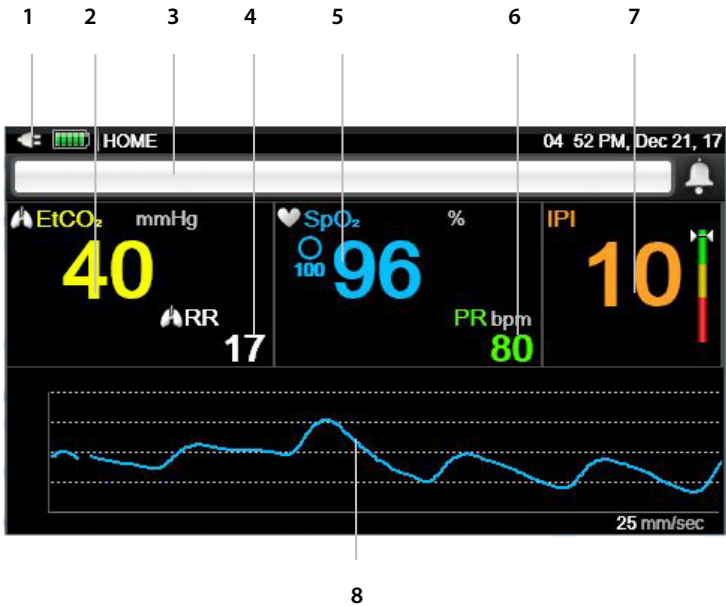
Figure 11. Home Screen Display #6



Label	Function	Description
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1	Power indicators	Indicates power level of batteries and connections status of power cord
2	EtCO ₂ patient data	Current patient etCO ₂ patient data
3	Message area	Area in which alarm and other messages appear
4	RR patient data	Current patient RR patient data
5	SpO ₂ patient data	Current patient SpO ₂ patient data
6	PR patient data	Current patient PR patient data
7	SpO ₂ (plethysmograph) waveform	Display of patient SpO ₂ values in waveform format

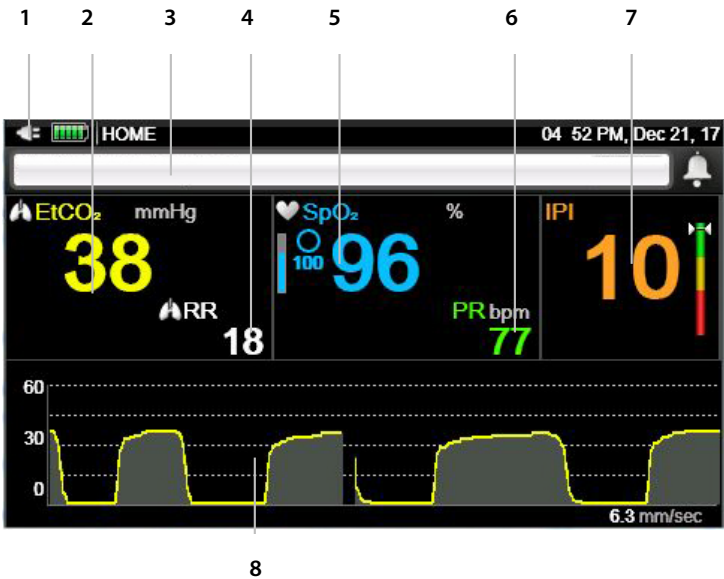
Figure 12. Home Screen Display #7



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	EtCO ₂ patient data	Current patient etCO ₂ patient data
3	Message area	Area in which alarm and other messages appear
4	RR patient data	Current patient RR patient data
5	SpO ₂ patient data	Current patient SpO ₂ patient data
6	PR patient data	Current patient PR patient data

7	IPI value	Integrated Pulmonary Index™ (IPI) algorithm value for current patient
8	SpO ₂ (plethysmograph) waveform	Display of patient SpO ₂ values in waveform format

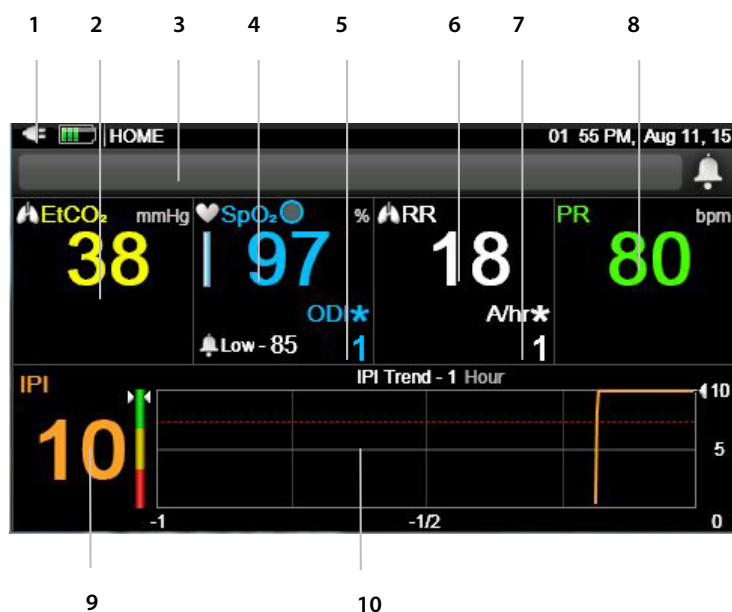
Figure 13. Home Screen Display #8



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	EtCO ₂ patient data	Current patient etCO ₂ patient data
3	Message area	Area in which alarm and other

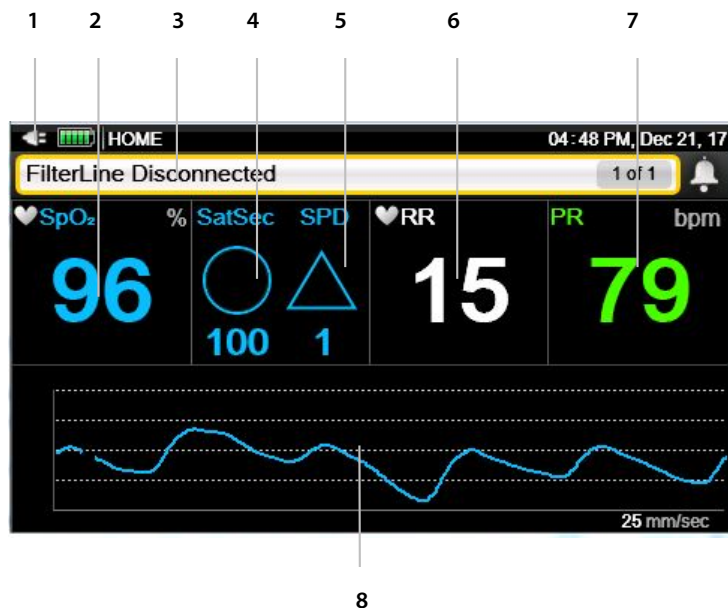
		messages appear
4	RR patient data	Current patient RR patient data
5	SpO ₂ patient data	Current patient SpO ₂ patient data
6	PR patient data	Current patient PR patient data
7	IPI value	Current patient Integrated Pulmonary Index™ (IPI) algorithm value
8	CO ₂ waveform	Display of patient CO ₂ values in waveform format

Figure 14. Home Screen Display #9



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	EtCO ₂ patient data	Current patient etCO ₂ patient data
3	Message area	Area in which alarm and other messages appear
4	SpO ₂ patient data	Current patient SpO ₂ patient data
5	ODI patient data	Current patient Oxygen Desaturation Index data
6	RR patient data	Current patient RR patient data
7	A/hr patient data	Current patient Apneas per hour data
8	PR patient data	Current patient PR patient data
9	IPI patient data	Current Integrated Pulmonary Index™ (IPI) algorithm data
10	IPI trend waveform	Integrated Pulmonary Index™ (IPI) algorithm value for current patient as a trend in waveform format

Figure 15. Home Screen Display #10



Label	Function	Description
1	Power indicators	Indicates power level of batteries and connections status of power cord
2	SpO ₂ patient data	Current patient SpO ₂ patient data
3	Message area	Area in which alarm and other messages appear
4	SatSec patient data	Current patient SatSec patient data
5	SPD patient data	Current patient OxiMax™ SPD alert patient data
6	RR patient data	Current patient RR patient data; note the heart icon, indicating that this

		value is derived from pulse oximetry
7	PR patient data	Current patient PR patient data
8	SpO ₂ (plethysmograph) waveform	Display of patient SpO ₂ values in waveform format

The user may change the display screen as follows:

1. When the device is turned on, the Display Default Home Screen #1 will appear.
2. On the Home screen, click the right directional arrow on the **Enter** button.
3. The Default Home Screen #2 will now appear.
4. Click the right directional arrow on the **Enter** button again to cycle through the Home screen options.
5. The chosen Home Screen will be displayed until the monitor is turned off. When the monitor is turned on again, the Display Default Home Screen #1 will appear.
6. To choose a different Home Screen Display option as a default, so that it remains the default even after the device is turned off, use the Institutional Defaults option; reference [11.1.7 Home Screen Configuration](#) on page 269.

2.3.6. Monitor Turn-off

To terminate operation of the monitor, take the following steps:

1. Remove the CO₂ sampling line and SpO₂ sensor from the patient and, if using a patient case, close the patient case.
2. Press the **On/Off** button for 2-3 seconds to turn off the monitor.

2.3.7. Screen Navigation

To move around the screen:

1. Use the directional arrows to move around the screen.
2. When the screen shows a choice of options/parameters which may each be adjusted to display a list of choices, a frame will appear around the selected section. Click **Enter** to select the option/parameter for which you want to make changes. You may now use the up and down arrows to make your selection.

Figure 16. Sample Parameters Adjustment Screen



Note the inactive arrows on the left and the active arrows with the shape change on the right.

Letters and numbers may be entered as follows:

1. When the cursor is located in a section of the screen in which letters or numbers may be added, click **Enter** and a keyboard will appear on the screen.
2. Navigate within the keyboard using the directional arrows. When you have moved the cursor to the letter/number which you want to input, click **Enter**.
3. When the data input is completed, navigate to **Done** and click **Enter**.

For setting time, date and other changes, reference [2.3.8 Date, Time, Language, and Other Options](#), below.

2.3.8. Date, Time, Language, and Other Options

Date and time may be changed as follows:

1. Click the **Menu** button on the front panel.
2. The first option in the menu is **Setup>Alarm Setup**. Click the down arrow on the **Enter/Navigation** button to navigate to **System Setup** and click **Enter**.
3. Use right arrow to navigate to **Date and Time** and click **Enter**.
4. Select the first item you want to update and click **Enter**.
5. Using directional arrows, navigate to the value you want to display and click **Enter**. Make the desired change and click **Enter**.
6. To make additional changes, navigate to the next item you want to update, and repeat.
7. When you are done, navigate to the **Apply** window and click **Enter**.

Languages and other options may be changed as follows:

1. Click the **Menu** button on the front panel.
2. The first option in the menu is **Setup>Alarm Setup**. Click the down arrow on the **Enter/Navigation** button to navigate to **System Setup** and click **Enter**.
3. Use the right arrow and the down arrow to navigate to the desired item and click **Enter**.
4. Using directional arrows, navigate to the value you want to display and click **Enter**. Make the desired change and click **Enter**.

5. To make additional changes, navigate to the next item you want to update, and repeat.
6. Return to the home screen using the Home button to implement the changes.

Table 2. Display Options Available on System Screen

Parameter	Default	Options
Date and Time	NA	NA
Language	English	English, French, German, Spanish, Italian, Dutch, Swedish, Norwegian, Japanese, Russian, Chinese, Korean, Finnish, Polish, Portuguese
Event Marking Mode	Detailed	Quick, Detailed
Graphical Trend Display	4 Hour	1,2,4,8,12, Hour
Tabular Trend Increment Display	30 sec	1,5,15,30 sec, 1,5,15,30 Min, 1 hour
Graphical Trend Display Mode	Recorded	Calendar, Recorded
Home Trend Display View	1 Hour	1,2,4, Hour
Brightness Index	Auto	Auto, 1.2.3.4.5.6.7,8,9,10

2.3.8.1. Keyboard locking

The monitor keyboard may be locked by the user, perhaps to avoid accidental changes by the patient or for another reason. Take the following steps to lock the keyboard:

1. Press the **Enter** key for more than two seconds to lock the keyboard. An on-screen message will indicate that the keyboard is locked.
2. To unlock the keyboard when it is locked, press the **Enter** key for more than two seconds. An on-screen message will indicate that the keyboard is unlocked.

2.3.8.2. Screen timeouts

After one minute of no user interaction, all set-up type screens, except screens in the service mode, will return to the screen that they were on last. If a setup screen times out with a change having been in process, no change is made.

2.3.9. Capnostream™35 Monitor Operational Check Sheet

To get the monitor up and running quickly and smoothly, follow the list of instructions below:

- ☐ 1. Unpack the monitor

Remove the monitor and the accessories from the box.

Check that the items on the enclosed packing list are included.

- ☐ 2. Install the battery pack

Refer to [3.2.2 Removable Battery Pack Installation](#) on page 53 for installation instructions.

- ☐ 3. Turn on the monitor

Plug an AC adapter into the 12VDC port on the right side panel of the monitor.

Plug the adapter cable into the mains AC supply.

The battery charging and power indicator at the front of the monitor will turn on.

Press the Power ON/OFF button  on the front panel to turn on the monitor. The screen will light up, showing that the monitor is turned on.

- ☐ 4. Change the date, time or language

- ☐ 5. Set the Patient Type
- ☐ 6. To toggle between the home screen options, click the right/left button on the front panel navigational pad.
- ☐ 7. Check Alarm Limits (**Menu>Setup>Alarm Setup>Select Alarm>Select Limit**)
- ☐ 8. Open a Patient Case (**Menu>Actions>Patient Admit**)
- ☐ 9. Connect a Sampling Line

Open the CO₂ input connector shutter on the right side panel of the monitor and connect the appropriate sampling line to the CO₂ input port at the top of this section.

Connect the sampling line to the patient as described in the Directions for Use supplied with the sampling line. The sampling line connector should be screwed clockwise into the monitor CO₂ port until it can no longer be turned, to ensure that it is connected securely to the monitor. This will assure that there is no leak of gases during measurement at the connection point and that measurement accuracy is not compromised.

- ☐ 10. Connect a SpO₂ Sensor

Connect the SpO₂ extension cord firmly to the monitor SpO₂ sensor port on the left side panel of the monitor, and then connect the appropriate SpO₂ sensor to the extension cord.

Connect the SpO₂ sensor to the patient as described in its Directions for Use.

- ☐ 11. Once either sensor or both sensors are connected to the monitor, it is ready for operation. Patient data should begin appearing on the screen within a few seconds.
- ☐ 12. View Trends if desired (**Menu>Trend> Graphical Trend or Tabular Trend**)
- ☐ 13. Set Up Data Transfer as required (if applicable; connectivity accessories must be purchased separately) (reference [5.4 Data Transfer](#) on page 191).

3. Installation

This chapter describes the physical components of the monitor and how to set up the monitor so it is ready for use.

3.1. Unpacking and Inspecting the Monitor

Unpack the monitor and check all the components before performing any further procedures.

To unpack and inspect the monitor:

1. Carefully remove the monitor and the accessories from the box.
2. Check that the items on the enclosed packing list are included:
 - a. Monitor
 - b. Mains electrical power cord (AC cable)
 - c. AC adapter with DC cable
 - d. Removable battery pack
 - e. Operator's manual
 - f. Monitor quick guide
 - g. CD with additional documentation (this manual in additional languages)

3. Inspect each component.

If any component is damaged or missing, contact your local representative.

Please note that in some models the DC power cable is provided with a two-color connector. When using such a cable, the connector should be attached to the device with the lighter color and the arrow facing upwards.

When unpacking the monitor, dispose of packaging waste according to local regulations for the disposal of packaging waste.

3.2. Batteries

3.2.1. Batteries

The device includes an internal battery, which is not accessible by the user, and a removable battery pack. Both are lithium-ion batteries.



WARNING:

Charge the monitor fully before your first use of the monitor, by attaching the monitor to AC power with the removable battery installed. See 3.2.5 Battery and AC Connections Indicators for more information.



WARNING:

The unit should always be operated with the removable battery installed in order to provide back-up power in the event of a momentary or temporary power outage.



Caution:

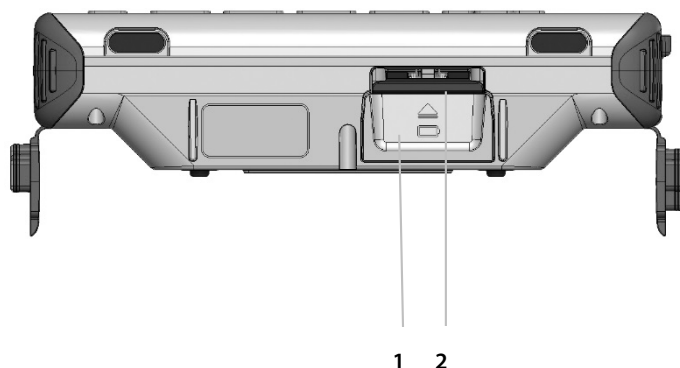
Prolonged storage of the device above 60° C might cause performance degradation of the batteries.

3.2.2. Removable Battery Pack Installation

The battery pack may be installed as follows:

1. Slide the battery pack into the battery pack compartment at the bottom of the monitor, with label facing downwards, towards the back panel of the monitor.
2. Push the battery pack in all the way, until a click is heard. The battery pack is now installed.
3. The device is shipped with a battery that is not fully charged. Before first use, charge the battery until the indicators indicate it is fully charged.

Figure 17. Monitor Bottom, with Battery Pack Installed



Label	Function	Description
1	Battery pack	Battery pack installed in monitor
2	Black lever	Black lever used to open the battery pack compartment

3. To remove or switch the battery pack, open the battery pack compartment of the monitor by pushing the black lever up towards the front of the monitor.
4. The battery pack will release itself from the compartment. Pull the battery pack out all the way.



Caution:

There should always be a removable battery installed in the device. If the removable battery is not installed, the unit will operate properly on AC power, and will operate on the internal battery for a very limited period (for battery exchange/hot swap purposes), but if AC power is lost for any reason, the monitor will work only for a limited period, until the internal battery's charge is used up.



Caution:

Only the battery pack provided with this monitor shall be used for the monitor. Other batteries may not operate correctly.



Caution:

The battery pack supplied with this monitor should not be used with other devices.

The battery icon indicates the charge level of the removable battery.

Reference [3.2.5 Battery and AC Connections Indicators](#) on page 55.

3.2.3. Battery and Power Usage

If power is lost when the monitor is operating from AC power, it automatically switches to the removable battery pack for power. If the removable battery pack is empty, the monitor includes an internal battery which can supply power for a limited period while the user inserts another removable battery pack or attaches the monitor to AC power.

The battery charging and power indicator at the front of the monitor () indicates the battery/power status, as described in [3.2.5 Battery and AC Connections Indicators](#) on page 55.

The monitor can work using the removable battery pack for approximately 3 hours. The monitor can work using the permanent internal battery for 20 minutes. It is recommended to use the internal battery only for battery switching periods and not for more extended use.

Both batteries will charge when the unit is plugged in. Charging time is up to 5 hours when the unit is turned off, and up to 8 hours when the unit is turned on. The removable battery may also be charged in the external battery charger (see [8.2 Available Accessories](#) on page 227).

The battery icon will show the battery pack's approximate charge level. A Battery Low alarm appears when approximately 30 minutes of battery charge remains.

3.2.4. Battery Hot Swap

The monitor includes an internal permanent battery pack, to maintain battery power during battery switching while the monitor is turned on but not attached to AC power. Remove the removable battery pack as described above and insert another battery pack.

When the monitor is attached to AC power, both the internal permanent and the external removable battery packs will charge.

3.2.5. Battery and AC Connections Indicators

The battery pack charge level and AC power connections should be confirmed before each use.

Recharge the removable battery pack when the **Battery Low** alarm appears on the display screen. To recharge the battery, make sure

that the monitor is plugged into the AC mains or use the external battery charger.

The battery charging and power indicator at the front of the monitor () indicates the battery/power status, as follows:

- The indicator is green when the monitor is connected to AC power and both batteries are fully charged.
- The indicator is orange when the monitor is connected to AC power and both batteries are currently being charged.
- The indicator is red when the monitor is connected to AC power and both batteries are not being charged as the result of a malfunction.
- The indicator is off (no light emitted from the indicator) when the monitor is not connected to AC power.

An on-screen indicator () will appear whenever the removable battery is installed, indicating the current status of the battery. This indicator will flicker when the removable battery charge level is very low (reference [3.2.3 Battery and Power Usage](#) on page 54).

An on-screen indicator () will appear if the internal battery is not charged, or if it maintains a small charge only. In this case, a hot swap of the removable battery should not take place, and the device should be attached to AC power to charge the internal battery. If the indicator appears even after the device is attached to AC power, the device will need to be serviced; contact Capnographytechnicalsupport@medtronic.com.

For normal operation, always check that the battery/power indicator is green or orange during monitor use. This will ensure the battery is charged during use and the monitor is prepared in case of a power outage. In the case of a patient transfer, the unit can be unplugged and transferred with the patient. Care should be taken to reconnect the monitor to the AC mains following the transfer.

3.2.6. Handling the Battery Pack



Caution:

Do not immerse the removable battery pack in water; it may malfunction.



Caution:

Recharge the removable battery pack only in the monitor or the external battery charger (reference [8.2 Available Accessories](#) on page 227) to avoid possible heating, burning or rupture of the battery pack.

3.2.7. Storing the Battery

The removable battery should be stored outside the device. The battery pack has an automatic discharge feature. You must periodically check the charge level of the battery pack. The battery may be stored as follows:

The battery pack must be stored in a cold, dry area, not inside the monitor. Its charge decreases over time. To restore the battery pack to full power, recharge the battery before use.

Optimum storage for a removed battery is room temperature. The batteries must be stored at the following temperature ranges:

- Less than 1 month: -20 to +50°C
- 1 month to 3 months: -20 to +40°C
- 3 months to 1 year: -20 to +20°C

3.2.8. Disposing of the Battery



Caution:

Do not dispose of the battery pack in fire; it may explode.

Follow local governing ordinances and recycling instructions regarding disposal or recycling of batteries.

3.2.9. Internal Battery Pack

The internal battery pack is not removable and should not be handled by the user.

3.2.10. Monitor Mounting Plate

The monitor may be attached to a 75mm VESA™* standard mounting plate by attaching the VESA™* adapter assembly to the back panel of the monitor. For more information, see the Mounting Adapter Kit directions for use.

TM* Trademark of its respective owner.

3.2.11. Operation in Helicopter Transport

The device was tested for use in helicopters according to the parameters listed in 10.9 General Characteristics on page 241. The performance of the device was not tested in fixed-wing aircraft.

When installing the device in a helicopter for use during transport, please note the following:

The device shall be mounted using a VESA™* mount and the monitor mounting plate, positioned so that the bottom of the device is parallel to the floor of the helicopter. The device shall be placed in a position where the controls are easily reached by the caregiver, and the screen may be viewed clearly by the caregiver.

3.3. Periodic Maintenance

If your institution has a periodic maintenance database, log the monitor in this database for its periodic regular maintenance and its calibration procedure.

Regular maintenance is required every 24 months, starting from the installation date.

Calibration is required after the first 1,200 hours of use (or 12 months, whichever comes first) and thereafter every 4,000 hours of use (or 12 months, whichever comes first). The number of hours remaining until calibration will appear on the monitor's Service Screen. For more details about calibration and other maintenance procedures, reference [6 Preventive Maintenance](#) on page 207.

4. Operation

4.1. Turning on the Monitor

This section explains how to turn on the monitor.



Caution:

The monitor is intended only as an adjunct in patient assessment. It must be used in conjunction with clinical signs and symptoms.



Caution:

Use only Microstream™ etCO₂ consumables and Nellcor™ SpO₂ sensors, to ensure that the monitor functions properly.



Note: Monitor should be fully charged before first use.

Take the following steps to turn on the monitor:

1. Attach the AC adapter to the 12VDC connector port on the right side of the monitor.
2. Attach the power cord to the AC adapter.
3. Plug the power cord into an external power source.
4. The battery/power indicator on the front of the monitor should be either green or orange once the monitor is

plugged in. If necessary, the monitor can work for a limited period of time on battery power; reference [3.2.3 Battery and Power Usage](#) on page 54 for details.

5. Press the **On/Off** button () at the front of the monitor. The monitor will turn on and the screen will light up. The yellow and red alarm lights will light up briefly and a beep will be emitted by the monitor speaker.



Caution:

If the battery charging and power indicator at the front of the monitor () is red or off, the monitor is running on battery power only and will stop operating when the battery is discharged.



Caution:

If the yellow and red alarm lights do not light up or there is no sound from the speaker, the monitor should not be used and should be sent for servicing.

6. The default mode for the device is the EMS-enabled mode, in which the monitor will immediately be ready for monitoring, once a CO₂ sampling line and an SpO₂ sensor are attached.
7. If EMS mode is disabled on the device, the monitor will display the following on-screen message when turned on:

Figure 18. Trend Memory Message



8. In this case, click **No** to continue monitoring the same patient, or click **Yes** if you are now monitoring a new patient and want to clear any information regarding the previous patient to avoid mix-ups. The monitor is now ready to begin monitoring, once a CO₂ sampling line and an SpO₂ sensor are attached. If you want to enable or disable EMS mode, reference [4.8.3 EMS Mode](#) on page 78.
9. Before attaching the monitor to a patient, check the date and adjust to the current date if required, following the instructions in [2.3.8 Date, Time, Language, and Other Options](#) on page 44.

4.2. Monitor Preparation for a Patient



Caution:

If any monitor response does not seem appropriate, do not use the monitor. Instead, contact your local Service Center or Capnographytechnicalsupport@medtronic.com.

The following steps describe the procedure for preparing the monitor for a patient.

1. Connect the sampling line, the SpO₂ sensor, or both to the monitor, following the instructions in this manual. To connect the sampling line and sensor, reference [4.10.2 FilterLine™ Sampling Line Connection](#) on page 85 and [4.11.2 Connecting an SpO2 Sensor to the monitor](#) on page 92.
2. Once either or both the SpO₂ sensor and the sampling line are connected to the monitor, it is ready for operation.
3. It is possible to use either the Capnography function (etCO₂) or the Pulse Oximetry function (SpO₂) individually. If you only want to operate one function, connect ONLY the component for that function, and the monitor will operate normally.

The monitor is for use with one patient at a time only.

4.2.1. Patient Types

There are six different patient types recognized by the monitor, listed below.

- Infant/Neonatal: for patients from birth to the age of one year
- Pediatric 1-3 yrs: for patients aged one to three years
- Pediatric 3-6 yrs: for patients aged three to six years
- Pediatric 6-12 yrs: for patients aged six to twelve years.
- Adult 12-22: for patients aged 12 to 22 years
- Adult 22+: For patients aged 22 years and up

The patient type is displayed at the top left hand corner of the menu screen.

The original default patient type on the monitor is ADULT 22+. The default patient type may be changed in Institutional Defaults

(reference 11.1.1 Changing Institutional Defaults on page 251.) The device will revert to the default patient type every time the trend is cleared. (Stopping a case will clear the trend memory and revert to the default patient type.) Other actions, including boot-up in which the trend is not cleared, or opening a new case, will retain the current patient type.



WARNING:

The characteristics of a breath are calculated differently for the different patient types. Setting the correct patient type is therefore very important. Incorrect setting will result in inaccurate monitoring of the patient's respiration, and could lead to incorrect alarm limits or produce incorrect patient IPI data.

To change the patient type:

1. Click the **Menu** button on the front panel.
2. The first option in the menu is **Setup>Alarm Setup**.
3. On the Setup Menu screen, navigate to the Patient Type Icon (on the upper left of the Menu screen) using the directional arrows and click **Enter** to open the drop-down menu.

Figure 19. Patient Type Drop-down List



4. Using the directional arrows, select the relevant patient type.

5. Select **Enter** and, if requested, **Confirm**, using the **Enter** button. If there is a contradiction between the patient type and patient age, check your patient data and change as required.

4.3. Patient Cases and Patient ID Numbers

The monitor may be used for one patient at a time. It is recommended that all patients should be recorded as patient cases, as described below.

1. Click the **Menu** button on the front panel.
2. On the **Menu** screen, navigate to and select **Actions>Patient Admit**.
3. Change the automatically generated patient ID number (which includes the current date according to the device and other information - see Table 20) to a number that will identify the patient in your system, if desired. The patient may also be identified with a case number, also inputted on this screen.
4. Input patient data as required. Note that a patient case may be opened even if you do not add patient data.
5. Navigate to and click the **Start Case** button on the left. A window will notify you that a case has been started, and the text on that button will change to **Stop Case**.
6. To stop a case that has already been started, navigate to the same screen and click the **Stop Case** button. Note that stopping a case will clear the trend memory and cause the device to revert to the default patient type.

4.4. Patient Events

When scanning patient history in the monitor, it is often useful to have a record of patient events that could have influenced the recorded readings. Events are added to the patient trend record as follows:

1. Click the **Menu** button on the front panel.
2. On the **Menu** screen, navigate to and select **Actions>Events**. The Event screen will open. Reference [Figure 20. Event Marking Screen](#) on page 68.
3. The Medication Event list will be seen on the main screen; if you want to add a Medication Event, navigate to the relevant event and click **Enter**. If you want to add a PaCO₂, Patient or Intervention Event, navigate to the **PaCO₂**, **Patient** or **Intervention** buttons and click **Enter**. For PaCO₂, navigate to and set the date, time and number of PaCO₂ using the **Enter** button. For a Patient or Intervention Event, navigate to and select the desired event.
4. If you want to add a different event, navigate to and click the **Text Entry** button and add free text, up to 20 characters, and date and time for this event. This date and time may be in the past.
5. Once you have inputted Event data, clicking **Enter** will return you to the Home screen. If you decide not to input Event data, click **Back** or **Home** to return to the Home screen.

Trend recording will continue as before; entering an event does not change the trend recording status.

Figure 20. Event Marking Screen



The default event marking options in the monitor are as follows:

Table 3. Event Markings

MEDICATION	PATIENT	INTERVENTION
Fentanyl	EATING	OXYGEN
Versed	DRINKING	SUCTION
Midazolam	Coughing	Adj Airway
Morphine	Ambulating	Narcan
Demerol	Chest PT	Romazicon
Propofol	Turned	Neb tx
Surfactant	Snoring	Stimulated
Medication-Other	Patient-Other	CO ₂ Insufflation
		ABG
		Intervention-Other

New events may be added and any of these event markings may be changed, using Institutional Settings. Reference [11.1.1 Changing Institutional Defaults](#) on page 92 for more information.

4.5. Alarm and Pulse Volumes and Alarm Delay

The alarm volume can be made louder or softer for patient alarms and pulse tone. The pulse tone can also be turned off using the SpO₂ menu. By default, the pulse tone is turned off.

4.5.1. Alarm Volume

Follow the steps listed below to make the alarm volume louder or softer.

1. Click the **Menu** button on the front panel.
2. Navigate to and select the **Setup>Alarm Volume** screen. Click **Enter** and select the desired volume using the directional arrows. Select **Home** to return to the Home screen while retaining any changes you have made.
3. If the monitor is set to the Alarm Volume default setting, the set volume will remain active when the monitor is turned off.

Other alarm volume options settable in Institutional Defaults are **Maximum** and **Audio Off**. To change the Alarm Volume Institutional Settings, reference [11.1.6.1 Main Monitor settings](#) on page 266.

4.5.2. Pulse Tone and Button Click Volume

The pulse tone and button click sound can be turned on or off as follows:

1. Click the **Menu** button on the front panel.
2. Navigate to and select **Setup > SpO₂ Setup** on the Menu Screen.
3. Navigate to **Pulse Tone** and click **Enter** to view options. The default value is **Off**. Navigate to **On** and click **Enter** to turn on the pulse tone and button click sound.
4. This will return you to the SpO₂ Setup screen. Select **Home** to return to the Home screen.

The pulse tone and button click volume can be set as follows:

1. The volume can be adjusted whether or not the Pulse Tone has been turned on as described above.
2. Click the **Menu** button on the front panel.
3. Navigate to the **Setup > Alarm Volume** screen. Click the down arrow to navigate to the Pulse Volume section of the screen. Click **Enter** and select the desired volume using the directional arrows. Select **Home** to return to the Home screen while retaining any changes you have made.

4.5.3. Alarm Volume Default Options



WARNING:

The Audio off option (which will create permanent audio alarm off until reversed) should be set in the Institutional Defaults only in a situation in which the caregiver is also monitoring the patient by other means. When installing the monitor as part of hospital's remote monitoring system, continuous wireless transmission of data/alarm signals is not guaranteed. It is therefore advised that, when setting permanent audio off mode while using the monitor with a remote monitoring system, extra care for operator-patient proximity is in place in order to assure operator awareness to the onset of alarm signals.

4.5.4. Alarm Delay

Alarm delay may be set for all patient alarms except for the No Breath and SatSeconds™ alarms, using the following steps:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **Service** on the Setup Menu Screen.
3. Enter Service password and select **Done**.
4. Navigate to **Institutional Defaults>Alarms>Alarm Delay**.
5. Navigate to and select **Enter** on the relevant alarm.
6. Navigate and select to the desired alarm delay. Click **Enter**.
7. Select **Home** to return to the Home Screen.
8. The device will ask you to save settings and shut down. Click **Shutdown** to shut down and then restart to retain the settings change you have just made.

If SatSeconds™ is selected and the SatSeconds™ value triggers an SpO₂ alarm, any SpO₂ delay that has been set will be superseded and the alarm will occur despite the SpO₂ delay.

4.6. Use of Scavenging System

When using the monitor with anesthetics, nitrous oxide or high concentrations of oxygen, connect the gas outlet to a scavenger system. The gas output is a barbed style connector intended for 3/32 inch ID tubing.

The gas outlet may be seen in [Figure 3. Monitor Right View](#) on page 23.

4.7. Use of Pump Off Mode

Use the Pump Off mode whenever performing suction or lavage. During Pump Off mode, pump activity is suspended to protect the monitor from drawing in liquids which could cause a malfunction.

In the Pump Off mode, the CO₂ module pump is switched *OFF* for a preset time to prevent liquids from entering the monitor.



WARNING:

If at any time the monitor displays the **FilterLine Blockage** message, replace the sampling line.

Set the pump mode as follows:

1. The monitor must be currently monitoring CO₂ in order to be placed in Pump Off mode.
2. Click the **Menu** button on the front panel.
3. Navigate to and select **Actions** on the menu screen and **Pump Off** on the Actions menu screen. Click **Enter**.
4. The monitor will indicate, in the CO₂ area and the message area, the time remaining until the pump will automatically turn back on.
5. The standard Pump Off period is 15 minutes. This may be changed in the CO₂ Setup screen (temporarily, until the monitor is turned off; reference [4.10.4 Adjustable CO₂ Parameters](#) on page 88), or in the Institutional defaults screen (permanently; Reference [11.1.1 Changing Institutional Defaults](#) on page 92.)
6. To turn the CO₂ pump back on before the set time period has passed, click the menu button on the front panel and then navigate to and select **Pump Off for xx:xx – Exit Now**.

7. To extend the timer for another Pump Off period, beyond the current period, click the menu button on the front panel and then navigate to and select **Extend Timer**. Clicking the **Extend Timer** button will begin another Pump Off period of the same length as the set Pump Off period.

While the pump is off, CO₂ is not monitored and no CO₂ waveform, etCO₂, or respiration rate number values are displayed. SpO₂ and pulse rate monitoring continues.

4.8. Additional Monitor Modes

4.8.1. Demo Mode

Access the Demo Mode as follows:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **Service** on the Setup Menu Screen.
3. Enter Service password and select **Done**.
4. Navigate to and select **Demo Mode**. Navigate to and select the desired demo patient type option. Click **Enter**.
5. After a brief waiting period, the monitor will move to Demo Mode. An on-screen message indicating Demo Mode can be seen on the upper right, next to the time and date, in black text on a yellow background.
6. To exit Demo Mode, turn off the monitor. When the monitor is restarted, it will no longer be in Demo Mode.



Note:

No patient monitoring will take place while the monitor is in Demo Mode, even if a sampling line and sensor are attached to the monitor. To return to monitoring, turn off the monitor and restart it.



Note:

The SPD alert feature will not be available in the Demo Mode.

When a device is in Demo Mode, the trend data in the device will not be erased. However, if trend data is viewed while the device is in Demo Mode, the trend data displayed will be the Demo data. When the device is turned off and turned on again, the device will show the previous patient's trend data, if the user has not cleared it (by choosing **Yes** when the device asks whether to clear trend data). For the period of time during which the Demo mode was viewed on the screen, no data will be available, since no data is recorded while the device is in Demo mode.

4.8.2. Parameter Standby Mode

The monitor may be placed in a separate parameter standby mode for capnography and/or for pulse oximetry. The purpose of this option is to enable the user to turn off this the alarm that sounds when a sampling line/sensor is removed from the monitor at will. Once the monitor is in Parameter Standby Mode, that mode can be activated as required.

By default, Parameter Standby Mode is disabled.

In both Standard Mode (when Parameter Standby Mode is not enabled) and Parameter Standby Mode, removing a sampling line or SpO₂ sensor/extension cable from the monitor will cause a message to appear on the screen (**FilterLine Disconnected** or **SpO₂ Sensor Disconnected**, as relevant) and cause an alarm to sound. Removing the SpO₂ sensor from the patient will set off an audible alarm and an

on-screen message. The purpose of this alarm is to prevent unauthorized disconnection of a sampling line/SpO₂ sensor from the monitor, perhaps by patients or patients' visitors.

When there is no CO₂ sampling line (for capnography) or no SpO₂ sensor (for pulse oximetry) connected to the monitor, the relevant Parameter Standby Mode may be activated (as described below). Once it is activated, there will be no audible alarms for the relevant parameter.

Parameter Standby Mode may be enabled as follows:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **Service** on the Setup Menu Screen.
3. Enter Service password and select **Done**.
4. Select **Institutional Defaults>Monitor**.
5. Navigate to Parameter Standby Mode and click **Enter**. Set Parameter Standby Mode to **Enabled** and click **Enter**.
6. Select **Home** to return to the Home Screen. The device will ask you to save settings and shut down
7. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made. Once Parameter Standby Mode has been enabled, it may be activated as follows:

1. Remove the sampling line and/or SpO₂ sensor from the monitor or remove the SpO₂ sensor from the patient.
2. Press the audio alarm paused hard key at the front of the monitor () for more than two seconds to enter the Parameter Standby Mode.
3. The monitor will emit the Standby pattern beep when the monitor is successfully placed in Parameter Standby Mode,

and the monitor screen will indicate the following messages in the waveform areas (and in the message areas, alternately with other relevant messages): **CO₂ Standby, SpO₂ Standby.**

If a CO₂ sampling line is connected to the monitor, the Parameter Standby Mode for CO₂ will not open even if the audio alarm paused key is pressed as required. Likewise, if an SpO₂ sensor is connected to the monitor and to a patient, the Parameter Standby Mode for SpO₂ will not open even if the audio alarm paused key is pressed as required. This is to prevent entering this mode while a patient is being monitored. Pressing the audio alarm paused key will therefore start Parameter Standby Mode for both CO₂ and SpO₂ if both are disconnected, or for one if only one is disconnected.

Please note the following:

- CO₂ standby will be exited automatically when a CO₂ sampling line is attached to the monitor.
- SpO₂ standby will be exited automatically when an SpO₂ sensor is attached to the monitor and to a patient.

Thus, if the user is unaware that the monitor is in the Parameter Standby mode, simply reattaching a sampling line to the monitor or SpO₂ sensor to the patient will cause the monitor to exit the Parameter Standby mode and monitoring to resume.

During Standby Mode, the current value of any parameter in standby will be displayed as a double dash in the relevant area on the home screen.

Table 4. Message and Alarm Status during Different Parameter Standby Situations

Feature	Parameter Standby Situations
---------	------------------------------

	Status when Parameter Standby mode disabled	Status when Parameter Standby mode enabled but not activated	Status when Parameter Standby mode enabled and Parameter Standby activated
FilterLine disconnected (from monitor) on-screen message / SpO ₂ sensor disconnected (from monitor) on-screen message	Yes	Yes	Yes
FilterLine disconnected (from monitor) alarm / SpO ₂ sensor disconnected (from monitor) alarm	No	Yes	No
SpO ₂ Sensor not on Patient Medium Priority audible alarm	Yes	Yes	No
SpO ₂ Sensor not on Patient on- screen message	Yes	Yes	Yes
CO ₂ Standby message / SpO ₂ Standby message	No	No	Yes

Feature	Parameter Standby Situations		
	Status when Parameter Standby mode disabled	Status when Parameter Standby mode enabled but not activated	Status when Parameter Standby mode enabled and Parameter Standby activated
High Priority (patient) alarms	Yes	Yes	No(for the parameter in standby), since the patient is not being monitored
Flashing red and yellow LEDs on the front panel during high Priority (patient) alarms	Yes	Yes	No (for the parameter in standby), since the patient is not being monitored
Storage or transfer to remote stations of high priority (patient) alarms	Yes	Yes	No (for the parameter in standby), since the patient is not being monitored

4.8.3. EMS Mode

When the EMS mode is enabled, the following change will occur in the monitor's functioning:

- After powerup, the monitor immediately opens the home screen, with no message regarding clearing trend memory for a new patient.

By default, the EMS Mode is enabled.

Disable the EMS Mode as follows:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **Service** on the Setup Menu Screen.
3. Enter Service password and select **Done**. Select **Institutional Defaults>Monitor>Factory Defaults**.
4. Navigate to **EMS Mode** and click **Enter**.
5. Set to EMS Mode to **Disabled** and click **Enter**.
6. Select **Home** to return to the Home Screen. The device will ask you to save settings and shut down; do so.
7. To enable the EMS mode, follow steps 1, 2, 3, 4 and 5 above. Set EMS Mode to **Enabled** and click **Enter**.
8. Select **Home** to return to the Home Screen. The device will ask you to save settings and shut down; do so.

4.8.4. Reminder Signal

When the monitor is set to Audio Off (Permanent Audio Alarm Off), a reminder signal, which is a single beep, may be provided every two minutes to indicate that the monitor is in a Permanent Audio Alarm Off state.

By default, this Reminder Signal is **Disabled**.

Enable the reminder signal as follows:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **Service** on the Setup Menu Screen.
3. Enter Service password and select **Done**.
4. Select **Institutional Defaults>Monitor**.

5. Navigate to **Reminder Signal** and click **Enter**. Set Reminder Signal to **Enabled** and click **Enter**.
6. Select **Home** to return to the Home Screen. The device will ask you to save settings and shut down; do so.
7. When you restart, the Reminder Signal will be enabled.

4.9. Monitor Screen Menu Reference Chart

The chart below shows the menu flow paths for navigating through the different screens of the monitor.

Capnostream™35 Portable Respiratory Monitor



4.10. Capnography with the Monitor

4.10.1. Microstream™ EtCO₂ Consumables



Caution:

Before use, carefully read the Microstream™ etCO₂ consumables Directions for Use.



Caution:

Use only Microstream™ etCO₂ consumables to ensure the monitor functions properly.



Caution:

Microstream™ etCO₂ consumables are designed for single patient use, and are not to be reprocessed. Do not attempt to clean, disinfect or blow out the sampling line as the monitor can be damaged.



Caution:

Dispose of Microstream™ etCO₂ consumables according to standard operating procedures or local regulations for the disposal of contaminated medical waste.



Caution:

When monitoring with capnography during sedation, please note that sedation may cause hypoventilation and CO₂ waveform distortion or disappearance. Waveform attenuation or disappearance is an indicator that the status of the patient's airway should be assessed.



Caution:

When monitoring patients during upper endoscopy, partial blockage of the oral airway due to endoscope positioning may cause periods of low readings and rounded waveforms. The occurrence will be more pronounced with high oxygen delivery levels.



Caution:

If CO₂ insufflation is performed during CO₂ monitoring, the etCO₂ values will accordingly rise very significantly and this may result in device alarms and abnormally high waveforms until the CO₂ is evacuated from the patient.



WARNING:

Loose or damaged connections may compromise ventilation or cause an inaccurate measurement of respiratory gases. Securely connect all components and check connections for leaks according to standard clinical procedures.



WARNING:

If too much moisture enters the sampling line (i.e., from ambient humidity or breathing of unusually humid air), the message Clearing FilterLine will appear in the monitor message area. If the sampling line (FilterLine™ sampling line) cannot be cleared, the message **FilterLine Blockage** will appear in the CO₂ waveform display section on the Home screen and in the monitor message area. (If there is no waveform display, the message will appear only in the message area.) Replace the sampling line once the FilterLine Blockage message appears.



WARNING:

The sampling line may ignite in the presence of O₂ when directly exposed to laser, ESU devices, or high heat. When performing head and neck procedures involving laser, electrosurgical devices or high heat, use with caution to

prevent combustion of the sampling line or surrounding surgical drapes.



Note:

When connecting a sampling line to the monitor, screw the sampling line connector clockwise into the monitor CO₂ port until it can no longer be turned, to ensure that it is connected securely to the monitor. This will assure that there is no leak of gases during measurement at the connection point and that measurement accuracy is not compromised.



Note:

Following connection of the CO₂ sampling line, check that CO₂ values appear on the monitor display.



Note:

CO₂ sampling lines used with the monitor are marked as to the upper limit of oxygen that may be provided with the sampling line. At levels of oxygen provision higher than those marked on the sampling line packaging, dilution of CO₂ readings may occur, leading to lower CO₂ values.

Microstream™ etCO₂ consumables are available in a number of varieties, depending on the patient size and type and other considerations. The following considerations should be taken into account when choosing the correct consumable for a patient.

- Whether the patient is intubated or non-intubated
- Whether the patient is on mechanical ventilation
- Duration of use
- Patient's size and weight

- Whether the patient is breathing through his nose, his mouth, or alternating between oral and nasal breathing

A list of Microstream™ etCO₂ consumables appears in [8.1 Microstream™ EtCO₂ Consumables](#) on page 225. For more information about FilterLine™ sampling lines or additional sizing and packaging options for these products, contact your local representative, or see <http://www.medtronic.com/covidien/en-us/products/capnography/filterline-etco2-sampling-lines.html>.

Select the appropriate sampling line and connect it to the monitor before attaching it to the patient's airway. Be sure to follow Microstream™ etCO₂ consumables' Directions for Use for proper connection.



Note:

The generic term FilterLine™ sampling line or sampling line, used in this manual, is interchangeable with any of the Microstream™ etCO₂ consumables.

4.10.2. FilterLine™ Sampling Line Connection

Before monitoring a patient with capnography, the appropriate sampling line (FilterLine™ sampling line) must be connected to the monitor and to the patient.

Connect the sampling line as follows:

1. Open the sampling line input connector shutter and connect the appropriate sampling line. Screw the sampling line connector into the monitor clockwise until it can no longer be turned.
2. Connect the sampling line to the patient as described in the Directions for Use supplied with the sampling line.

3. When the sampling line is connected, the monitor will immediately begin to search for breaths, but it will not indicate a **No Breath** condition nor alarm for any capnography-related alarms before any valid breaths have occurred.

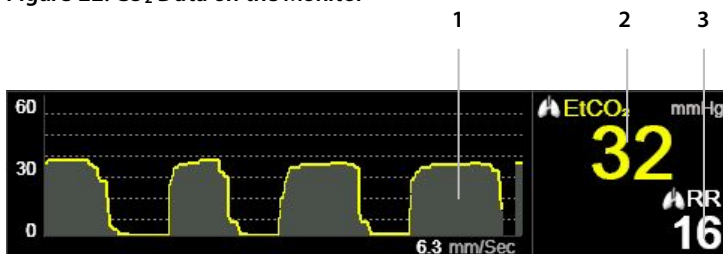
4.10.3. CO₂ Data Displayed by the Monitor

EtCO₂ monitoring is available for adult, pediatric (all types) and neonatal patients.

The monitor Home screen displays real time CO₂ data. The displayed data includes:

- Real time etCO₂ values along with selected unit(for available units, reference [10.6 Microstream™ Capnography](#) on page 237)
- Respiration rate (RR) in breaths per minute, as derived from capnography
- CO₂ Waveform (on some screens)

Figure 22. CO₂ Data on the Monitor



Label	Function
1	CO ₂ Waveform
2	EtCO ₂ value
3	Respiration Rate value



Note:

Note the lung icon that appears next to the EtCO₂ and RR texts, indicating that this data is derived from capnography, which measures patient respiration.

Additionally, the monitor can display CO₂ data in trend form, showing time, date, etCO₂, RR, alarms, and events. For more information about trend display, reference [4.15 Trends](#) on page 161.

If the numeric home screen is chosen, the CO₂ waveform will not appear. Instead, etCO₂ data will appear in a large font, to enable easy viewing, even at a distance. For the CO₂ section of the numeric home screen, reference [Figure 8. Home Screen Display #3](#) on page 32. For other home screen types, reference [2.3.5 Monitoring Display Screen Options](#) on page 26.

For all types of patients, the etCO₂ numeric displayed on the screen is the maximum value of CO₂ over the last 20 seconds, updated once a second. An etCO₂ alarm will occur based on the etCO₂ value displayed on the screen.

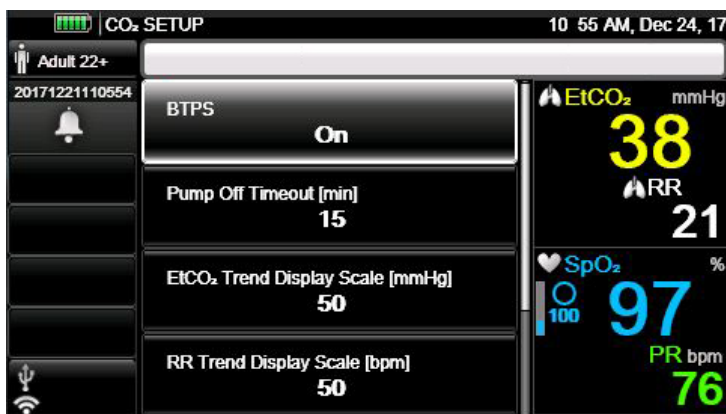
In high-altitude environments, etCO₂ values may be lower than values observed at sea level, as described by Dalton's law of partial pressures. When using the monitor in high-altitude environments, it is advisable to consider adjusting etCO₂ alarm settings accordingly.

Measuring mode corrects the etCO_2 value for BTPS setting (Body Temperature, Pressure, Saturation) which assumes that alveolar gases are saturated with water vapor. BTPS should remain on for patient measurement. For other purposes, it may be turned off in the CO_2 Setup screen.

4.10.4. Adjustable CO_2 Parameters

The monitor provides the option of adjusting some parameter settings used for CO_2 measurement to suit your patients, your institution's requirements, or other needs. To change these settings on a temporary basis, until the monitor is turned off, follow the procedure below. To set changes as institutional defaults so that the settings will remain in effect even after the monitor is turned off, reference [11.1.1 Changing Institutional Defaults](#) on page 251.

Figure 23. CO_2 Setup Screen



CO_2 parameter settings may be changed as follows:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **CO_2 Setup** on the Setup Menu Screen. Click **Enter**.

3. Navigate to the desired parameter and click **Enter**. Select the desired value and click **Enter** to return to the CO₂ Setup screen. Select **Home** to return to the Home screen.
4. To make changes to the CO₂ waveform, select **CO₂ Waveform Setup** on the Setup Menu Screen. Click **Enter**.
5. In either case, navigate to the desired parameter and click **Enter**. Select the desired value and click **Enter** to return to the CO₂ Setup screen. Select **Home** to return to the Home screen.
6. For permanent changes to these parameters, change the parameters in Institutional Defaults. For more information reference [11.1.1 Changing Institutional Defaults](#) on page 251.

Table 5. CO₂ Parameters

Parameter	Choices	Factory Default
BTPS*	On, Off	On
PumpOff Timeout (min)	5, 10, 15, 30	15
EtCO ₂ Trend Display Scale (mmHg)	50, 100, 150	50
RR Trend Display Scale (bpm)	50, 100, 150	50
A/hr Visual Alert Level	1-99	10
CO ₂ Mode	EtCO ₂	EtCO ₂

* BTPS denotes the standard correction used during measurement for body temperature, pressure, and saturation. BTPS should be set to ON during all measurement procedures. The monitor automatically turns off the BTPS correction during calibration procedures and turns it on again following these procedures. There is no need for the user to make any changes to the BTPS setting.

If the operator changes the capnography-based respiration rate trend scale settings in the CO₂ setup screen described above, the RR trend scale settings in the SpO₂ setup screen, which refers to Nellcor™ RR, will automatically be changed to match those settings.

Table 6. CO₂ Waveform Setup Parameters

Parameter	Choices	Factory Default
Sweep Speed Adult/Pediatric (mm/sec)	1,2,3,6.3,12.5,25	6.3
Sweep Speed Infant/Neonatal (mm/sec)	1,2,3,6.3,12.5,25	6.3
CO ₂ Waveform Scale (mmHg)	20, 60, 120, 150, Auto	60

Sweep speed is the speed at which the waveform completes one cycle of the graph to cover the entire display screen once.

4.10.5. Monitoring CO₂ during MRI Scanning



WARNING:

Do not use the FilterLine™ H Set Infant/Neonatal set or the VitalLine™ H Set Infant/Neonatal set during magnetic resonance imaging (MRI) scanning. Using the FilterLine™ H Set Infant/Neonatal set during MRI scanning could cause an artefact on the MRI image.



WARNING:

During MRI scanning, the monitor must be placed outside the MRI suite. When the monitor is used outside the MRI

suite, etCO₂ monitoring can be implemented by attaching the FilterLine™ XL sampling line, to provide extended length.



Caution:

Use of a CO₂ sampling line with H in its name (indicating that it is for use in humidified environments) during MRI scanning may cause interference. The use of non-H sampling lines is advised. For a list of sampling lines, reference [8.1 Microstream™ EtCO₂ Consumables](#) on page 225.

Non-invasive etCO₂ monitoring during magnetic resonance imaging (MRI) can be accomplished with the monitor, a FilterLine™ XL sampling line, and an appropriate CO₂ sampling line.

The monitor may be used during MRI scanning, as follows:

1. Place the monitor outside the MRI suite. There must be a hole in the wall of the suite (approximately 10 cm. diameter).
2. Connect the FilterLine™ XL sampling line to the monitor and guide the FilterLine™ XL sampling line through the hole in the wall of the MRI suite.
3. Attach the FilterLine™ XL sampling line to the patient.

Due to the extended length of the FilterLine™ XL sampling line, there may be an increased delay time and thus a slower response time.

To purchase the FilterLine™ XL sampling line, contact your local representative.

4.11. Pulse Oximetry with the monitor

4.11.1. Pulse Oximetry Features

Pulse oximetry is used to continuously monitors a patient's oxygen saturation (SpO₂), based on the difference in the absorption of red and infrared light (spectrophotometry) by oxyhemoglobin and deoxyhemoglobin; and changes in the volume of arterial blood in tissue during the pulse cycle (plethysmography), and hence, light absorption by that blood.

Pulse oximetry can also be used to measure respiration rate for patients, creating the Nellcor™ RR value, under the following conditions: capnography is not available, the Nellcor™ RR feature is available on the device, a sensor that supports Nellcor™ RR is in use, and the patient type is Adult 22+. Under these conditions, the pulse-oximetry-derived Nellcor™ RR will be seen on the device, as seen in

Figure 6. Home Screen Display #1, with Nellcor™ RR and OxiMax™ SPD Alert on page 29. For more information, reference 4.11.3 Nellcor™ Respiration Rate on page 93.

4.11.2. Connecting an SpO₂ Sensor to the monitor

Before monitoring a patient with pulse oximetry, the appropriate SpO₂ sensor must be connected to the monitor and to the patient.

Connection of the SpO₂ Sensor is performed as follows:

1. Connect the SpO₂ extension cord firmly to the monitor SpO₂ sensor port on the left side panel, and then connect the appropriate Nellcor™ SpO₂ sensor to the extension cord.

2. Connect the Nellcor™ SpO₂ sensor to the patient as described in its Directions for Use, using a Nellcor™ SpO₂ sensor extension cable.
3. When the SpO₂ sensor is plugged into the extension cable and connected to the monitor, the monitor will immediately begin to search for a pulse. It will indicate **SpO₂ Sensor Not On Patient** until the time that the sensor is placed on the patient. This is classified as a Medium Priority Alarm, and in this case, before the sensor is connected to a patient for the first time, will generate an on-screen message only, and not an audible alarm. (Once the SpO₂ sensor is connected to a patient and then removed, the **SpO₂ Sensor Not On Patient** alarm will be both visual and audible, which is standard behavior for Medium Priority alarms.)
4. To begin monitoring, place the sensor on the patient as described in the sensor's Instructions for Use.

4.11.3. Nellcor™ Respiration Rate

The parameter provides caregivers with respiration rate information on patients.

Pulse oximetry-derived respiratory rate (Nellcor™ RR) interprets modulations of the photoplethysmograph (PPG) in response to hemodynamic changes during the respiratory cycle. The PPG signal is consistently modulated by cardiovascular, respiratory and autonomic nervous system interactions throughout the respiratory cycle. The respiratory-specific alterations – including the effect of respiration on venous return, intrathoracic pressure, and diastolic and stroke volume - provide the physiological foundation for the extraction of respiratory rate from the PPG signal. This information provides an oximetry-based respiratory rate during spontaneous respiration. In comparison, capnography provides a respiratory rate through measurement of changes in CO₂ concentration at the mouth or nose, and calculates respiration rate based on detected breath-to-breath intervals in the CO₂ waveform. Both types of respiration rate provide an alarm for high and low respiration rates, while capnography

additionally provides a No Breath (apnea) alarm if the patient's breathing stops. The clinician should select the type of respiration rate monitoring most suitable for the specific patient's needs, keeping in mind that the range of the Nellcor™ RR is 4-40 bpm, while capnography provides a range of 0-150 bpm. Likewise, the clinician should note that, when using Nellcor™ RR, the risk associated with unrecognized obstructive apnea would be similar to that of other pulse oximeters.

Use of Nellcor™ RR does not require modification of other monitoring system settings. The operator may continue to use the monitoring system as before for monitoring SpO₂ and pulse rate, while also monitoring for respiration rate. The parameter does not in any way replace SpO₂ and pulse rate data, but augments it.

The Nellcor™ respiration rate (RR) is available under the following conditions only:

1. The Capnostream™35 module in use must support Nellcor™ RR. To determine if your device supports Nellcor™ RR, see the Service Entry screen (Click the **Menu** button on the front panel of the monitor and then **Setup>Service**). Scroll down to see the features data. If the Service Entry screen so indicates, the device supports Nellcor™ RR. Reference [6.2 Service Entry and Service Mode Screens](#) on page 207.
2. The Nellcor™ RR feature should not be disabled in the Institutional Defaults: SpO₂ screen. By default, the Nellcor™ RR feature is enabled.
3. Respiration rate derived from capnography must not be currently available on the device. Therefore, Nellcor™ RR will be available only if no CO₂ sampling line is connected to the device, or if the device has a CO₂ error and cannot display capnography.
4. Caregivers must use a Nellcor™ adult respiratory sensor (SKU 10068119) to display Nellcor™ RR. Reference [Table 7. Nellcor™ SpO₂ Sensor Models and Patient Sizes](#) on page 108.

If another Nellcor™ sensor is used, SpO2 and pulse rate data will appear, but Nellcor™ RR data will not be available.

5. Nellcor™ RR is for use only for adult patients. Thus, it is available only for patients who are indicated with the Adult 22+ patient type.

Nellcor™ RR provides alarms for high and low respiration rate, just as the capnography-derived respiration rate does. The parameter is measured in breaths per minute and has an operating range of 4 to 40 breaths per minute, with an accuracy of ± 1 breath per minute, relative to a capnography-based reference. For default RR High and RR Low alarm limits, reference [4.14.11 Alarm Limits - Factory Defaults](#) on page 160. As with the capnography-derived respiration rate alarm limit settings, these settings are user-adjustable. Reference [11.1.1 Changing Institutional Defaults](#) on page 251 for changing alarm limits.

Note the heart icon that appears on the home screen next to the Nellcor™ RR data, which indicates that this data is derived from pulse oximetry. When the lung icon appears next to the RR text, this indicates that the RR data is derived from capnography.

In the trend screens, likewise, a lung icon next to the RR text will indicate that the RR data is derived from capnography, and a heart icon next to the RR text will indicate that the data is derived from pulse oximetry. If data in a given period on a trend screen is derived from both capnography and pulse oximetry (that is, for some of the time in that time period from each one), both icons will appear.



WARNING:

Respiration Rate is not intended for use as an apnea monitor. During apnea, Respiration Rate may post a non-zero number.



WARNING: The Nellcor™ pulse oximetry-based respiration rate and SPD features are intended for use in hospitals, hospital-type facilities and during intra-hospital

transport only. These features are not intended for use in out-of-hospital Emergency Medical Service applications.



Caution:

Accuracy of Respiration Rate was established using bench-top testing and clinical studies in 30 healthy volunteers and 75 hospitalized patients. Hospital studies did not necessarily include all patient conditions found in hospitals and hospital-type settings. These clinical study results may not generalize to all patient conditions. Use caution in patient populations in which a displayed respiration rate value outside of the stated accuracy specification could present a serious risk or hazard.



Caution:

Respiration Rate provides an indicator of central ventilatory drive and not a direct indication that air is moving through the upper airway. Always consider clinical signs and symptoms when assessing the patient and before intervening in response to a respiration rate alarm.



Caution:

Respiration Rate should be used only in conjunction with the SpO₂ and pulse rate information provided by the monitoring system. Reliance on respiration rate information alone is not recommended. Health care personnel should always consider clinical signs and symptoms when assessing the patient and before intervening in response to a respiration rate alarm.



Caution:

Periods of continuous patient motion may impact device performance. During prolonged periods of motion, the Nellcor™ respiration rate may not report a respiration rate value. In situations where respiration rate is not reported, and is the only measurement affected by signal quality, the monitoring system will indicate "RR Interference."



Caution:

The operating range for Respiration Rate is 4 to 40 breaths per minute. Use on patients with respiration rates outside this range may result in inaccurate displayed respiration rate values.



Caution:

Respiration Rate should not be used on patients with significantly irregular cardiac rhythms (defined as three or more events of irregularity observed within 30 seconds) because the presence of these irregular cardiac rhythms may cause inaccurate respiration rate values or the loss of displayed respiration rate information. Safety and effectiveness of Respiration Rate in patients with significantly irregular cardiac rhythms have not been established. Use an alternate means of monitoring ventilatory status for patients with significant cardiac dysrhythmia or an implanted pacemaker.



Caution:

Safety and effectiveness of Respiration Rate in pediatric and neonatal patients have not been established.



Caution:

Safety and effectiveness of Respiration Rate in pregnant or lactating women have not been established.



Caution:

Safety and effectiveness of Respiration Rate in patients on mechanical ventilation have not been established.



Caution:

An SpO₂ alarm may be the first indication of hypoventilation.



Caution:

A low SpO₂ alarm limit of at least 90% is recommended when monitoring patients on supplemental oxygen.



Caution:

Use of the Respiration Rate parameter does not change the need to set threshold limits appropriate to the patient being monitored.



Caution:

Nellcor™ respiration rate performance has not been established during cases of extreme motion (for example seizure) or poor perfusion (for example shock). In cases of extreme motion or poor perfusion other forms of respiration rate monitoring are recommended.

4.11.4. OxiMax™ SPD Alert Management Technology

The OxiMax™ SPD alert (SPD) method detects repetitive occurrences of desaturation followed by resaturation in adult patients. These patterns are indicative of repetitive reductions in airflow through the upper airway and into the lungs. Relative reductions in a patient's minute ventilation over a period of time may cause a progressive drop in alveolar partial pressure of oxygen, leading to arterial desaturation.

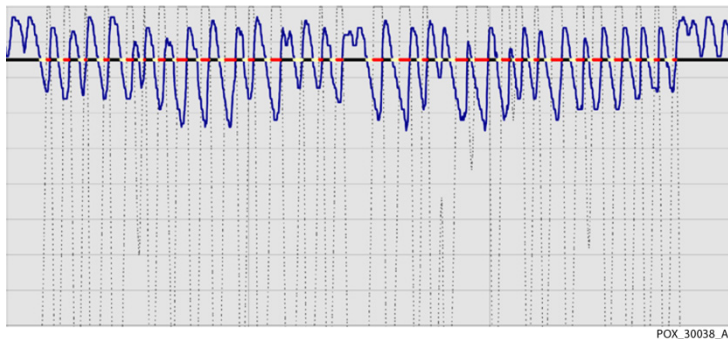
If these decreases in ventilation are repetitive, they generate distinct patterns in the saturation trend. Patterns of repetitive desaturation often develop gradually over time, increasing in severity. Detection of patterns indicates that a patient might be suffering progressively severe decrements in airflow that may increase in acuity if left untreated.

Patterns of desaturation are multiple, sequential occurrences of a desaturation followed by a resaturation. The SPD parameter qualifies

patterns of desaturation resulting from such repetitive reductions in airflow based on specific characteristics.

The SPD parameter qualifies these patterns of desaturation over a period of six (6) minutes. Depending on the sensitivity setting for SPD, patterns that persist may result in an SPD alert, alerting the caregiver to the condition.

Figure 24. Clinically Significant Desaturation Patterns



Thus, the SPD parameter provides information about:

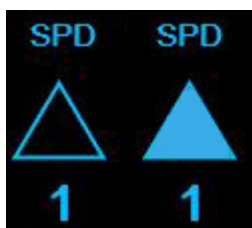
- The severity of the desaturation event (the depth of the desaturation during the event) and the extent of the following resaturation
- The regularity of the desaturation events (how often the pattern repeats)
- The slope of the desaturation/resaturation trends that form the events

The SPD indicator is a triangle in the SpO₂ section of the screen. When no patterns of desaturation have been detected, the indicator is seen as a triangle frame only. When the SPD parameter detects patterns of desaturation in the SpO₂ trend in adults, the triangle icon fills from bottom to top as patterns become more severe. The triangle empties from top to bottom as the patterns become less severe. When the icon fills completely, an alarm sounds. As desaturations occur, the frame will fill from the bottom; first filling one third of the

triangle, then two thirds and finally filling the entire triangle. When the indicator reaches capacity, indicating the SPD limit has been reached, the triangle indicator will be full, an audible alarm will sound, and an alarm message will appear.

The OxiMax™ SPD alert icon as it appears on the Home Screen #10 is seen in [Figure 25. The SPD Icon](#), below. In other screens in which the OxiMax™ SPD alert feature appears, this icon is shown next to the SpO2 data, in a smaller format and without the text SPD above it. Note that the icon at left is seen when the SPD feature is enabled but an SPD alert has not occurred, while the icon at right appears as it would while an SPD alert is occurring, when the full blue triangle will appear alternately with the yellow icon seen in [Figure 26. Home screen with SPD Alert](#) on page 104.

Figure 25. The SPD Icon



The OxiMax™ SPD alert feature is available only under the following conditions:

1. The Capnostream™35 module in use must support the OxiMax™ SPD alert feature. To determine if your device supports the OxiMax™ SPD alert feature, see the Service Entry screen (Click the **Menu** button on the front panel of the monitor and then **Setup>Service**). Scroll down to see the features data. If the Service Entry screen so indicates, the device supports the OxiMax™ SPD alert feature. Reference [6.2 Service Entry and Service Mode Screens](#) on page 207.
2. The SPD feature should not be turned off in the Alarm Setup screen or in the Institutional Defaults Alarms screen. By default, the SPD feature is turned on.

3. Capnography must not be currently available on the device. Therefore, the OxiMax™ SPD alert feature will be available only if no CO2 sampling line is connected to the device, or if the device has a CO2 error and cannot display capnography.
4. The OxiMax™ SPD alert feature is for use only for adult patients. Thus, it is available only for patients who are indicated with the Adult 22+ patient type.

Caregivers can select from three alarm sensitivity settings: 1 (most sensitive), 2 (medium sensitivity) or 3 (least sensitive), with 1 resulting in more alarms and 3 resulting in fewer alarms. The default setting of one (1) is the most sensitive to desaturation patterns and results in more frequent alarms. For less frequent alarms, use a less sensitive setting of two (2) or three (3). The rate at which the SPD icon fills depends on the SPD sensitivity setting.



WARNING:

The Nellcor™ pulse oximetry-based respiration rate and SPD features are intended for use in hospitals, hospital-type facilities and during intra-hospital transport only. These features are not intended for use in out-of-hospital Emergency Medical Service applications.



WARNING:

Supplemental oxygen will attenuate patterns of desaturation. A patient's respiratory compromise can be proportionally more severe before patterns appear in the saturation trend. Remain vigilant when monitoring a patient on supplemental oxygen.



WARNING

Only use the SPD parameter with adult patients (patient type Adult 22+) for in-hospital use. Do not use on Adult 12-22, pediatric or neonate patients. In Adult 12-22, all pediatric, and Infant/Neonatal modes, the SPD parameter remains OFF.



Caution:

Use of the SPD parameter does not change the need to set threshold limits appropriate to the patient being monitored.



Caution:

Do not modify any other SpO2 alarm settings while using the SPD parameter.



Note:

Unrecognized repetitive reductions in airflow through the upper airway occur in some clinically significant scenarios. Patients exhibiting sleep apnea symptoms were used in studies to validate the OxiMax™ SPD alert parameter. The presence of repetitive reductions in airflow was scored using a standard diagnostic polysomnogram. Study results indicate SPD is a sensitive marker in detecting repetitive reductions in airflow.



Note:

Prior to changing SPD sensitivity settings, clear all alarms.



Note:

Unrecognized repetitive reductions in airflow through the upper airway occur in some clinically significant scenarios. Patients exhibiting sleep apnea symptoms were used in studies to validate the OxiMax™ SPD alert parameter. The presence of repetitive reductions in airflow was scored using a standard diagnostic polysomnogram. Study results indicate SPD is a sensitive marker in detecting repetitive reductions in airflow.

4.11.4.1.1. Enabling and Setting up the OxiMax™ SPD Alert Feature

The OxiMax™ SPD alert feature is automatically enabled if your device supports this feature, for adult (patient type Adult 22+) patient only, as long as a CO2 sampling line is not attached to the device or capnography is not currently available.

To turn on or off or set the level of the SPD alert until the device is turned off, navigate to and click **Menu>Setup>Alarm Setup>SPD Sensitivity**. Click **Enter** and use the up and down arrows to set the sensitivity or turn the feature off. In this case, the device will return to the default (in which the SPD feature is turned on with a sensitivity level of 1) once the device is turned on again.

To turn on or off or set the level of the SPD alert permanently (until changed), follow the instructions below:

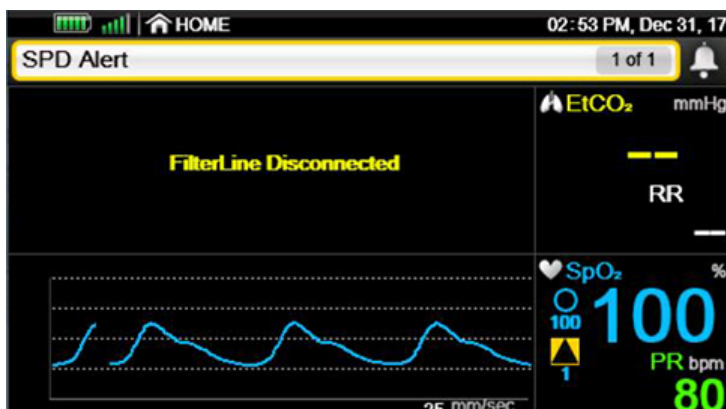
1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **Service** on the Setup Menu Screen.
3. Enter Service password and select **Done**. Select **Institutional Defaults>Alarms>Factory Defaults**.
4. Navigate to **SPD Sensitivity** and click **Enter**.
5. Change the SPD Sensitivity level to the desired level or to **Off** and click **Enter**.
6. Select **Home** to return to the Home Screen. The device will ask you to save settings and shut down; do so. When you restart the device, the changes you have made will appear.

The default level of the SPD alert is 1; this is the highest sensitivity level. Setting the SPD level to a lower sensitivity means that the SPD alert will alert the caregivers to patient desaturations only after a higher incidence or increased severity of desaturations.

4.11.4.1.2. SPD Alerts and Dismissing SPD Alerts

Once the patient's desaturation patterns exceed the SPD limit, an SPD ALERT message will flash. Reference [Figure 26. Home screen with SPD Alert](#), below. Caregivers should view the data to examine SatSeconds™ and SPD patterns in the trend data history.

Figure 26. Home screen with SPD Alert



The SPD audible alarm may continue to sound for a long period after it is first triggered – up to a number of minutes. The device provides an option to dismiss the SPD alert only, after it has sounded, to reduce the interference that a continually sounding alarm may cause to caregivers' tasks.

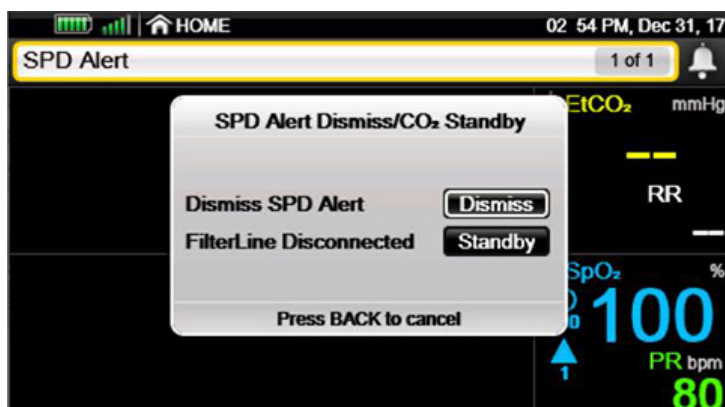
Should the caregiver silence an SPD alert, this resets the index that tracks repetitive patterns of desaturation and silences and dismisses only the SPD alert. Note that once this alert is dismissed, only that specific instance will be dismissed; if an event that triggers an SPD alert occurs again, it will sound again.

To dismiss an SPD audible alarm, follow the instructions below:

1. This action may be taken only when an SPD audible alarm is currently active. It cannot be taken in advance in order to prevent alarms or dismiss future alarms.
2. Press the audio alarm paused hard key at the front of the monitor () for more than two seconds.

3. A pop-up screen will ask you if the action you want is to or dismiss an SPD audible alarm or to enter Parameter Standby mode for capnography. (Reference [Figure 27. The SPD Dismiss Window](#), below.) (If the device is already in CO₂ Standby Mode, the SPD alert will be dismissed without this window.) The cursor will be on the **Dismiss** button; click **Enter**. The SPD alert will now be dismissed and it will no longer sound. An on-screen message will indicate: **SPD Dismissed**.
4. If you navigate to the **Standby** button on this screen and click **Enter**, you will enter the CO₂ standby mode (since there is no CO₂ sampling line attached to the device) and the SPD alert will continue to exist and continue to sound.

Figure 27. The SPD Dismiss Window



4.11.5. Nellcor™ SpO₂ Sensors



WARNING:

Before use, carefully read the sensor Directions for Use, including all warnings, cautions, and instructions.



WARNING:

Shock hazard — Do not immerse or wet the sensor.



WARNING:

Do not use any monitoring system, sensor, cable, or connector that appears damaged. Remove any damaged equipment from service for inspection by a qualified service technician.



WARNING:

Do not use a damaged sensor or interface cable. Do not use a sensor with exposed optical components.



WARNING:

Do not spray, pour, or spill any liquid on the monitor, its accessories, connectors, switches, or openings in the chassis, since this may cause damage to the monitor.



WARNING:

Use only Covidien-approved sensors and interface cables when connecting to the sensor port. Connecting any other cable or sensor influences the accuracy of sensor data, which may lead to adverse results. Verify the compatibility of the pulse oximetry probe, cable and monitor before use to ensure accuracy and safe use.



WARNING

Use only Covidien-approved interface cables with the monitor. Use of another interface cable will adversely impact performance. Do not attach any cable intended for computer use to the sensor port.



WARNING:

Disconnect the SpO₂ sensor from the patient during magnetic resonance imaging (MRI) scanning. Objects containing metal can become dangerous projectiles when subjected to the strong magnetic fields created by MRI equipment. Also, induced currents could potentially cause burns. For use of the monitor during MRI scanning for

capnography only, reference 4.10.5 Monitoring CO₂ during MRI Scanning on page 98.



WARNING:

Tissue damage can be caused by incorrect application or use of a pulse oximetry sensor. Inspect the sensor site as directed in the Instructions for Use. Do not apply the sensor too tightly or by using excessive pressure. Do not wrap the sensor, apply supplemental tape, or leave the sensor too long in one place. If the skin under the sensor becomes irritated, change the location of the sensor.



Caution:

Single-patient use sensors and adhesive sensors are intended for single-patient use only. Do not transfer a single-use sensor or an adhesive sensor from one patient to a second patient.

The sensors used with this monitor can be categorized as surface devices contacting skin for a limited duration of time. The sensors have passed the recommended biocompatibility testing and are therefore in compliance with ISO 10993-1.

4.11.6. Nellcor™ SpO₂ Sensor Selection

When selecting a sensor, consider the patient's weight and activity, the adequacy of perfusion, the available sensor sites, and the anticipated duration of monitoring. Use the recommended sensor's Directions for Use to guide sensor selection or contact Covidien or a local Covidien representative. Reference [Table 7. Nellcor™ SpO₂ Sensor Models and Patient Sizes](#) on page 108. Sensor length information is included in the Directions for Use for each sensor.

The Nellcor™ interface cable (DOC-10, 10 ft/3m or DEC-4, 4 ft/1.2m) connects the monitoring system with the Nellcor™ sensor. Do not attach any cable to the sensor port that is intended for computer use.

Use only Covidien-approved sensors and interface cables when connecting to the sensor port.

Table 7. Nellcor™ SpO₂ Sensor Models and Patient Sizes

Nellcor™ Sensor	SKU	International SKU, if available	Patient Size
Nellcor™ Premie SpO ₂ Sensor, non-adhesive (Single-patient use)	SC-PR*	SC-PR-I	<1.5 kg
Nellcor™ Neonatal SpO ₂ Sensor, non-adhesive (Single-patient use)	SC-NEO*	SC-NEO-I	1.5 to 5 kg
Nellcor™ Adult SpO ₂ Sensor, non-adhesive (Single-patient use)	SC-A*	SC-A-I	>40 kg
Nellcor™ Adult-Neonatal SpO ₂ Sensor with Wraps (Reusable with adhesive)	OXI-A/N*		<3 or >40 kg
Nellcor™ Pediatric-Infant SpO ₂ Sensor with Wraps (Reusable with adhesive)	OXI-P/I*		3 to 40 kg
Nellcor™ Pediatric SpO ₂ Sensor, Two Piece (Sterile, single-use only)	P		10 to 50 kg
Nellcor™ Neonatal-Adult SpO ₂ Sensor, Two Piece (Sterile, single-use only)	N		<3 or >40 kg
Nellcor™ Adult SpO ₂ Sensor, Two Piece (Sterile, single-use only)	A		> 30 kg
Nellcor™ Neonatal-Adult SpO ₂ Sensor (Sterile, single-	MAXN*	MAX-N-I	<3 or >40 kg

Nellcor™ Sensor	SKU	International SKU, if available	Patient Size
use only)			
Nellcor™ Infant SpO ₂ Sensor (Sterile, single-use only)	MAXI*	MAX-I-I	3 to 20 kg
Nellcor™ Pediatric SpO ₂ Sensor (Sterile, single-use only)	MAXP*	MAX-P-I	10 to 50 kg
Nellcor™ Adult SpO ₂ Sensor (Sterile, single-use only)	MAXA*		>30 kg
Nellcor™ Adult XL SpO ₂ Sensor (Sterile, single-use only)	MAXAL*	MAX-AL-I	>30 kg
Nellcor™ Adult SpO ₂ Nasal Sensor (Sterile, single-use only)	MAXR*		>50 kg
Nellcor™ Forehead SpO ₂ Sensor (Sterile, single-use only)	MAXFAST*		≥10 kg
Nellcor™ Adult SpO ₂ Sensor, Reusable (Nonsterile)	DS-100A*		>40 kg
Nellcor™ SpO ₂ Sensor, Multisite Reusable (Nonsterile)	D-YS*		>1 kg
Nellcor™ SpO ₂ Ear Clip, Reusable (Nonsterile)	D-YSE*		>30 kg
Nellcor™ Pediatric SpO ₂ Clip, Reusable (Nonsterile)	D-YSPD*		3 to 40 kg
Nellcor™ Adult Respiratory Sensor, SpO ₂ , RR (Single-	10068119		>30kg

Nellcor™ Sensor	SKU	International SKU, if available	Patient Size
patient use)			

*suitable for use in EMS typical environments, refer to Degree of protection against matter and water ingress in 10.11 Compliance on page 243.

4.11.7. Nellcor™ Sensor Features

Nellcor™ sensor features are different for sensors at a different revision level and by sensor type (adhesive, recycled, and reusable). The revision level of a sensor is located on the sensor plug.

4.11.7.1. Biocompatibility testing

Biocompatibility testing has been conducted on Nellcor™ sensors in compliance with ISO 10993-1, Biological Evaluation of Medical Devices, Part 1: Evaluation and Testing. Nellcor™ sensors have passed the recommended biocompatibility testing and are therefore in compliance with ISO 10993-1.

4.11.8. Performance Considerations

This section contains information for assisting users to optimize the performance of pulse oximetry in the monitor.

Prior to initial installation in a clinical setting, a qualified service technician should verify the performance of the monitor per the Service Manual.

4.11.8.1. Monitoring system constraints



WARNING:

Do not utilize for measurements outside the display levels listed for the monitor while monitoring patients.

Pulse Rate — The monitoring system only displays pulse rates between 20 and 250 bpm (beats per minute). Detected pulse rates above 250 bpm appear as 250. Detected pulse rates below 20 appear as a zero (0).

Saturation — The monitoring system displays saturation levels between 1% and 100% (in percent format).

4.11.9. Nellcor™ Sensor Performance Considerations



WARNING:

Pulse oximetry readings and pulse signal can be affected by certain ambient environmental conditions, sensor application errors, and patient conditions.



Caution:

Failure to cover the sensor site with opaque material in high ambient light conditions may result in inaccurate measurements.

4.11.9.1. Inaccurate sensor measurement conditions

A variety of conditions can cause inaccurate sensor measurements.

- Incorrect application of the recommended sensor
- Placement of the recommended sensor on an extremity with a blood pressure cuff, arterial catheter, or intravascular line

- Ambient light
- Failure to cover the sensor site with opaque material when operating under high ambient light conditions
- Excessive patient activity
- Dark skin pigment
- Intravascular dyes or externally applied coloring, such as nail polish or pigmented cream
- Signal Loss
- Loss-of-pulse signal can occur for several reasons.
 - Recommended sensor applied too tightly
 - Inflation of a blood pressure cuff on the same extremity as the attached sensor
 - Arterial occlusion proximal to the recommended sensor
 - Poor peripheral perfusion

4.11.9.2. Recommended usage

Select an appropriate recommended sensor, apply it as directed, and observe all warnings and cautions presented in the Directions for Use accompanying the sensor.

Clean and remove any substances such as nail polish from the application site. Periodically check to ensure that the sensor remains properly positioned on the patient.

High ambient light sources such as surgical lights (especially those with a xenon light source), bilirubin lamps, fluorescent lights, infrared heating lamps, and direct sunlight can interfere with the performance of an SpO₂ sensor. To prevent interference from ambient light,

ensure that the sensor is properly applied, and cover the sensor site with opaque material.

If patient movement presents a problem, try one or more of the following remedies:

- Verify that the sensor is properly and securely applied.
- Move the sensor to a less active site.
- Use an adhesive sensor that tolerates some patient motion.
- Use a new sensor with fresh adhesive backing.

If poor perfusion affects performance, consider using the MAX-R sensor; it obtains measurements from the nasal septal anterior ethmoid artery, an artery supplied by the internal carotid. This sensor may obtain measurements when peripheral perfusion is relatively poor.

4.11.9.3. Patient conditions

Application issues and certain patient conditions can affect the measurements of the monitoring system and cause the loss of the pulse signal.

Anemia — Anemia causes decreased arterial oxygen content.

Although SpO₂ readings may appear normal, an anemic patient may be hypoxic. Correcting anemia can improve arterial oxygen content.

The monitoring system may fail to provide an SpO₂ reading if hemoglobin levels fall below 5 gm/dL.

Dysfunctional hemoglobins — Dysfunctional hemoglobins such as carboxyhemoglobin, methemoglobin, and sulfhemoglobin are unable to carry oxygen. SpO₂ readings may appear normal; however, a patient may be hypoxic because less hemoglobin is available to carry oxygen. Further assessment beyond pulse oximetry is recommended.

The following additional possible patient conditions may also influence measurements:

- Poor peripheral perfusion
- Excessive patient activity
- Venous pulsations
- Dark skin pigment
- Intravascular dyes, such as indocyanine green or methylene blue
- Externally applied coloring agents (nail polish, dye, pigmented cream)
- Defibrillation

4.11.9.4. Reducing EMI (Electromagnetic Interference) during use



WARNING:

EMI disruption can cause erratic readings, cessation of operation, or other incorrect functioning.



WARNING:

The monitoring system is intended for use by healthcare professionals only. It may cause radio interference or may disrupt the operation of nearby equipment. Mitigation for such disruption may require re-orienting or relocating the monitoring system or shielding the location.



Note:

This monitor has been tested and found to comply with the limits for medical devices related to IEC 60601-1-2: 2007. These limits are designed to provide reasonable protection against harmful interference in a typical medical installation.

Because of the proliferation of radio frequency transmitting equipment and other sources of electrical noise in health care environments (for example, electrosurgical units, cellular phones, mobile two-way radios, electrical appliances, and high-definition television), it is possible that high levels of such interference due to close proximity or strength of a source might result in disruption of monitoring system performance. Reference [Table 9. Electromagnetic Immunity Compliance and Guidance, Not Life-Supporting Equipment](#) on page 122. The monitoring system is designed for use in environments in which electro-magnetic interference might obscure the client's pulse. During such interference, measurements may seem inappropriate or the monitoring system may not seem to operate correctly. EMI disruption can cause erratic readings, cessation of operation, or other incorrect functioning. If this occurs, survey the site

of use to determine the source of this disruption, and take the listed actions to eliminate the source.

Turn equipment in the vicinity off and on to isolate the interfering equipment. Reorient or relocate the interfering equipment.

Increase the separation between the interfering equipment and the monitoring system. The monitoring system generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with these instructions, may itself cause harmful interference with other susceptible devices in the vicinity.

4.11.10. Nellcor™ Sensor Technology

Use Nellcor™ sensors, which are specifically designed for use with the monitor. Identify Nellcor™ sensors by the Nellcor™ logo on the plug. All Nellcor™ sensors contain a memory chip carrying information about the sensor which the monitor needs for correct operation, including the sensor's calibration data, model type, troubleshooting codes, and error detection data.

Any monitoring system containing OxiMax™ technology uses calibration data contained in the sensor in calculating the patient's SpO₂. With sensor calibration, the accuracy of many sensors is improved, since the calibration coefficients can be tailored to each sensor.

Contact Covidien or a local Service Center for a Nellcor™ oxygen saturation accuracy specification grid listing all of the Nellcor™ sensors used with the monitoring system. A soft copy is available at www.covidien.com.

4.11.10.1. SatSeconds™ alarm management parameter

The monitoring system monitors the percentage of hemoglobin binding sites saturated with oxygen in the blood. With traditional

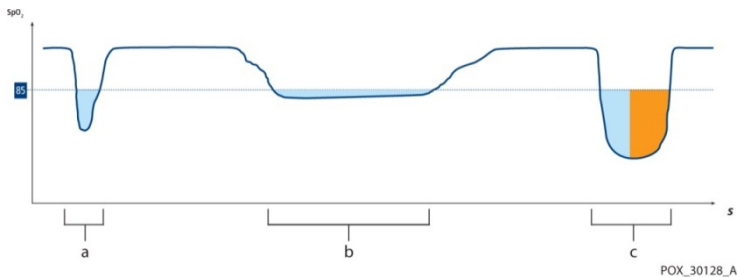
alarm management, upper and lower alarm limits are set to alarm at specific SpO₂ levels. When the SpO₂ level fluctuates near an alarm limit, the alarm sounds each time it violates the alarm threshold. SatSeconds™ monitors both degree and duration of desaturation as an index of desaturation severity. Thus, the SatSeconds™ parameter helps distinguish clinically significant events from minor and brief desaturations that may result in nuisance alarms.

4.11.10.1.1. SpO₂ events sequence

Consider a series of events leading to a violation of the SatSeconds™ alarm limit. An adult patient experiences several minor desaturations, then a clinically significant desaturation.

See the sketch below to view a chronological sequence of SpO₂ events and how they will appear on the monitor.

Figure 28. Series of SpO₂ Events

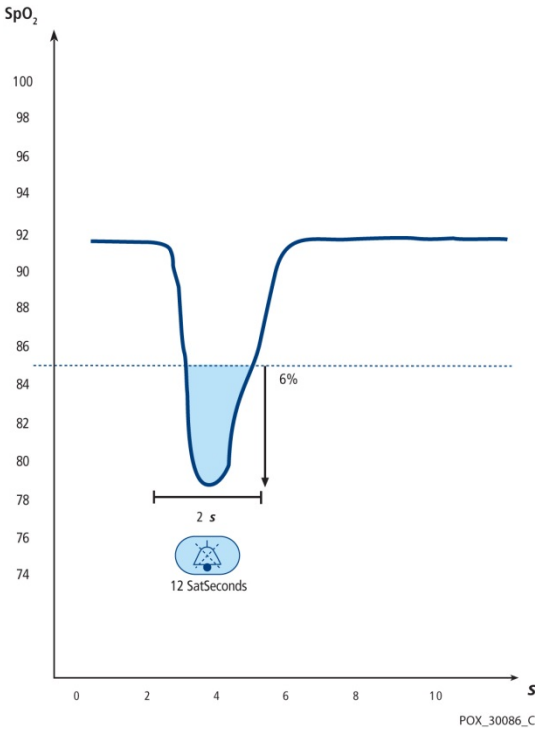


4.11.10.1.2. First SpO₂ event

The SatSeconds™ value is calculated by multiplying the amount of desaturation below the threshold value by the time (in seconds) of the desaturation. Consider the first event. Suppose the SpO₂ alarm limit is set to 85 and the SatSeconds™ alarm limit is set to 25. The patient's SpO₂ drops to 79% and the duration of the event is two (2) seconds before saturation again exceeds the lower alarm threshold of 85%. In this scenario, the amount below threshold is 6 (85-79) and the number of SatSeconds™ is therefore 12 (6x2).

Because the SatSeconds™ alarm limit is set to 25 and the actual number of Sat-Seconds equals 12, there is no audible alarm.

Figure 29. First SpO₂ Event: No SatSeconds™ Alarm

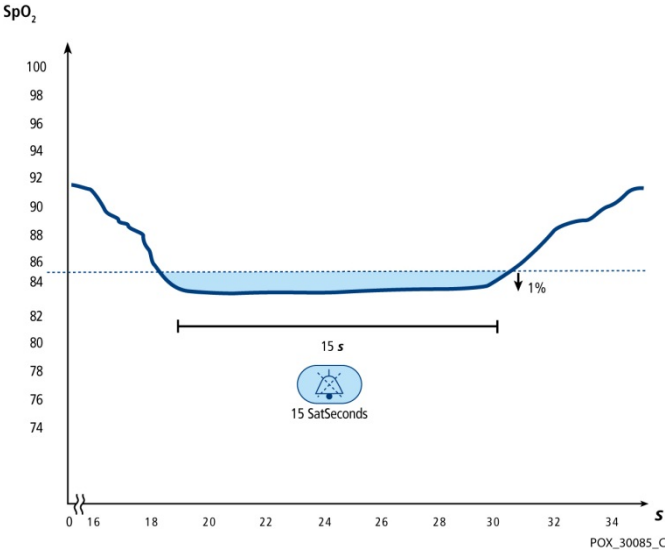


4.11.10.1.3. Second SpO₂ event

Consider the second event. Suppose the SatSeconds™ alarm limit is still set to 25. The patient's SpO₂ drops to 84% and the duration of the event is 15 seconds before saturation again exceeds the lower alarm threshold of 85%.

Because the SatSeconds™ alarm limit is set to 25 and the actual number of Sat-Seconds equals 15 (1x15), there is no audible alarm.

Figure 30. Second SpO₂ Event: No SatSeconds™ Alarm

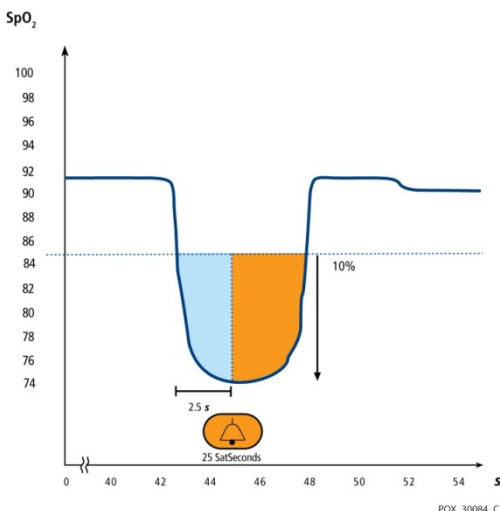


4.11.10.1.4. Third SpO₂ event

Consider the third event. Suppose the SatSeconds™ alarm limit is still set to 25. During this event, the patient's SpO₂ drops to 75%, which is 10% below the lower alarm threshold of 85%. Since the patient's saturation does not return to a value over the lower alarm threshold within 2.5 seconds, an alarm sounds.

At this level of saturation, the event cannot exceed 2.5 seconds without invoking a SatSeconds™ alarm.

Figure 31. Third SpO₂ Event: Triggers SatSeconds™ Alarm



4.11.10.1.5. The SatSeconds™ Safety Net

The SatSeconds™ “Safety Net” is for patients with saturation levels frequently below the limit, but not staying below the limit long enough for the SatSeconds™ alarm time setting to be reached. When three or more limit violations occur within 60 seconds, an alarm sounds even if the SatSeconds™ alarm time setting has not been reached.

By default, the SatSeconds™ alarm option is 100 for adult and pediatric patients and OFF for infant/neonatal patients. The SatSeconds™ alarm option may be changed in Institutional Defaults; reference [11.1.4 Monitor Alarms Settings](#) on page 254.

4.11.10.1.6. SpO₂ Alarms and SatSeconds™

The monitor uses Nellcor™ SatSeconds™ alarm technology to help reduce the number and frequency of false SpO₂ alarms.

A SatSeconds™ alarm indicator () in the SpO₂ section of the screen indicates the SatSeconds™ alarm status. When the SatSeconds™ alarm is Off, only the indicator appears. If the SatSeconds™ alarm is on, its alarm value will appear under the icon. With the SatSeconds™ alarm on, the SatSeconds™ circle icon fills in the clockwise direction as the alarm management system detects SpO₂ readings outside of the limit setting. The circle icon empties in counterclockwise direction when SpO₂ readings are within limits. When the icon fills completely, the SpO₂ alarm sounds. For more information about the SatSeconds™ technology, reference [4.11.10.1 SatSeconds™ alarm management parameter](#) on page 116. If the SatSeconds™ is selected and the SatSeconds™ value triggers an SpO₂ alarm, any SpO₂ delay that has been set will be superseded and the alarm will occur despite the SpO₂ delay.

4.11.11. Nellcor™ Pulse Oximetry with OxiMax™ Technology

Essential Performance

This monitoring system has the capacity to detect physiological alarm conditions using SpO₂ accuracy, pulse rate accuracy and alarm limit conditions.

Table 8. Pulse Oximetry Performance

Type	Values
Measurement Ranges	
SpO ₂ saturation range	1% to 100%
Pulse rate range	20 to 250 beats per minute (bpm)
Perfusion range	0.03% to 20%
Respiration rate range	4 to 40 breaths per minute (bpm)

Measurement Accuracy	
Pulse rate accuracy	20 to 250 beats per minute (bpm) ± 3 digits (including under low perfusion); with motion, 48 to 127 bpm ± 5 digits
SpO ₂ saturation accuracy	70% to 100% ± 2 to ± 3 digits
Respiration rate accuracy	± 1 breath per minute, relative to a capnography-based reference.
Operating Range and Dissipation	
Red Light Wavelength	Approximately 660 nm
Infrared Light Wavelength	Approximately 900 nm
Optical Output Power	Less than 15 mW
Power Dissipation	52.5 mW

Table 9. Electromagnetic Immunity Compliance and Guidance, Not Life-Supporting Equipment

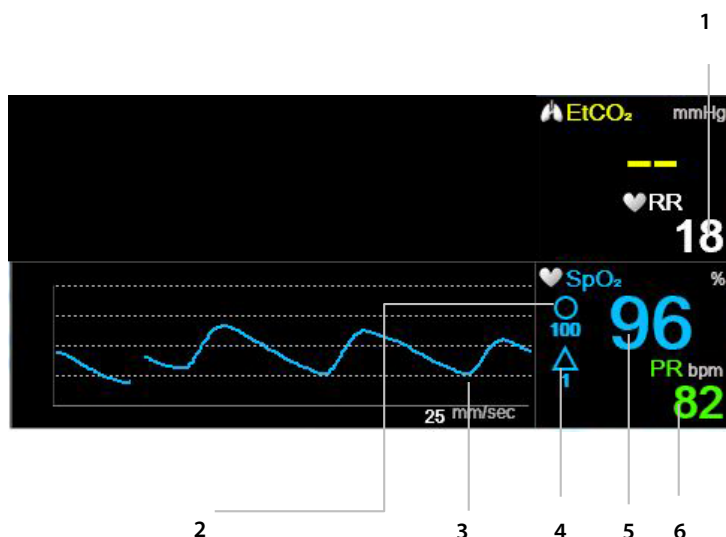
Immunity Test	IEC/EN 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
	Frequency of Transmitter		Equation for Separation Distance (d)
Conducted RF	3 Vrms 150 kHz	3 Vrms 150 kHz	$d = 1.2 \sqrt{P}$ 150 kHz to 80

IEC/EN 61000-4-6	80 MHz	80 MHz	MHz
Radiated RF IEC/EN 61000-4-3	3 V/m 80 MHz 800 MHz	3 V/m 80 MHz 800 MHz	$d = 1.2 \sqrt{P}$ 80 MHz to 800 MHz
	3 V/m 800 MHz 2.5 GHz	3 V/m 800 MHz 2.5 GHz	$d = 2.3 \sqrt{P}$ 800 MHz to 2.5 GHz

4.11.12. SpO₂ Data Displayed by the Monitor

The monitor Home screen displays real time SpO₂ data. The displayed data includes:

- SpO₂ numeric value
- Pulse Rate
- SpO₂ waveform or bar Indicator indicating arterial pulse beat. This is also referred to as the plethysmograph.

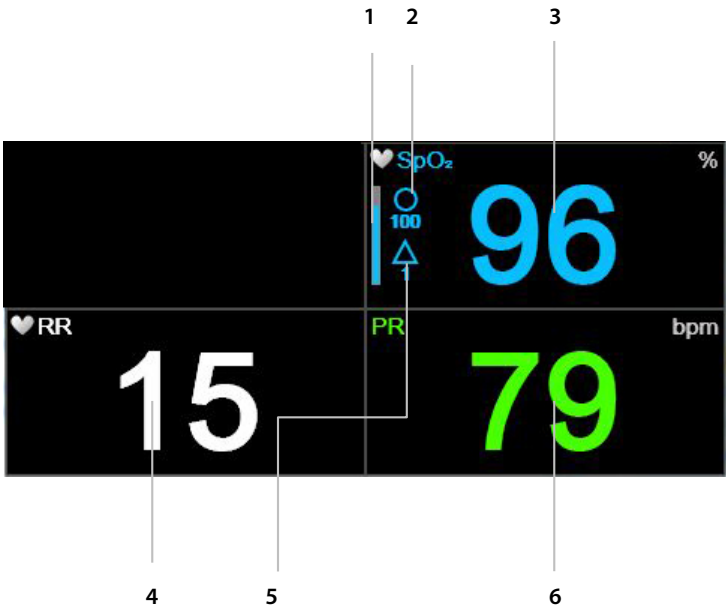
Figure 32. SpO₂ Data on Waveform Screen


Label	Function	Description
1	Nellcor™ RR value	Current patient Nellcor™ RR value
2	SatSec value	Current patient SatSec value
3	SpO ₂ waveform	SpO ₂ waveform, which indicates the patient's arterial pulse beat
4	SPD value	Current patient SPD value
5	SpO ₂ value	Current patient SpO ₂ value
6	PR value	Current patient pulse rate (PR) value

On home screens with SpO₂ waveforms, the patient's arterial pulse beat is indicated by the plethysmograph waveform on the SpO₂ section of the screen. The plethysmographic waveform is non-normalized, using real-time sensor signal to reflect relative pulsatile strength.

When home screens with bar Indicator are chosen, the patient's arterial pulse beat is indicated by the rhythm of the blue vertical bar on the SpO₂ section of the screen. The peak of this bar will coincide with the peak of the patient's arterial pulsation. Reference [Figure 33. SpO₂ Data in Numeric Screen](#), below.

Figure 33. SpO₂ Data in Numeric Screen



Label	Function	Description
1	SpO ₂ bar indicator	Blue vertical bar, the rhythm of which indicates the patient's arterial pulse beat
2	SatSec value	Current patient SatSec value
3	SpO ₂ value	Current patient SpO ₂ value
4	Nellcor™ RR value	Current patient Nellcor™ RR value
5	SPD value	Current patient SPD value
6	PR value	Current patient pulse rate (PR) value

Additionally, the monitor can display SpO₂ data in trend form, showing time, date, SpO₂, pulse rate (PR), alarms, events.

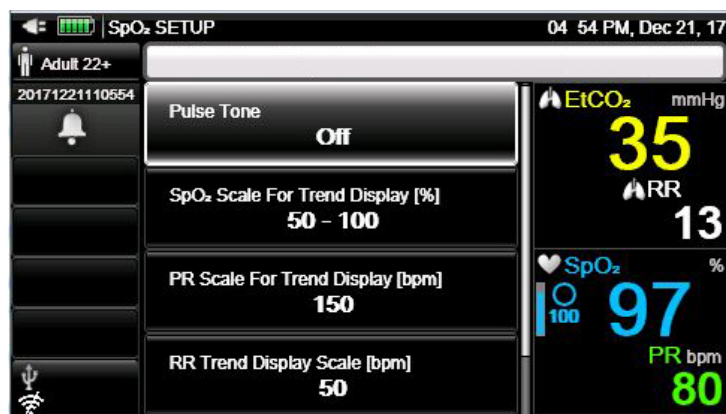
When the SpO₂ high or low alarm limits are exceeded, the affected reading will flash to alert the attending health care professional to the specific reading that is affected.

4.11.13. Adjustable SpO₂ Parameters

The monitor provides the option of adjusting some parameter settings used for SpO₂ measurement to suit your patients, your institution's requirements, or other needs. To change these settings on a temporary basis, until the monitor is turned off, follow the procedure below. To set changes as institutional defaults so that the settings will remain in effect even after the monitor is turned off, reference [11.1.1 Changing Institutional Defaults](#) on page 251.

SpO₂ Parameter Settings may be changed as follows:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **SpO₂ Setup** on the **Setup Menu** screen. Click **Enter**. The **SpO₂ Setup** screen will open.
3. Navigate to the desired parameter and click **Enter**. Select the desired value and click **Enter** to return to the **SpO₂ Setup** screen. Select **Home** to return to the Home screen.
4. To make changes to the SpO₂ waveform (Sweep Speed), navigate to and select **SpO₂ Waveform Setup** on the Setup Menu Screen. Click **Enter**. The SpO₂ Setup screen will open.
5. Navigate to the Sweep Speed and click **Enter**. Select the desired value and click **Enter** to return to the SpO₂ Setup screen. Select **Home** to return to the Home screen.
6. For permanent changes to these parameters, change the parameters in Institutional Defaults. For more information reference [11.1.1 Changing Institutional Defaults](#) on page 251.
7. The SatSeconds™ alarm may be turned on using Institutional defaults; reference [11.1.1 Changing Institutional Defaults](#) on page 251.

Figure 34. SpO₂ Setup ScreenTable 10. Adjustable SpO₂ Parameters

Parameter	Choices	Factory Default
Pulse Tone	On/Off	Off
SpO ₂ Scale for Trend Display (%)	0-100, 50-100	50-100
PR Scale for Trend Display (bpm)	150, 300	150
RR Trend Display Scale* (bpm)	50, 100, 150	50
ODI Visual Alert Level*	1-99	10
Sweep Speed Adult/Pediatric	3, 6.3, 12.5, 25	25
Sweep Speed Infant/Neonatal	3, 6.3, 12.5, 25	25

*Only available for patients indicated as patient type Adult 22+

Sweep speed is the speed at which the waveform completes one cycle of the graph to cover the entire display screen once.

4.11.14. SpO₂ Alarm Limit Message

When the SpO₂ alarm limit is set below 85%, a message reading **SpO₂ Low Alarm Limit: XX** will appear in the header area (as this is an advisory, it will only appear if there are no alarms occurring at the same time), indicating the level of the SpO₂ LOW alarm limit. The level of the SpO₂ LOW alarm limit will also appear on the SpO₂ section of the screen, if the SpO₂ LOW alarm limit is lower than 85%.

4.12. Integrated Pulmonary Index™ Value

4.12.1. Integrated Pulmonary Index™ Algorithm: Introduction

The Integrated Pulmonary Index™ algorithm (henceforth referred to as IPI) is a numerical value which integrates four major parameters measured by the monitor in order to provide a simple indication of the patient's overall ventilatory status. The integrated parameters are etCO₂, RR, SpO₂, and PR. Only these four parameters are used to calculate IPI; other parameters are not taken into account.

IPI is calculated using the current values of these four parameters and their interactions, based on known clinical data. IPI can thus provide an early indication of a change in ventilatory status which may not be shown by the current value of any of these four parameters individually. The IPI is designed to provide additional information regarding patient status, possibly before etCO₂, RR, SpO₂, or PR values reach levels of clinical concern.

The IPI is an adjunct to, and is not intended to replace, vital sign monitoring.

A technical note containing details regarding the Integrated Pulmonary Index™ algorithm is available from Covidien.

Since the index uses data from the monitoring of both CO₂ and SpO₂, it will only be available when both parameters are available.

The range of the index is 1-10; index values should be understood as seen in the table below.

Table 11. IPI Values

Index Range	Patient Status
10	Normal
8-9	Within normal range
7	Close to normal range; requires attention
5-6	Requires attention and may require intervention
3-4	Requires intervention
1-2	Requires immediate intervention

The interpretation of the patient's IPI score may change in different clinical environments. For example, patients with specific respiratory difficulties (in contrast to normally healthy patients who are being monitored during sedation or pain management) may require a lower IPI Low Alarm threshold to reflect their impaired respiratory capacity.

The Integrated Pulmonary Index™ algorithm is available for all three groups of pediatric patients (1-3 years, 3-6 years, and 6-12 years), and for adult patients. It is not available for Neonatal/Infant patients (patients up to the age of one year), and thus will not appear on screens for Neonatal/Infant patients.

4.12.2. Integrated Pulmonary Index™ Algorithm Warnings



WARNING:

Ensure that the patient type is correctly selected before beginning monitoring of a patient. Choosing an incorrect patient type could produce incorrect patient IPI data.



WARNING:

When an IPI Low Alarm is triggered for a patient, medical staff should review the patient's status to determine if a change in medical care is required.

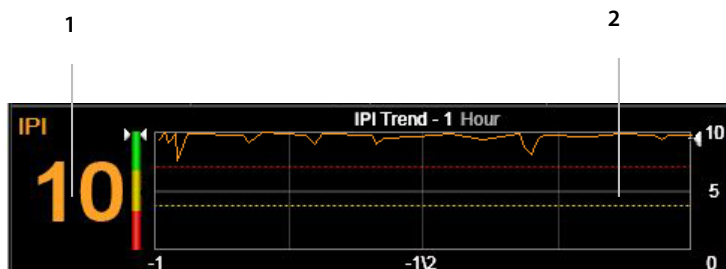
4.12.3. Integrated Pulmonary Index™ Algorithm Display

The Integrated Pulmonary Index™ algorithm appears on Home screen #3 and Home screen #9 both as a numeric value and in a trend graph. In some of the other home screens it appears only as a numeric value.

The IPI option may be disabled from the Institutional Defaults screen; see below for more information. In the Neonatal measurement mode, the IPI option is automatically disabled.

For, the IPI trend graph (seen on the monitor home screen), reference [Figure 35. IPI Trend Graph](#), below.

Figure 35. IPI Trend Graph



Label	Function
1	IPI value
2	IPI trend waveform

4.12.4. Integrated Pulmonary Index™ Algorithm Options

The IPI zoom level may be changed as follows:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **Trend Waveform Setup** on the Setup Menu Screen. Click **Enter**.
3. Navigate to and select **IPI Trend Zoom (Hour)**. Click **Enter**. Navigate to the desired Zoom level.
4. Click **Enter** and then select **Home** to return to the home screen.

Table 12. Adjustable IPI Options

Parameter	Choices	Factory Default
IPI Trend Zoom (Hour)	1 hour, 2 hour, 4 hour	1 hour

Use the Institutional Defaults option to Enable/Disable the IPI alarm, or to change IPI Display options on a more permanent basis;

reference [11.1 Institutional Defaults](#) on page 251.

4.13. Apneas per Hour (A/hr) and the Oxygen Desaturation Index (ODI)

4.13.1. A/hr and ODI: Introduction

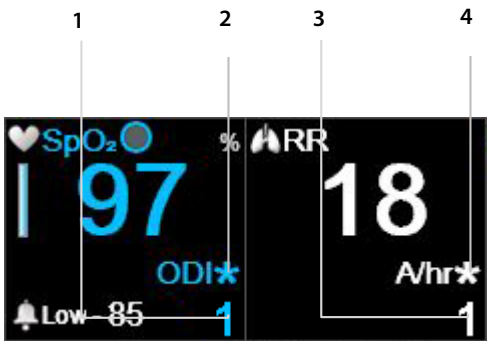
The Apneas per Hour (A/hr) (also known as ASA, Apnea Saturation Alert) and Oxygen Desaturation Index (ODI) algorithm is part of the Smart Capnography™ technology family of innovative algorithms. Smart Capnography™ technology simplifies the use of CO₂ monitoring on Microstream™ technology enabled products, to improve patient safety and clinical workflow.

A/hr and ODI provide an easy way to help with the identification and quantification of apnea and oxygen desaturation events during a patient's hospital stay.

A/hr and ODI report apnea and oxygen desaturation events and calculates the associated Apnea per Hour (A/hr) and Oxygen Desaturation Index (ODI). By using A/hr and ODI, clinicians are able to recognize ventilation and oxygenation abnormalities while patients are being monitored during their hospital stay with the monitor. A/hr and ODI are for adult patients only (age 22 and older). Thus, the patient must be indicated with the patient type Adult 22+ in order to obtain A/hr and ODI data.

A/hr and ODI are displayed in real time on one of the home screens (Figure 14. Home Screen Display #9 on page 39) and on easy-to-read Vent and O₂ Desat Report screens. In the home screen display, an asterisk next to the ODI or A/hr text indicates that the patient has reached the A/hr visual alert level or the ODI visual alert level. The level for both of these indicators can be set in the Institutional Defaults>Parameters screen; see 11.1 Institutional Defaults on page 251.

Figure 36. ODI and A/hr on the Home Screen



Label	Function
1	ODI value
2	ODI visual alert
3	A/hr value
4	A/hr visual alert

4.13.2. Apneas per Hour

An apnea is a period of time during which breathing stops or is greatly reduced. The Apnea per Hour (A/hr) calculated by the monitor provides a count of the number of pauses of at least 10 seconds in breathing the patient experienced over the past hour.

On the Vent. and Desat Reports A and B, A/hr is the average Apneas per Hour over the selected time period of 1,2, 4, 8 and 12 hours. On that screen, the number of apnea events of ≥ 10 seconds; 10-19 seconds; 20-30 seconds; and >30 seconds over the selected time period are also shown.

This A/hr value is used to help indicate to the caregiver the regularity of the patient's breathing.

In the monitor, apnea per hour is a count of the number of times in an hour that breathing ceased for ten seconds or more. Since this is a per-hour count, dashed lines will appear on the screen until one hour has passed since the beginning of monitoring.

The A/hr visual alert indicates to the caregiver that the A/hr count in any one hour period during the last 12 hours exceeded a set number. The asterisk will appear next to the A/hr only when an A/hr Visual Alert is triggered, and it is updated once every 10 minutes. The visual alert on the screen indicates to the caregiver that the Vent. and O2 Desat Report B (reference [5.2 Reports](#) on page 176) should be viewed to learn more about the patient's status.

A/hr is currently available for adult patients only and is intended for patients age 22 and older. It is not available for infants (age 0-1) or for patients under age 22.

4.13.3. Oxygen Desaturation Index (ODI)

The Oxygen Desaturation Index (ODI) indicates the "dips" in SpO_2 – that is, the number of times that the SpO_2 value dropped 4% or more from baseline and returned to baseline in 240 seconds or less. (This refers to percent of oxygen saturation, not percent of the patient's current SpO_2 rate). The baseline is created when a consistent and stable (in the range of ± 1 SpO_2 point [%]) SpO_2 value is detected over a 20-second period. This baseline, updated once a second, will be the rounded maximum SpO_2 over these 20 seconds. If a valid baseline

cannot be established based upon the definition above, then the previous baseline is kept.

A lower ODI (that is, fewer such instances) indicates more stability of the patient's oxygen saturation. Since this index is a per-hour count, dashed lines will appear on the screen until one hour has passed since the beginning of monitoring.

On the Vent. and Desat Reports A and B, ODI is average per hour "dips" in SpO₂ over the selected time period of 1, 2, 4, 8 and 12 hours.

The ODI visual alert indicates to the caregiver that the ODI count in any one hour period during the last 12 hours exceeded a set number. The asterisk will appear next to the ODI only when an ODI Visual Alert is triggered, and it is updated once every 10 minutes. The visual alert on the screen indicates to the caregiver that the Vent. and O₂ Desat Reports A and B (reference [5.2 Reports](#) on page 176) should be viewed to learn more about the patient's status.

ODI is currently available for adult patients only and is intended for patients age 22 and older. It is not available for infants (age 0-1) or for patients under age 22.

4.13.4. Monitoring with A/hr and ODI

When monitoring patients with A/hr (also known as ASA, Apnea Saturation Alert) and ODI, take the following issues into consideration:



Caution:

Please note that A/hr and ODI do not report hypopnea events.



Caution:

Apnea per Hour (A/hr) and Oxygen Desaturation Index (ODI) do not represent and should not be interpreted as an apnea

hypopnea index (AHI) as reported by formal polysomnography studies.



Caution:

Apnea per Hour (A/hr) and Oxygen Desaturation Index (ODI) do not represent and should not be used to diagnose sleep disordered breathing.



Caution:

Alarms and a noisy environment may affect A/hr and ODI values. See the Caution regarding recommendations on changing monitor settings for a sleeping patient below.



Caution:

A/hr and ODI are reported by the monitor throughout the monitoring period; however, the monitor cannot discern if the patient is actually sleeping.



Caution:

If a patient removes a sensor, the monitor may indicate apneas when no such apneas took place.



Caution:

The administration of opioid analgesia and sedatives may cause respiratory depression, which will result in transient apnea and O₂ desaturation events that will be reflected in A/hr and ODI values.



Caution:

Please read all user information to ensure complete understanding of A/hr.

Please note that the patient type is used to calculate the A/hr and ODI. For this reason, it is important to select the patient type correctly. For the same reason, changing the patient type (from adult to pediatric, for example) will clear the A/hr and ODI data stored for

the current patient. A/hr and ODI headings will not appear for infant/neonatal or pediatric patients.

When using the monitor for A/hr and ODI with a sleeping patient, it is recommended that the monitor be attached to a central monitoring station where alarms will be heard. Once this is done, the alarm sound on the monitor at the patient's bedside can be disabled, so as not to disturb the patient's sleep. The audible alarms are silenced through

System>Service>input Service password (reference [13 Appendix B: Monitor Service](#) Password on page 287)>**Inst Defaults>Monitor**. In the list of options on this screen, change the **Alarm Volume** to **Audio Off**. This should only be done if the monitor is under constant surveillance via connection to a central station (or another means of surveillance), so that patient alarms are noted by caregivers while the alarm sound is turned off on the bedside monitor.

4.14. Alarms and Messages

4.14.1. Alarms Introduction

The monitor triggers alarms related to patient condition as well as equipment errors. Alarms alert the health care provider that the patient's condition is beyond predefined limits, or indicate a malfunction or operating condition of the monitor hardware.

The monitor contains three levels of alarms and advisories, each defined by a set of audible and/or visual indications. Each alarm has a default alarm priority level, but can be set to another level at the discretion of the institution using Institutional Defaults. The levels of alarms and advisories are as follows:

- High Priority Alarms
- Medium Priority Alarms
- Advisories
- Silent Advisories

The following table describes how the alarms are indicated.

Table 13. Alarm Indications

Alarm Type	Indicators			
	Audible	Numerics	Messages	Indicator Light
High Priority (Patient) Alarms	High priority beep pattern repeated every 12 seconds	Alternating red background and red frame for numeric value	Appear in message area; certain messages appear also in waveform area	Flashing red indicator
Medium Priority Alarms	Triple beep repeated every 10 seconds	Alternating yellow background and yellow frame for numeric value	Appear in message area; certain messages appear also in waveform area	Flashing yellow indicator
Advisories	Single beep	NA	Appear in message area; certain messages appear also in waveform area	No indicator light

Alarm Type	Indicators			
	Audible	Numerics	Messages	Indicator Light
Silent Advisories	No audible alarm	NA	Appear in message area; certain messages appear also in waveform area	No indicator light

Some messages are displayed in the waveform area as well as in the message area; these messages are listed in [Table 17. Waveform Area Messages](#) on page 157.

4.14.2. Alarm Display

In order to view the visual alarm indicators, the intended position of the operator is in front of the monitor display screen. The intended position of the operator in order to hear audible signals is any position near the monitor (in a 360° range).

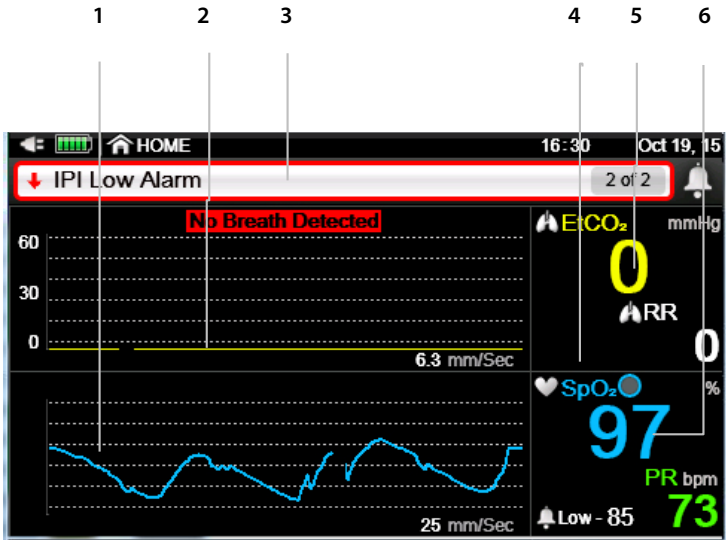
Alarms are indicated, when they occur, on the relevant section of the home screen in the following ways:

- Indicated in text format in message area (with a directional arrow indicating whether the parameters has crossed a high or low alarm limit and the alarm priority, with red for high priority alarms and yellow for medium priority alarms). Additionally, the dark grey box at the right side of the alarm message area will indicate, for example, 1 of 3, indicating that the alarm displayed is the first of the three currently active alarms. This is useful if there is more than one active alarm at one time.
- Indicated in numeric area, with a directional arrow indicating high or low alarm and the alarm priority, with red

for high priority alarms and yellow for medium priority alarms, and a red alternating background/frame indicating a high priority alarm or a yellow alternating background/frame indicating a medium priority alarm.

An example of a screen with an alarm appears below.

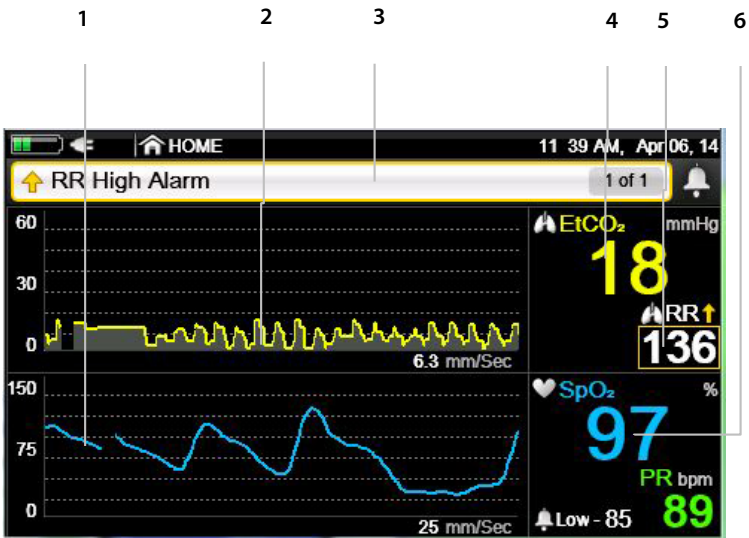
Figure 37. High Priority Alarm Example (No Breath)



Label	Function	Description
1	SpO ₂ waveform	Current patient SpO ₂ waveform
2	CO ₂ waveform	Current patient CO ₂ waveform
3	Alarm indicated in message area	Alarm indicated in text format in message area, with red frame indicating a high priority alarm (other high priority alarms will also show a red directional arrow in the message area)
4	Numeric area	For high priority alarms other than No Breath, the alarm is

		indicated in numeric area, with red arrow and red background for patient value indicating a high priority alarm
5	CO ₂ numeric data	Current patient CO ₂ numeric data
6	SpO ₂ numeric data	Current patient SpO ₂ numeric data

Figure 38. Medium Priority Alarm Example



Label	Function	Description
1	SpO ₂ waveform	Current patient SpO ₂ waveform
2	CO ₂ waveform	Current patient CO ₂ waveform
3	Alarm indicated in	Alarm indicated in text format in message area, with yellow

	message area	frame and yellow arrow indicating a medium priority alarm
4	CO ₂ numeric data	Current patient CO ₂ numeric data
5	Alarm indicated in numeric area	Alarm indicated in numeric area, with yellow arrow and yellow frame for patient value indicating a medium priority alarm
6	SpO ₂ numeric data	Current patient SpO ₂ numeric data

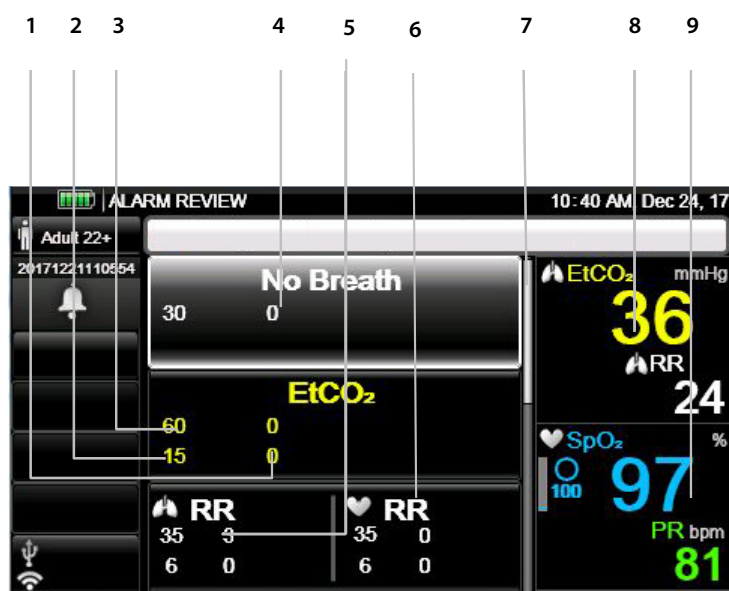
4.14.3. Alarm Review Screen

The monitor also provides the opportunity to view an Alarm Review screen, so the user can see at a glance the number of alarms the patient has incurred over the previous hour. The Alarm Review screen lists patient alarms (except for SatSeconds™; reference list in [Table 14. High/Medium Priority Alarms](#) on page 148) and indicates both the current upper and lower limits for each alarm and the number of each alarm which has occurred over the last hour. The screen only shows three alarms at a time; scroll down using the scroll bar at right to view additional alarms. An example of the alarm review screen appears below.

Please note that the screen shows both capnography-based respiration rate and Nellcor™ respiration rate, on the same screen. The number of alarms for each type of RR within the last hours will be shown on this screen.

The Alarm Review screen can be viewed whether or not you have opened a patient case.

Figure 39. Alarm Review Screen



Label	Description
1	Number of EtCO ₂ Low alarms in the previous hour indicated here
2	Current lower alarm limit for EtCO ₂
3	Current upper alarm limit for EtCO ₂
4	Number of No Breath alarms in the previous hour indicated here
5	Capnography-based RR data (upper and lower alarm limits and number of upper and lower alarms in the last hour)
6	Nellcor™ RR data (upper and lower alarm limits and number of upper and lower alarms in the last hour)

7	Scroll here to view data on additional alarms
8	Real-time EtCO ₂ patient data
9	Real-time SpO ₂ patient data

The Alarm Review screen may be opened as follows:

1. Click the **Menu** button on the front panel.
2. Navigate to and select **Report** on the Menu Screen and **View Report** on the Setup Menu Screen. Select the **Alarm Review** screen. Click **Enter**.
3. The **Alarm Review** screen will appear. If the alarm that you want to view is not visible, scroll down using the scroll bar at right of the main window to view the rest of the alarms.

4.14.4. Audio Alarm Paused or Off

Alarms may be paused or turned off in one of two ways:

- Temporary Audio Alarm Paused
- Permanent Audio Alarm Off

Alarms may be turned off permanently (until reversed) only through Institutional Defaults.

In both cases, alarm paused or off affects only the audible alarms. Visual alarms are not affected.

Temporary audio alarm paused may be set for the device, as follows:

1. Press the **Audio Alarm Paused** button at the front of the monitor.
2. This will pause the audio alarms for a 2-minute period only. The alarm indicator will show a broken red X () through the indicator, indicating that the alarm is temporarily paused.

Permanent audio alarm off may be set for the device, as follows:

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the **Menu** screen. Navigate to and select **Service** on the **Setup Menu** screen.
3. Input Service password and click **Done**.
4. Navigate to and select **Institutional Defaults>Monitor>Factory Defaults**.
5. Navigate to and select **Alarm Volume** and select **Audio Off**.
6. Select **Home** to return to the Home screen. The device will ask you to save settings and shut down.
7. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made. Restart the monitor.
8. The alarm indicator on the Home screen will now show an unbroken red X through it (X), indicating that the monitor is currently in a state of permanent audio alarm off.
9. To turn off Permanent Audio Alarm Off, repeat the process, setting the **Alarm Volume** to **Maximum** or **Last Setting**. Select **Home**; at this point the Permanent Audio Alarm Off status will no longer be in force; you will be required to restart the monitor.

When the monitor is set to permanent audio alarm off in this manner, the Audio Alarm Paused button will serve as a toggle key for audio alarm off; pressing it will turn the audible alarms on or off.



WARNING:

Do not pause or turn off the audible alarms until you verify that the patient is being monitored by other means, such as an audible alarm at a central station.

4.14.5. Reminder Signal

When alarms are silenced permanently through Institutional Defaults, a Reminder Signal will provide a single beep every two minutes, to meet regulatory requirements.

The Reminder Signal may be disabled in Institutional Defaults, by clicking **Menu Button>Setup>Service**. Input Service password and click **Done** and then select **Institutional Defaults>Monitor>Set Reminder Signal to Disabled**.

4.14.6. Message Priorities

When more than one alarm message appears in the message area, they will appear sequentially, with the following priority: First High Priority messages will be shown, then Medium Priority messages, and then Advisories. A grey box indicating the number of active messages and their order appears on the right side of the message area, so that each message will be accompanied by the indication 1 of 3, or 2 of 4, etc. If there are alarm messages, advisories and silent advisories will not be shown until the alarm messages are no longer shown.

4.14.7. Alarm Delay

An alarm delay may be set for the all patient alarms except for No Breath.

If an alarm delay is set, the alarm will not sound or display until the set delay period has passed. If the condition no longer exists by the time the delay period has passed, the alarm will not sound or display at all. Likewise, although the trend display will display the correct data, it will not show an alarm until the delay period has passed.

For all of these alarms, the delay period may be 0, 10, 15, 20, or 30 seconds. By default, no delay periods are set for any alarms.

To set alarm delay, reference [11.1.4.4 Alarm Delay settings](#) on page 260.

4.14.8. Types of Alarms

4.14.8.1. High/medium priority alarms

Table 14. High/Medium Priority Alarms

Message	Description	Corrective Action	Default Priority	Priority Options
Patient Alarms				
No Breath Detected xxx Seconds	No valid breath has been detected for xxx seconds	Patient requires immediate medical attention.	High	High
EtCO ₂ High Alarm	The etCO ₂ is above the upper alarm limit	Patient requires immediate medical attention.	High	Medium, High
EtCO ₂ Low Alarm	The etCO ₂ is below the lower alarm limit	Patient requires immediate medical attention.	High	Medium, High
RR High Alarm	The RR is above the upper alarm limit	Patient requires immediate medical attention.	High	Medium, High
RR Low Alarm	The RR is below the lower alarm limit	Patient requires immediate medical attention.	High	Medium, High

Message	Description	Corrective Action	Default Priority	Priority Options
SpO ₂ High Alarm	The SpO ₂ is above the upper alarm limit	Patient requires immediate medical attention.	High	Medium, High
SpO ₂ Low Alarm	The SpO ₂ is below the lower alarm limit	Patient requires immediate medical attention.	High	Medium, High
Pulse Rate High Alarm	The pulse rate is above the upper alarm limit	Patient requires immediate medical attention.	High	Medium, High
Pulse Rate Low Alarm	The pulse rate is below the lower alarm limit	Patient requires immediate medical attention.	High	Medium, High
Nellcor™ Respiration Rate (RR) High Alarm	The Nellcor™ respiration rate is above the upper alarm limit	Patient requires immediate medical attention.	High	Medium, High
Nellcor™ Respiration Rate (RR) Low Alarm	The Nellcor™ respiration rate is below the lower alarm limit	Patient requires immediate medical attention.	High	Medium, High
IPI Low Alarm*	The IPI™ value is below the lower alarm limit.	Patient requires immediate medical attention.	High	Medium, High

Message	Description	Corrective Action	Default Priority	Priority Options
SPD alert	Saturation pattern is detected	Patient requires medical attention.	Medium	Medium, High

Technical (Equipment) Alarms				
CO ₂ Error	Failure has occurred which prohibits the operation of the CO ₂ function.	Contact Covidien authorized personnel.	Medium	Medium, High
SpO ₂ Error	Failure has occurred which prohibits the operation of the SpO ₂ function.	Contact Covidien authorized personnel.	Medium	Medium, High
Pulse Not Found	No detectable pulse.	Patient requires immediate medical attention. Reposition sensor on patient.	Medium	Medium, High

FilterLine Blockage	FilterLine™ sampling line is kinked or clogged.	Disconnect and reconnect the sampling line (FilterLine™ sampling line). Check the airway adapter and if necessary, replace the sampling line.	Medium	Medium, High
Battery Low	Battery charge level is low and monitor will shut down soon.	Connect monitor to AC power.	Medium	Medium, High
SpO₂ Sensor Not On Patient	Sensor is off patient.	Place sensor properly on patient.	Medium	Medium, High
Replace SpO₂ Cable	SpO₂ cable should be replaced	Replace SpO₂ disposable sensor. If message continues to appear, replace reusable sensor or extension cable	Medium	Medium, High

Check SpO ₂ Sensor Connection	SpO ₂ sensor connection should be checked	Check SpO ₂ Sensor Connection	Medium	Medium, High
Communications Stopped**	Relevant communication has stopped. When used with a remote system, this message may be displayed with a different wording if so programmed by the host computer. The host computer may also initiate a message which would appear upon the stopping of communications.	Check communication status	Medium	Medium, High
System Reset, Check Settings	The system has reset and will revert to institutional default settings.	Reset your alarm limits and other settings to match the requirements of the patient being monitored.	Medium	Medium, High

FilterLine Disconnect	No sampling line is connected to the monitor.	Insert sampling line into CO ₂ port on monitor.	Medium	Medium
SpO ₂ Sensor Disconnect	No SpO ₂ sensor is connected to the monitor.	Insert SpO ₂ sensor into SpO ₂ port on monitor.	Medium	Medium
Temperature High	Monitor temperature is too high	Check ambient temperature.	Medium	Medium
Date is invalid. Set date and time	Date was set to an invalid date	Adjust date to display the current date	Medium	Medium

*The IPI LOW ALARM is an alert which is intended to indicate a change in patient status to the physician. When this alert appears, levels of other patient parameters should be evaluated.

** When set at Medium Priority, the Communications Stopped alarm will provide a visual alert only, with no audio alarm. When that alarm is set at High Priority, it will provide both a visual and an audio alert as usual for High Priority alarms.

4.14.8.2. Advisories

Table 15. Advisory Messages

Message	Description
No USB Device Found	A valid flash memory device is not connected to the USB port.
USB Drive Full	No room on the USB flash memory device.
No SD Card Found	No SD card is inserted in the monitor

Message	Description
SD Card Full	The current SD card is full
Pump Off Mode	Appears during Pump Off status
Network Disconnected	The configured network has been disconnected
No Signal, Scanning for networks	No signal strength for the configured network
Connecting to <Network name>	Device is connecting to the named network
Wi-Fi Connected to: <Network Name>	Device is connected to the named network
Static IP address conflict. Connection disabled	Configured Static IP address conflict exists. Connection has been disabled.

* When used with a remote system, this message may be displayed with a different wording if so programmed by the host computer. The host computer may also initiate a message which would appear upon the stopping of communications.

4.14.8.1. Silent Advisories

Table 16. Silent Advisory Messages

Message	Description
Clearing FilterLine	Sampling line (FilterLine™ sampling line) kinked or clogged with water. Appears during clearing time until sampling line is unclogged, or a blockage state is determined.

Message	Description
SpO₂ Weak. Reposition Error. SpO₂ Weak. Too Much Light. SpO₂ Weak. Try Ear Sensor. SpO₂ Weak. Try Nasal Sensor. SpO₂ Weak. Try Adhesive Sensor SpO₂ Weak. Secure Sensor Cable SpO₂ Weak. Try Using Headband SpO₂ Weak. Sensor Too Cold. SpO₂ Weak. Check Bandage. SpO₂ Weak. Nail Polish. SpO₂ Weak. Sensor Too Tight. SpO₂ Weak. Reposition Sensor SpO₂ Weak. Due To Interference. SpO₂ Weak. Clean Sensor Site. SpO₂ Interference	SpO₂ module detects a weak pulse and suggests possible causes.
CO ₂ Warm Up	CO ₂ module is preparing for operation.
CO ₂ Ready	Before the first measurement of CO ₂ , after the FilterLine™ sampling line is connected and before patient breath is detected, CO ₂ Ready replaces the CO ₂ Warm-up message.
Calibration Required	CO ₂ calibration is overdue.
Maintenance Required	CO ₂ maintenance is overdue.

Message	Description
SpO ₂ Extended Averaging Alarm	SpO ₂ extended averaging is now taking place
EtCO ₂ H XX, EtCO ₂ L XX	Displayed briefly on boot up and when alarm limit is changed
SpO ₂ Low Alarm Limit: xx	Displayed if the SpO ₂ Low alarm limit is set below 85%.
Performing Auto Zero	The monitor automatically performs a zeropoint calibration.
Demo Mode Prerecorded Data	Displayed during Demo Mode when no other message is displayed.
CO ₂ Monitoring Has Been Off For Hh:Mm	Displays the hours and minutes the pump has been turned off during PUMP OFF mode.
Remote System Connected*	The monitor is connected to a remote system. This message will only be present if enabled by the host computer, and may have different wording if so programmed by the host computer.
CO ₂ Standby	CO ₂ standby has been activated.
SpO ₂ Standby	SpO ₂ standby has been activated.
Keypad Locked	Keypad is locked
Keyboard unlocked	Keypad is unlocked
Report Transfer In Progress	Relevant communication is in progress
Report Transfer Aborted	Relevant communication has stopped

Message	Description
Service Mode	Service mode had been entered. No patient monitoring is taking place.
Oximetry RR Interference	Oximetry RR sensor connected, but interference is detected
Unable to Calculate Oximetry RR	Oximetry RR sensor connected, but Nellcor™ RR cannot be calculated
Calculating Oximetry RR	Oximetry RR sensor connected, and Nellcor™ RR is being calculated
SpO2 sensor does not support Oximetry RR	Incompatible pulse oximetry sensor connected
Oximetry RR Not Enabled in Service	All other requirements to display Nellcor™ RR are filled, but RR has not been activated in service menu

* When used with a remote system, this message may be displayed with a different wording if so programmed by the host computer. The host computer may also initiate a message which would appear upon the stopping of communications.

4.14.8.2. Waveform area messages

The following messages will appear in the relevant waveform area in addition to appearing in the message area.

Table 17. Waveform Area Messages

Message	Description
FilterLine Disconnected	No FilterLine™ sampling line is connected to the monitor
Clearing FilterLine	FilterLine™ sampling line kinked or clogged with water. Appears during clearing time until the FilterLine™ sampling line is unclogged, or a blockage state is determined.

Message	Description
FilterLine Blockage	FilterLine™ sampling line is kinked or clogged
Performing Auto Zero	Autozero is currently being performed; CO ₂ is not available during Autozero
CO ₂ Error	Failure has occurred which prohibits the operation of the CO ₂ function.
CO ₂ Standby	CO ₂ standby has been activated
SpO ₂ Sensor Not On Patient	SpO ₂ sensor is off patient.
Pulse Not Found	No detectable pulse
SpO ₂ Sensor Disconnected	SpO ₂ sensor is disconnected; SpO ₂ is not available
SpO ₂ Standby	SpO ₂ standby has been activated
SpO ₂ Error	Failure has occurred which prohibits the operation of the SpO ₂ function.
No Breath xxx Seconds	No breath has been detected for xxx seconds

4.14.9. Changing Alarm Limits

In high-altitude environments, etCO₂ values may be lower than values observed at sea level, as described by Dalton's law of partial pressures. When using the monitor in high-altitude environments, it is advisable to consider adjusting etCO₂ alarm settings accordingly.

Alarm limits may be changed as follows:

To change alarm limits on a temporary basis (until the monitor is turned off) for the current patient type:

1. Click the **Menu** button on the front panel.
2. **Setup>Alarm Setup** will be the first choice on the **Menu** screen. Click **Enter**.
3. Navigate to the relevant alarm and click **Enter**. The up-and-down indicator arrows on the screen will become active. Using the directional arrows, select the desired alarm limit and click **Apply**.
4. Change additional limits if desired.
5. Select **Home** to return to the Home screen.

To change alarm limits on a permanent basis (that is, also after the monitor is turned off and on):

1. Click the **Menu** button on the front panel.
2. **Setup** will already be selected on the Menu screen. Navigate to and select **Service** on the **Setup Menu** screen.
3. Input Service password and click **Done**.
4. Navigate to and select **Institutional Defaults>Alarms>Factory Defaults**.
5. Select the relevant type of alarm limits and click **Enter**.
6. Select the relevant alarm limit and click **Enter**. A window for these changes will appear on the right side of the screen. Using the directional arrows, move to the desired screen section, select the desired alarm limit, and click **Enter**. Alarms may also be disabled from this screen.
7. Change additional limits if desired.
8. Select **Home** to return to the Home screen.

9. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

When the SpO₂ alarm limit is set below 85%, a message reading **SpO₂ Low Alarm Limit: xx** will appear in the header area, indicating the level of the SpO₂ Low alarm limit.



Note: A potential hazard may exist if different alarm pre-sets are used for the same or different equipment in any single area.

4.14.10. Testing Alarm Settings

In order to test the **No Breath** alarm, establish a display of normal breathing on the monitor. Once normal breathing is displayed, remove the sampling line from the test subject to create a no breath situation. The monitor should then display a **No Breath** alarm.

In order to test the pulse oximetry **SpO₂ Sensor Not on Patient** alarm, establish a display of SpO₂ values on the monitor. Once SpO₂ values are displayed, remove the sensor from the test subject to create a **SpO₂ Sensor Not on Patient** situation. The monitor should then display a **SpO₂ Sensor Not on Patient** alarm.

4.14.11. Alarm Limits - Factory Defaults

For the factory defaults for the alarm limits for different patient types, reference [Table 30. Factory Default Alarm Limits](#) on page 254.

4.15. Trends

4.15.1. Introduction

The monitor stores patient data that provides detailed information on the history of the patient during monitoring in a trend screens. The trend displays allow you to look at the patient history as part of the medical analysis to aid in patient assessment.

4.15.2. Trend Storage

Trends are automatically stored. The monitor shall store the previous 48 hours of monitored (turned-on) data regardless of whether there is a period of time during which the monitor is turned off. Data is saved once a second, except for the CO₂ waveform data, which is saved 20 times a second (the latter is saved only when recorded in real time using the Real-time waveforms or Real-time Full Continuous Transfer reports).

Data will be displayed according to the selected trend increment.

The trend data stores the following parameters:

- Current Date
- Current Time
- EtCO₂
- RR
- SpO₂
- PR
- IPI
- etCO₂ trend in waveform format*

- High/medium priority patient alarms (one per second, one per parameter)
- Equipment-caused events such as **Battery Low** or other monitor-related messages.
- Event markers input by the user, along with any event label (one per second)

*Waveform is shown in Graphical Trend.

During Pump Off mode no CO₂ data shall be recorded.

The RR displayed in the Trend screens will be the respiration rate that was recorded at that point in time; if the capnography-based RR was recorded, it will be displayed, and if the Nellcor™ RR was recorded, it will be displayed. A heart or lung icon next to the RR will indicate which type of RR is displayed in the trend screen. See [Figure 40. Graphical Trend Screen - 1 Hour Display](#) and [Figure 41. Tabular Trend Screen – 30 Second Display](#), both below.

4.15.3. The Trend Display Screens

Trend data is displayed in two different formats; graphical and tabular.

The Graphical Trend screen allows you to view the patient data over a longer time scale (1, 2, 4, 8, or 12 hours at a time) and scroll through the data looking for patterns, specific events or alarms. The Tabular Trend screen offers views of 1 sec, 5 sec, 15 sec, 30 sec, 1 min, 5 min, 15 min, 30 min, and 1 hour, enabling the user to see more details about patient status.

4.15.3.1. Graphical Trend display screen

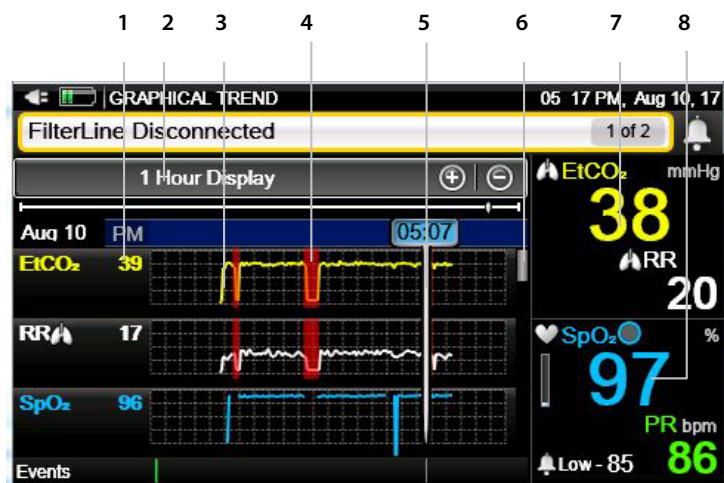
The graphical trend display screen may be viewed as follows:

1. Click the **Menu** button on the front panel.
2. Navigate to and select **Trend** on the Setup Menu screen.

3. Navigate to and select **Graphical Trend** on the Trend Menu screen. The screen will display the graphical trend currently being recorded in a 4-Hour display. A white indicator line with a blue time marker will indicate the selected time seen on the left side of the screen. A red line on the trend waveform will indicate an alarm at that point of time.
4. To change the time period being displayed, click the **Enter** button, and navigate to and select the desired Zoom options from the options that will appear on the screen. Click **Enter** again. The display will now show the desired display time period.
5. To view a different time period, use the directional arrows to adjust the time seen on the screen to the time that you would like to view. The white indicator line with blue time marker will indicate the location of the time indicator and the data at the left will show the data at that period in time.
6. To view additional parameters, click the down directional arrow and then the **Enter** button to scroll down to see those parameters not seen on the screen.

The numerical data seen on the right section of the screen will continue to display current data.

Figure 40. Graphical Trend Screen - 1 Hour Display



Label	Function	Description
1	EtCO ₂ , RR, and SpO ₂ patient data at cursor point	EtCO ₂ , RR and SpO ₂ patient values at cursor point in numeric format
2	Time range	Range of selected display time period
3	EtCO ₂ , RR, and SpO ₂ patient trend data	EtCO ₂ , RR and SpO ₂ patient values during the selected display time period shown in a waveform format; note the lung icon, indicating that the RR data on this screen is derived from capnography
4	Alarm marker	The red section on the trend waveform indicates a red high priority alarm occurring at that point in time; yellow markers would show a yellow medium priority alarm.
5	Cursor bar	Indicates time of cursor

		location; patient data for this time point is shown at left
6	Scroll bar	Bar used to scroll down to view other parameters (PR and IPI)
7	CO ₂ current patient data	Current patient CO ₂ values, displayed regardless of what trend data is being displayed on the screen
8	SpO ₂ current patient data	Current patient SpO ₂ values, displayed regardless of what trend data is being displayed on the screen

4.15.3.2. Graphical Trend screen search

Follow the steps below to search on the graphical trend display screen:

1. Click the **Menu** button on front panel of the monitor.
2. Navigate to and select **Trend** on the Setup Menu screen.
3. Navigate to and select **Search Graphical** on the Trend Menu screen. The screen will display the Trend Menu Search Date and Time screen for the graphical view.
4. On the **Trend Menu: Calendar Search Date and Time** screen, navigate to and select the exact date and time that you want to view on the screen.
5. When you have selected the desired time, navigate to the search button on the screen and select. If there is recorded data for the selected time point, the monitor will display the Graphical Trend screen view, at the selected time point.
6. If there is no recorded data for the selected time point, the monitor will inform you of this fact and the screen will

indicate that there are no records found for the selected time and request that you select another time value.

Table 18. Trend Increments and Zoom Levels

Trend Increment	Time Frame of Report
15 sec	1 hour
30 sec	2 hours
1 min	4 hours
2.5 min	8 hours
3.0 min	12 hours

4.15.3.3. Tabular Trend display screen

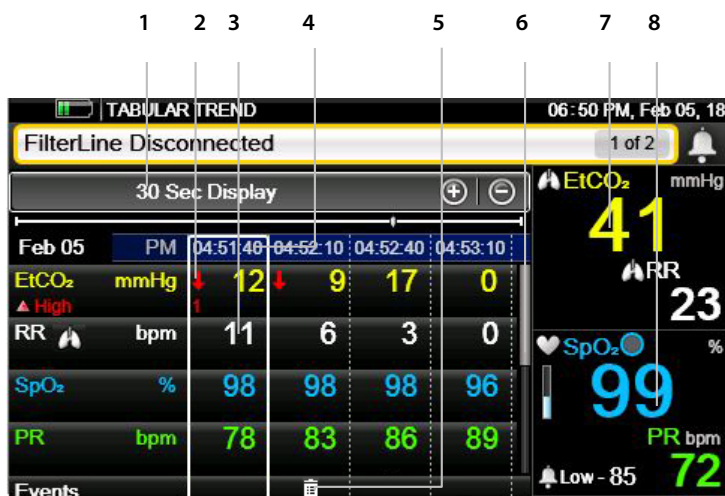
The tabular trend display screen may be viewed as follows:

1. Click the **Menu** button on front panel of the monitor.
2. Navigate to and select **Trend** on the Setup Menu screen.
3. Navigate to and select **Tabular Trend** on the Trend Menu screen. The screen will display the tabular trend currently being recorded in a 30-second display.
4. To change the time period being displayed, click the **Enter** button, and navigate to and select the desired zoom option from the **Zoom Options** that will appear on the screen. Click **Enter** again. The display will now show the desired display time period.
5. To view a different time period, use the directional arrows to adjust the time seen on the screen to the time that you

would like to view. The frame will indicate the chosen time period and the data at that period in time.

6. When the latest period of time recorded is reached at the right, the screen will indicate: **End of data**. When the time previous to the recorded data is reached at the left, the screen will indicate: **No data**.
7. To view additional parameters, click the down directional arrow and then the **Enter** button to scroll down to see those parameters not seen on the screen.
8. The numerical data seen on the right section of the screen will continue to display current data.

Figure 41. Tabular Trend Screen – 30 Second Display



Label	Function	Description
1	Time range	Range of selected display time period

2	Alarm marker	A red or yellow up or down arrow indicated a high or medium priority alarm taking place at that point in time, along with an alarm count. In the example shown, a high priority EtCO ₂ alarm has occurred. Note that the value shown is an average for the 30 second period of time; at one point during that 30-second period, the etCO ₂ value was under the Low etCO ₂ alarm limit, and a Low etCO ₂ High Priority alarm occurred.
3	EtCO ₂ , RR, SpO ₂ and PR patient data	EtCO ₂ , RR, SpO ₂ and PR patient values during the selected display time period shown for each time period; note the lung icon near the RR text,, indicating that the RR data on this screen is derived from capnography
4	Cursor data	Date and time of cursor location (the white frame marks the cursor location)
5	Event marker	An event marker on the events listing indicates an event occurring at that point in time.
6	Scroll bar	Bar used to scroll down to view other parameters
7	EtCO ₂ current patient data	Current patient etCO ₂ values, displayed regardless of what trend data is being displayed on the screen
8	SpO ₂ current patient data	Current patient SpO ₂ values, displayed regardless of what trend data is being displayed on the screen

4.15.3.4. TabularTrend screen search

Follow the steps below to search on the tabular trend screen:

1. Click the **Menu** button on front panel of the monitor.
2. Navigate to and select **Trend** on the Setup Menu screen.
3. Navigate to and select **Search Tabular** on the Trend Menu screen. The screen will display the Trend Menu Search Date and Time screen for the tabular view.
4. On the **Trend Menu: Calendar Search Date and Time** screen, navigate to and select the exact date and time that you want to view on the screen.
5. When you have selected the desired time, navigate to the search button on the screen and select. If there is recorded data for the selected time point, the monitor will display the Tabular Trend screen view, at the selected time point.
6. If there is no recorded data for the selected time point, the monitor will inform you of this fact and the screen will indicate that there are no records found for the selected time and request that you select another time value.

4.15.3.5. Events List screen

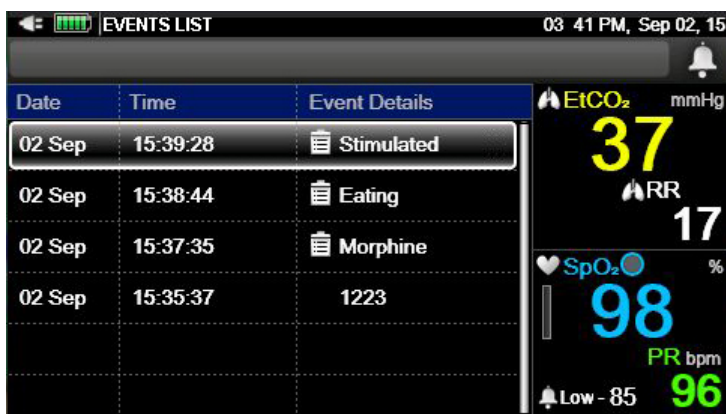
The events list display screen may be viewed as follows:

1. Click the **Menu** button on front panel of the monitor.
2. Navigate to and select **Trend** on the Setup Menu screen.
3. Navigate to and select **Events List** on the Trend Menu screen. The screen will display a list of events for the current patient case or a list of all events if there is no open patient case. Please note that this list includes events added by the user but not alarms. If there are no events, the list will contain no items.

4. To view additional events, if there are any, click the down directional arrow to scroll down to see those events not seen on the screen.

The numerical data seen on the right section of the screen will continue to display current data.

Figure 42. Events List



4.15.3.6. Events List screen search

Follow the steps below to search on the events list search screen:

1. Click the **Menu** button on front panel of the monitor.
2. Navigate to and select **Trend** on the Setup Menu screen.
3. Navigate to and select **Search Events** on the Trend Menu screen. The screen will display the Trend Menu Search Date and Time screen for events.
4. On the **Trend Menu: Calendar Search Date and Time** screen, navigate to and select the exact date and time that you want to view on the screen.
5. When you have selected the desired time, navigate to the search button on the screen and select. If there is recorded

data for the selected time point, the monitor will display the Events list at the selected time point.

6. If there is no recorded data for the selected time point, the monitor will indicate that there are no events found for that time point and display the events that occurred closest to the requested time point.

4.15.4. Choosing Trend Parameters

Reference [11.1.5.3 Trend Display Config settings](#) on page 264 to change the trend parameters seen on the screen or change the order of the parameters.

4.15.5. Clearing Trend Memory

If the operator is not using cases on the device for each new patient, it is recommended to erase trend memory when the monitor is switched to a new patient, in order to avoid mistaking the earlier data for the present patient's data. If the operator opens a new case for each patient on the device, the trend memory will automatically be erased when the case is exited. Note that the patient type will revert to default when the trend memory is erased in both situations.

4.15.6. Trend Display Mode

The monitor offers two types of trend display modes for the graphical display, Recorded and Calendar. The default is Recorded. In the Recorded mode, if the monitor is turned off and then on, the trend memory will record the time that these actions occur and display these actions in the record, but not display the missing time periods. In the Calendar Mode, if the monitor is turned off and then on, the trend will include the time during which the monitor is turned off in its trend recording period, and, if that time period is long enough to leave enough room on the screen, will mark those time periods with the text **Monitor Off** on the display.

In both modes, no data will be shown for those time periods during which the monitor was turned off, since not data is available. However, since the monitor can only record up to 48 hours of data, use of the Calendar Mode might limit the amount of useful data stored in the trend memory.

To change the Trend Mode, reference [11.1.5.1 Main Trend settings](#) on page 262.

4.15.7. Trends Configuration

4.15.7.1. Event marking mode

There are two types of event marking modes available.

- **Detailed Event Marking:** When the **Event** button is pushed, you can enter a specific description of the event from a table of user definable values (reference [4.4 Patient Events](#) on page 67).
- **Quick Event Marking:** Marks that an event occurred when the **Event** button is pushed, but does not give any details.

The default event marking mode for the monitor is **Detailed**.

To change the Event Marking Mode on a permanent basis (that is, also after the monitor is turned off and on), click the **Menu** button on the front panel of the monitor.

1. **Setup** will already be selected on the Menu screen.
2. Navigate to and select **Service** on the Setup Menu Screen.
3. Input Service password and click **Done**.
4. Navigate to and select **Institutional Defaults>Trend**.
5. Navigate to and select **Event Marking Mode** and click **Enter**.

6. Select the desired mode and select **Enter**.
7. Select **Home** to return to the Home screen.

4.15.7.2. Trend Increment Display

The increments seen on the tabular trend screen can be changed in **Institutional Defaults>Trend**. The options are 1, 5, 15, or 30 seconds, or 5, 15, or 30 minutes, or 1 hour.

The default value is 30 seconds.

4.15.7.3. A/Hr and ODI trends

A/Hr and ODI Trends are accessed using the **Reports** soft key and are covered in [5.2 Reports](#) on page 176.

5. Product Data Output

5.1. Report Options

The monitor provides the opportunity to view or store various reports. These reports include Ventilation and Desaturation reports, Alarm Review reports, and case and trend reports.

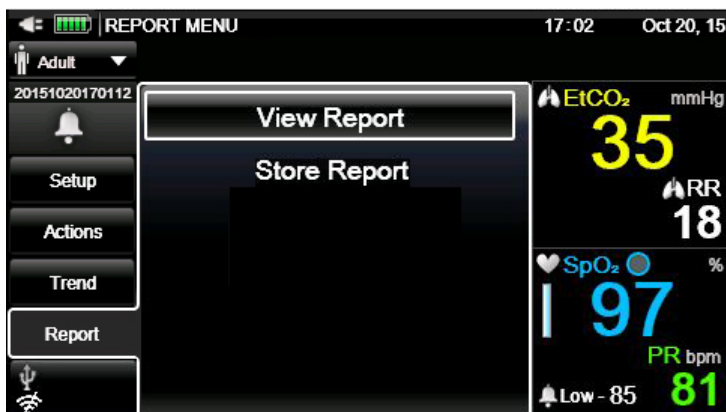
The various types of report options can be accessed in various ways; reference [Table 19. Report Options Access](#), below, for more details.

Table 19. Report Options Access

Report Access	Report Option	Notes
View Report	Vent. And Desat reports A and B Parameter Statistics report Alarm Review report	
Store Report	Tabular Case Graphical Case Tabular Trend Graphical Trend Realtime Continuous Tabular Realtime Continuous Waveform Real Time Full Continuous Transfer Current Alarm Limits	All stored reports can be printed on an external device; reports saved in Html format may be printed by this method.

For the Report Selection screen, reference [Figure 43. Report Selection](#), below.

Figure 43. Report Selection



5.2. Reports

The monitor provides the option to view four types of reports:

- Vent and Desat Report A
- Vent. and Desat Report B
- Parameter Statistics
- Alarm Review

The Vent. and Desat report can only be viewed if a patient case has been opened. If no patient case is open, a message indicating **Open a New Patient Case.** will appear and these reports will not be available.

To open a patient case, reference [4.3 Patient Cases and Patient ID Numbers](#) on page 66.

To scroll down on these reports, navigate to the main window of the report, click **Enter** and then use the up and down arrows to scroll.

On all of these report screens, real-time data of the current patient is seen on the right side of the screen.

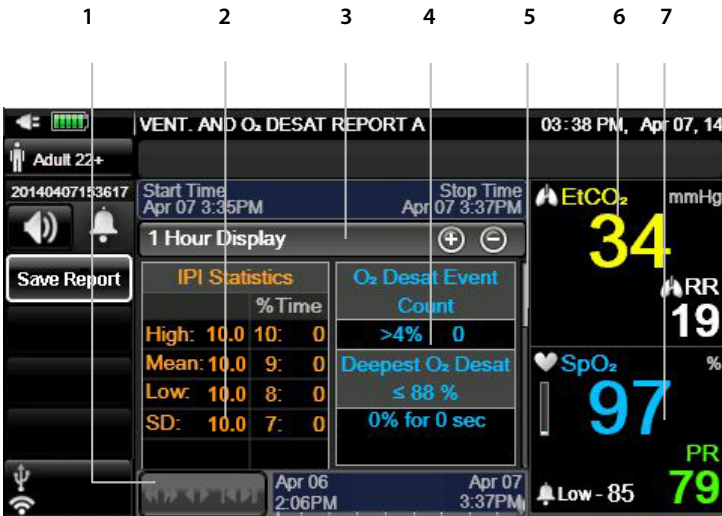
A report may be viewed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. Navigate to and select **Report>View Report** on the Setup Menu Screen.
3. Select the desired report and click to view.

A Vent. and Desat report can only be viewed if a patient case has been opened. If the open patient does not have IPI or Apnea and Desaturations data (for example, a neonatal patient), the Vent. and Desat Reports will not be available. An Alarm Review report may be viewed even if a patient case has not been opened.

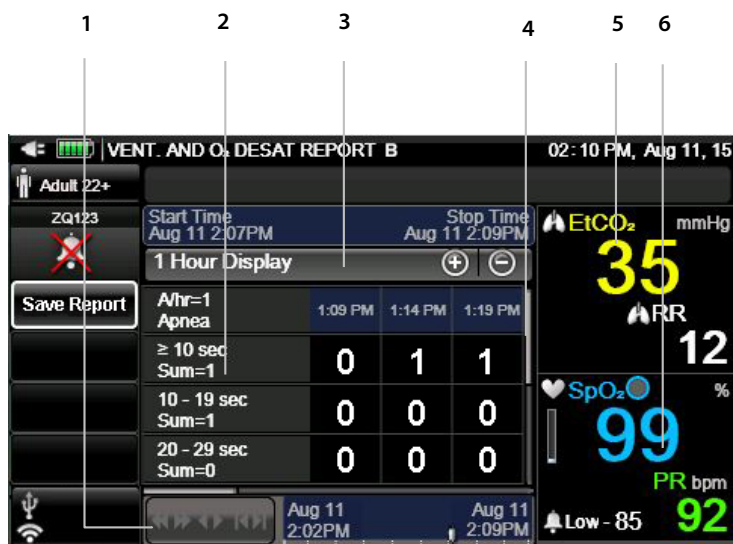
On the Alarm Review screen, real-time data of the current patient is seen on the right side of the screen.

Figure 44. Vent and Desat Report A



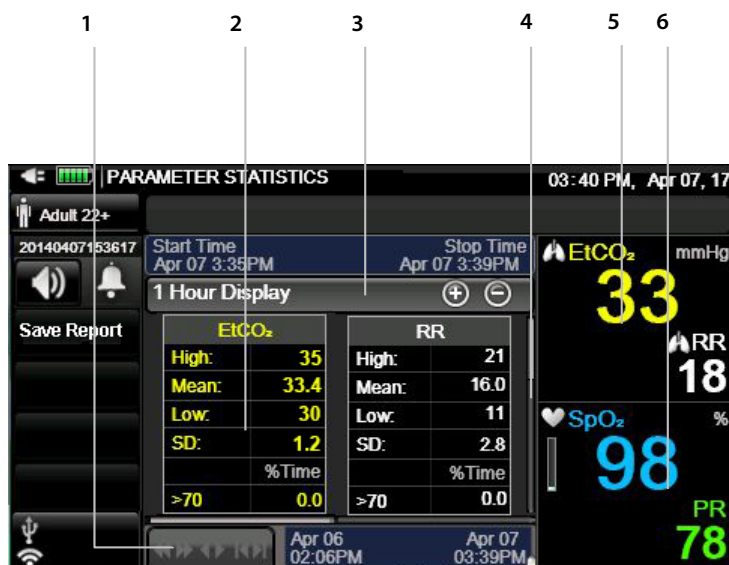
Label	Function	Description
1	Scroll bar (horizontal)	Bar used to scroll right to view additional time periods
2	IPI Statistics data	Patient IPI Statistics data
3	Time range	Range of selected display time period
4	O ₂ Desat and Apnea Event Counts data	Patient O ₂ Desat and Apnea Event Counts data
5	Scroll bar (vertical)	Bar used to scroll down to view other parameters
6	Real-time etCO ₂ data	Current patient real-time data
7	Real-time SpO ₂ data	Current patient real-time data

Figure 45. Vent and Desat Report B



Label	Function	Description
1	Scroll bar (horizontal)	Bar used to scroll right to view additional time periods
2	Apnea and Desat data	Patient Apnea and Desat data
3	Time range	Range of selected display time period
4	Scroll bar (vertical)	Bar used to scroll down to view other parameters
5	Real-time etCO ₂ data	Current patient real-time data
6	Real-time SpO ₂ data	Current patient real-time data

Figure 46. Parameter Statistics Report



Label	Function	Description
1	Scroll bar (horizontal)	Bar used to scroll right to view additional time periods
2	Parameter data	Patient parameter data: the RR data which appears will be whichever type of RR data is available in the patient record
3	Time range	Range of selected display time period
4	Scroll bar (vertical)	Bar used to scroll down to view other parameters
5	Real-time etCO ₂ data	Current patient real-time data
6	Real-time SpO ₂ data	Current patient real-time data

Figure 47. Alarm Review Report



Label	Function	Description
1	No Breath data	Patient No Breath data
2	EtCO ₂ data	Patient EtCO ₂ data
3	SpO ₂ data	Patient SpO ₂ data
4	Scroll bar (vertical)	Bar used to scroll down to view other parameters
5	Real-time etCO ₂ data	Current patient real-time data
6	Real-time SpO ₂ data	Current patient real-time data

Table 20. Ventilation and Desaturation and Parameter Statistics Report Types

Report Name	Fields Included	Time Frame of Report
Vent. and Desat Report A	IPI statistics, Apnea Event Count and Desaturation Event Count	1, 2, 4, 8, or 12 hours, as selected
Vent. and Desat Report B	Apnea Event Count and Desaturation Event Count	
Parameter Statistics	EtCO ₂ and RR data, including High, Mean, Low, SD, and %Time for various values	
Alarm Review	Current alarm limits and number of each type of alarm in the last hour for the current patient.	1 hour

5.3. Data Output

The monitor can export stored and current data to external devices by the following methods:

Data may be transferred to a USB flash memory device or micro SD card for later transfer to a computer and printing using an external printer. Reference [Figure 3. Monitor Right View](#) on page 23 and [Figure 4. Monitor Left View](#) on page 25 for the location of the USB and micro SD card ports.



Caution:

Ensure that there are no viruses on the USB flash memory

device or micro SD card prior to interfacing with the monitor.



Caution:

Note that case stop will erase the trend memory, so any required trend download should be performed before case stop.

A case or trend report can be stored (downloaded) as follows:

1. Place the USB flash memory device in the USB flash memory device port.
2. Click the **Menu** button on the front panel of the monitor.
3. Navigate to and select **Report** on the Menu screen.
4. Navigate to and select **Store Report** on the Report Menu Screen.
5. Navigate to and select the report you want to store. If a case is not currently active, you will not be able to store a case report.
6. If the type of report permits more than one type of report (HTML or Text) navigate to and select the desired report. Some reports are available only in one format; for example, Graphical Trend and Graphical Case Reports are available in BMP format only.
7. Navigate to and select **Choose Device**. Navigate to and select the device on which you want to store the report.
8. Navigate to and select **Save Report**.
9. To store another report, repeat the process.
10. To stop storing a report (this is generally used for real-time reports, which will continue downloading onto the storage device unless they are stopped), click the **Active Reports** button.

11. Navigate to and select the report that you want to stop.
Navigate to and select the **Stop** button to stop downloading the report.

Selecting **Active Reports** during the download process will display a list of reports that are currently downloading. The list can contain more than one report if more than one download type occurs at the same time, or (even though only one report can be downloaded at one time to a USB device) more than one USB device is used (using a multiplier).

The following data transfer report types are available:

Table 21. Data Transfer Report Types

Report Name	Description	File type	Fields Included
Tabular Case	Stores the tabular trend report of selected patient ID, data shall be from current case start to current point in time. Case stop will erase the trend memory.	Txt, HTML	Name of report
			Patient ID
			Patient Type
			Case Number
			Patient Gender, Age, Weight
Tabular Case	Stores the tabular trend report of selected patient ID, data shall be from current case start to current point in time. Case stop will erase the trend memory.	Txt, HTML	Case Start Date and Time
			Report Generation Date and time
			DATE, TIME
			Patient Readings: etCO ₂ , RR, IPI, SpO ₂ , PR
			Patient Alarm Occurrences
Tabular Case	Stores the tabular trend report of selected patient ID, data shall be from current case start to current point in time. Case stop will erase the trend memory.	Txt, HTML	Apnea events and Desat events

Report Name	Description	File type	Fields Included
			count
			Equipment Advisory Message Occurrences: CO ₂ Not Available, SpO ₂ Not Available, Primary Battery Low, Secondary Battery Low
			Events (one per second, one per parameter)
Tabular Trend	Stores the tabular report of the entire trend available in the memory for the current Patient ID from previous stop case to current point in time. Case stop will erase the trend memory. This report shall include all the parameters available in trend storage based on enable/disable of the parameters. Parameters active for real time monitoring is not related to this	Txt, HTML	Name of report Patient ID Patient Type Report Generation Date and time Zoom Selected (in html reports only)
			DATE, TIME
			Patient Readings: etCO ₂ , RR, SpO ₂ , PR, IPI.
			Patient Alarm Occurrences
			Apnea events and Desat events count

Report Name	Description	File type	Fields Included
	report.		Equipment Advisory Message Occurrences: CO₂ Not Available, SpO₂ Not Available, Primary Battery Low, Secondary Battery Low Events (one per second, one per parameter)
Graphical Case	Stores the graphical trend report of the entire trend available in the memory selected Patient ID (data shall be from current case start to current point in time. Case stop will erase the trend memory.). Graph shall be plotted on the BMP page in zoom selected for graphical trend.	BMP	Name of report Patient ID Patient Type Case Number Patient Gender, Age, and Weight Case Start Time Zoom Selected Report Generation Date and time Start Time Stop Time Graphs for selected parameters
Graphical Trend	Stores the graphical report of all the trend or present displayed page in GRAPHICAL TREND SCREEN	BMP	Name of report Patient ID Patient Type Zoom Selected Report Generation

Report Name	Description	File type	Fields Included
	with selected zoom		Date and Time Start Time Stop Time Graphs for selected parameters
Realtime Continuous Tabular	Stores the tabular report of real time values of the current active parameters with 1 second resolution. Anytime there is a change in the available parameters due to patient type or CO ₂ mode change, the current file will be closed and a new file created.	Txt	Name of report Patient ID Patient Type Report Generation Date and Time
			DATE, TIME
			Patient Readings: etCO ₂ , RR, SpO ₂ , PR, IPI, A/hr, ODI
			Patient Alarm Occurrences
			Apnea events and Desat events count
			Equipment Advisory Message Occurrences: CO ₂ Not Available, SpO ₂ Not Available, Primary Battery Low, Secondary Battery Low
			Events (one per second, one per parameter)

Report Name	Description	File type	Fields Included
Realtime Continuous waveform	Stores in txt format the graphical report of CO ₂ waveform (with one data point every 50 ms)	Txt	Name of report Patient ID Patient Type Report Generation Date and time Data to build graph for CO ₂ waveform
Real-time Full Continuous Transfer	Stores the report containing real-time CO ₂ waveform plus real time values of current active parameters with 1 second resolution by repeating numerical values for every 50ms along with waveform.	Txt	Name of report (Realtime Full Continuous Transfer) Patient ID Patient Type Report Generation Date and time
			DATE, TIME
			Patient Readings (at trend storage resolution): etCO ₂ , RR, IPI, SpO ₂ , PR, A/hr, ODI
			Patient Alarm Occurrences
			Apnea events and Desat events count
			Equipment Advisory Message Occurrences: CO ₂ Not Available,

Report Name	Description	File type	Fields Included
			SpO ₂ Not Available, Primary Battery Low, Secondary Battery Low
			Events (one per second, one per parameter)
			Patient reading every 50 milliseconds – 20 times a second – (for creating CO ₂ waveform): CO ₂ wave
Current Alarm Limits	Stores the current selected alarm limits of the available/active parameters	Text, Html	Name of report Patient ID Patient Type Report Generation Date and time
			Unit, high limit and low limit for current relevant parameters For patients measured with standard etCO ₂ measurement: EtCO ₂ , RR, No Breath, SpO ₂ , PR, SatSec, IPI

Figure 48. Sample Html Report: Tabular Trend

Tabular Trend																								
Patient ID: 2000010102000																								
Patient Type: Adult																								
Report Generation Time: Jun 01, 14 12:48:55 PM																								
Zoom Selected: 30 sec																								
Date	Time	SpO ₂ Reading	SpO ₂ Alarm	FiO ₂ Reading	FiO ₂ Alarm	RR Reading	RR Alarm	HR Reading	HR Alarm	SpV _{O₂} Reading	SpV _{O₂} Alarm	PR Reading	PR Alarm	PI Reading	PI Alarm	CO ₂ Reading	CO ₂ Alarm	SpO ₂ Percent Alarm	% Sat	CO ₂ Raw Available	SpO ₂ Raw Available	Primary Alarm Low	Secondary Alarm Low	Events
Jun 01/14	11:16:45 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			3:10:04 AM Coughing 3:10:14 AM 75
Jun 01/14	11:17:00 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:17:15 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:17:30 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:17:45 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:18:00 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:18:15 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:18:30 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:18:45 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:19:00 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:20:45 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			3:20:59 AM Data Event
Jun 01/14	11:21:00 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:21:15 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:21:30 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:21:45 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:22:00 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:22:15 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:22:30 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:22:45 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:23:00 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:23:15 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:23:30 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:23:45 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			
Jun 01/14	11:24:00 AM	97	95	0	10	18	18	98	97	97	97	10	10	10	10	10	10	10	10	10	10			

5.3.1. Clinical Report Option

The device provides a feature for the production of reports that include both capnography-based respiration rate and Nellcor™ oximetry-based respiration rate, for use in clinical studies. If this feature is enabled, and both types of data are available, the Realtime Continuous Tabular and Real-time Full Continuous Transfer reports will include both types of RR.

To generate such reports, follow the instructions below.

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Setup Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Enter the Service password and select **Done**.
5. Navigate to and select **Institutional Defaults**.
6. Navigate to and select **Monitor**. Navigate to and select **Clinical Report**. Click **Enter**, navigate to **Enabled** and click **Enter**.

7. Click the **Home** button. The device will request that you shut down. Click **Enter** to shut down and the restart the device.
8. When you restart the device, ensure that the following conditions are in force in order to record the Realtime Continuous Tabular and Real-time Full Continuous Transfer reports as clinical reports, that is, as reports with both types of respiration rate:
 - The patient type must be Adult 22+.
 - Both a CO2 sampling line and an SpO2 sensor that supports Nellcor™ respiration rate must be attached to the device and the patient.
9. With both CO2 and SpO2 measurement taking place under these conditions, the display will show only capnography-based respiration rate, but both types of respiration rate will be shown in the Realtime Continuous Tabular and Real-time Full Continuous Transfer reports.

5.3.2. Printing reports

Report printing is done via an external printer after storing a report using a USB device or micro SD card as described in [5.4 Data Transfer](#), below.

5.4. Data Transfer

5.4.1. USB

The monitor provides both a standard USB port and mini USB port on the monitor. The standard USB port on the monitor is for use with a flash memory device only. The mini USB port on the monitor is for use of qualified service personnel only. Do not attempt to power the monitor via either USB port. To use the standard USB port, place the USB flash memory device into the port before beginning the download process.

The monitor recognizes flash memory drive devices manufactured by all major brands. It will recognize flash memory drives with the file systems FAT, FAT32, exFAT; it may not recognize other file systems, including NTFS. Some flash drives may not be able to attach to the port on the monitor as a result of mechanical incompatibility; check each flash drive to ensure that it fits into the port. A typical flash memory device is shown below.

Figure 49. Typical Flash Memory Device



The standard USB port on the monitor is for use with a flash memory device only. It is not a full service USB port. Do not attempt to connect the monitor to a computer via the USB port.

If the monitor does not detect the flash memory drive, remove and re-insert the flash memory drive. If the flash memory drive is still not detected, check that the drive being used is from a supported manufacturer.

If free disk space on the flash memory drive is less than 100 kb, writing to the USB disk drive is not allowed. Under this condition, if data transfer is already in progress, it will be aborted. Any new data transfer CANNOT be initiated under the low disk space condition.



Note: The USB flash memory drive must be inserted with care in the USB port, without using excessive force. If the flash memory drive cannot be inserted easily into the port, do not use the drive.



Note: Only one memory device (one flash memory drive or micro SD card) may be used at one time, and only one report can be downloaded at one time.



Note: Do not remove the USB flash drive from the device during download and upload procedures.

Table 22. USB File Naming Convention

Report Name	File Naming Conventions
Tabular Case	TCR_ PATIENT TYPE_ DATE_ TIME_ PATIENT ID_ MONITOR SN
Tabular Trend	TTR_ PATIENT TYPE_ DATE_ TIME_ PATIENT ID_ MONITOR SN
Graphical Case	GCR_ PATIENT TYPE_ DATE_ TIME_ PATIENT ID
Graphical Trend	GTR_ PATIENT TYPE_ DATE_ TIME_ PATIENT ID
Real-time Continuous Tabular	RCT_ PATIENT TYPE_ DATE_ TIME_ PATIENT ID_ MONITOR SN
Real-time waveforms	RCW_ PATIENT TYPE_ DATE_ TIME_ PATIENT ID_ MONITOR SN
Real-time Full Continuous Transfer	FCTR_ PATIENT TYPE_ DATE_ TIME_ PATIENT ID_ MONITOR SN
Current Alarm Limits	ALIM_ PATIENT TYPE_ DATE_ TIME_ PATIENT ID_ MONITOR SN

Table 23. USB Files Examples

Report Name	File Name
-------------	-----------

Tabular Case	TCR_ ADULT_060714_141453_20140607121546_SF17100536
Tabular Trend	TTR_ ADULT_060714_141453_20140607121546_SF17100536
Graphical Case	GCR_ ADULT_060714_141453_20140607121546
Graphical Trend	GTR_ ADULT_060714_141453_20140607121546
Real-time Continuous Tabular	RCT_ ADULT_060714_141453_20140607121546_SF17100536
Real-time waveforms	RCW_ ADULT_060714_141453_20140607121546_SF17100536
Real-time Full Continuous Transfer	FCTR_ ADULT_060714_141453_20140607121546_SF17100536
Current Alarm Limits	ALIM_ ADULT_060714_141453_20140607121546_SF17100536

To read patient data from an html report, simply download the file to a local folder and open the file in html. Txt files may be opened in Microsoft™ Excel™ software or a similar program to facilitate viewing and analyzing the data. If opening a file in Excel™ software, open Excel™ software first and then open the file from the program. In order to view the file with cell borders, select the entire document and navigate to and select the **All Borders** icon in the Excel™ file, to add cell borders to all cells.

Contact your local sales representative for more information.

5.4.2. SD

The monitor supports micro SD high-capacity SDHC cards up to 32GB. Use of these cards is similar to the use of a USB flash drive.

5.4.3. Wireless Communication

The monitor can work with three different types of WiFi networks, as follows:

- Open networks
- Secure networks (which require a password and/or a security key)
- Enterprise networks (which require a certificate and/or other security requirements)

Follow the instructions below to set up and turn on WiFi connectivity on the device for the different types of networks.



Caution:

Ensure that there are no viruses on the network to which you are connecting prior to interfacing with the monitor.



Note :

Hereby, Medtronic declares that this Radio Equipment Model number [Capnostream™35] is in compliance with Directive 2014/53/EU. The full text of the EU declaration of conformity is available at the following internet address: www.MedtronicCompliance.eu

5.4.3.1. Turning on WiFi Connectivity

WiFi connectivity must be configured first; reference [5.4.3.2 Configuring WiFi](#), below.

Once WiFi is configured to work with a network, clicking **On** as described below will turn on WiFi for the configured network.

To turn on WiFi connectivity, navigate to and select the connectivity icon on the main **Menu** screen (reference Connectivity Icon, below) to open the Connectivity screen.

Figure 50. Connectivity Icon; WiFi off at left, on at right



Navigate to **Wi-Fi Connection**, select, and click **On**. WiFi will be turned on, the text **WiFi connected to: *Network Name*** and a green check mark will appear on the screen, the WiFi icon will be in color, and the button will now show **Off**. Reference [Figure 51](#), below. If no networks are available, an on-screen message will inform you of this fact.

Figure 51. Menu Connectivity Screen showing WiFi connected



By default, WiFi is off. To turn it off when it is on, select and click the **Off** button.

Once connected, the name of the connected network will appear under the text **WiFi** in the middle of the Connectivity screen, and the WiFi signal strength indicator () will appear in the upper left-hand corner of all screens.

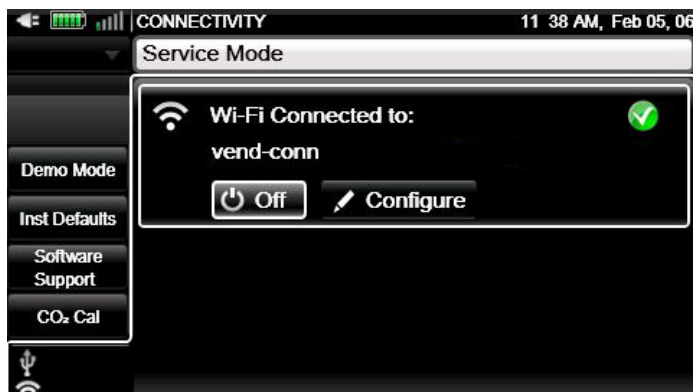
5.4.3.2. Configuring WiFi

The WiFi configuration step should be performed by technical support personnel, using the Service password in the Service screen.

Take the following steps to set up configuration of your WiFi connection:

1. **Setup** will already be selected on the Menu screen.
2. Navigate to and select **Service** on the Setup Menu Screen.
3. Input Service password and click **Done**.
4. Navigate to and select the connectivity icon on the service screen to open the Service Mode Connectivity screen. Reference [Figure 52](#), below.

Figure 52. Service Mode Connectivity Screen, with Connected Network



5. Navigate to the WiFi Connected to: section of the screen and click the **On** button.

6. Navigate to and click the **Configure** button in the WiFi Connected To: section of the screen.
7. The Service WiFi Configuration screen will open. Each available network will appear as a separate line on this screen. A lock icon indicates that the network requires a security key or a password, and a yellow star indicates that the network is the preferred network; a network will become the preferred network when it is added to the Network list by the user. The network strength will also be indicated on each line.
8. Navigate to the desired network on the list of available networks, using the directional arrows, and select. If the network does not require any security information, click **Connect** to connect to the network.
9. If the network requires a password, a pop-up screen will request that you input the required data. Fill in the **User Name** and **Password** fields and click **Connect** to add the network. If the connection succeeds, the system will add the network; if not, an on-screen message will inform you of this fact. If connection is not successful, check the password or try another network.
10. If the network requires a security key, a pop-up window will request that you add the security key. Do so, following the instructions on the screen to add the required information. Click **Connect** to add the inputted information. If the connection succeeds, the system will add the network; if not, an on-screen message will inform you of this fact. If connection is not successful, check the information or try another network.
11. If the network is an enterprise network, more information will be required. See the instructions in [5.4.3.2.1 Adding a network](#) on page 199 for more details.

When a network is successfully added and all required data is uploaded, the relevant network will show as connected in the WiFi Configuration screen, reference [Figure 53. WiFi Configuration](#), below.

[Figure 53. WiFi Configuration](#), below, is an example of a screen that shows a list of available networks. The currently connected network is marked by a green check mark icon and a yellow star.

Figure 53. WiFi Configuration



On this screen, other buttons provide additional functionality, as follows:

Off: Navigate to and click **Off** to turn off Wi-Fi connection for the device.

Add: Reference [5.4.3.2.1 Adding a network](#) on page 199

ReScan: To rescan for available networks to update the list of available networks, navigate to and click **Rescan** on the WiFi Network Configuration screen (reference [Figure 53. WiFi Configuration](#), above) to rescan for available networks.

Certificate: Reference [5.4.3.2.1 Adding a certificate](#) on page 201

5.4.3.2.1. Adding a network

If the desired network does not appear on the list, navigate to and click **Add** as seen in [Figure 53. WiFi Configuration](#), above, and follow the instructions on the screen.

Connections may be made to three types of networks; each will require a different type of security authorization, as follows:

1. Open networks. Input the network name and mark it as **Open**, and click **Enter** to mark this network as your preferred network. No security authorization is required.
2. Secure networks. Input the network name and mark it as **Shared, WPA Personal, or WPA2 Personal (as required)**. Input any required information, such as encryption type and security key, and click **Add** to connect to the network. If the inputted information is correct, this network will be marked as your preferred network.
3. WPA Enterprise or WPA2 Enterprise networks. Input the network name and mark it as **WPA Enterprise or WPA2 Enterprise (as required)**. Input any other required information, such as encryption type and username and password. Scroll down and select an EAP Type for the network; the choices are **PEAP** or **TLS**. With PEAP type networks, the server requires a certificate to authorize connection to the device. With TLS type networks, or if you choose to do so, both authorization of the device to the server and authorization of the server to the device are required for connection.
4. In both cases, upload the required certificate/s to a USB flash drive and attach that flash drive to the device. Click the **Validate Server** box if you want to or are required to add validation of the server to the device and click **Add certification**. Select the correct certificate/s (this will depend on the certificate that you are uploading) from the list of certificates on the flash drive which will appear on your screen. Navigate to the first required certificate and click **Add**. If the chosen certificate is valid, that certificate will be added. If another certificate is required, repeat the process.

5. Click the **Back** button and then navigate to and click **Add** to connect to the network. If all information is correct, the network will be added as the connected network.

5.4.3.2.1. Adding a certificate

You may also add a certificate for a selected network by clicking the Certificate button on the WiFi Configuration page (reference [Figure 53. WiFi Configuration](#), above, to open the Certifications Summary page. On this page, select the type of certificate that you want to upload; a Root CA certificate required by your network to identify the device to a protected network or another type of user certification which your network requires. In both cases, upload the certificate to a USB flash drive and attach the drive to your device. For a Root CA certificate, navigate to and click **Root CAs**, then navigate to the type of certificate and click **Enter**, and then navigate to **Add** and click **Add**. For the user certificate, navigate to and click **User Certifications** and then navigate to the type of certificate and click **Enter**, and then navigate to **Add** and click **Add**. If a certificate is added in this screen, it will not need to be added during the connection process.

5.4.3.2.2. Removing a network

To remove a network from the list of available networks, navigate to the relevant network on the list of available networks on the WiFi Network Configuration screen (reference [Figure 53. WiFi Configuration](#), above), using the directional arrows, and select. On the pop-up window that will open, click **Remove**. Once the user removes a network from the table, the focus button goes to the first network on the list.

5.4.3.3. Configuring IP Settings

Navigate to and select **Menu>Setup >Service >Input the Service password>Institutional Defaults>Monitor>Interfaces>IP Settings** to view MAC ID and other information regarding the connected network. Correct if required.

Once you are connected, select **Server Settings** for the desired network using **Menu>Setup>Service >Input the Service**

password>**Institutional Defaults>Monitor>Interfaces.** Input the Server IP address and Server Port of the network with which you want to interface.

Please note that configuring a network to work with the device can only be performed by a qualified service technician. Consult Capnographytechnicalsupport@medtronic.com for more information.

Table 24. Wireless Specifications

Issue	Value
Wireless certification	FCC ID 2AE5J-CS35001, Canada certification 20321-CS35001 (Modifications – C4PC)
Center frequency for wireless connection	2.4GHz (2.412GHz - 2.484GHz) 2.4-2.4835 GHz (maximum RF output power of 10mW [10 dBm]).
Wireless bandwidth	Bandwidth divided into 14 channels, where each channel has a 20MHz bandwidth
Wireless protocol specification	IEEE 802.11b/g/n
Max Tx power	< 10dBm
Rx sensitivity	-89 dBm, 11 Mbps, CCK (b) -76 dBm, 54 Mbps, OFDM (g) -73 dBm, 65 Mbps, OFDM (n)
Available modulation types	CCK and OFDM

The device can connect using three different types of Communication Protocols; the user can choose the desired protocol via: **Menu>Setup >Service >Input the Service password>Institutional Defaults>Monitor>Interfaces.** Select the desired Comm Protocol

Mode. Reference [11.1.6.2 Interface settings](#) on page 268; the list of protocols appears in that table.



WARNING

Only use Covidien-approved hardware or remote monitoring software for data port connectivity.



WARNING

Use the appropriate configuration information to ensure proper connectivity.



WARNING

If there are changes to the network with which you are connecting the device, verify that changes are reflected in the device to ensure connectivity.

5.4.4. Vital Sync™ System

The Vital Sync™ virtual patient monitoring platform is Covidien's EMR connectivity and remote continuous patient monitoring software solution that allows clinicians to remotely view patient information from multiple device categories on web enabled devices and send this information to electronic medical records and Clinical Information Systems.

The monitor may be used for data transfer to a Covidien Vital Sync™ system. This option permits regular, real-time transfer of data from the monitor to a Vital Sync™ system.

Before beginning the connection process, ensure that the following equipment is available:

- Server with Vital Sync™ system installed and WiFi capability
- Capnostream™35 monitor

Data is transferred from Capnostream™35 monitor to the Vital Sync™ system via WiFi; therefore connecting the monitor to the system is

done using the WiFi configuration procedure described in [5.4.3.2 Configuring WiFi](#) on page 197.

Take the following steps to configure the WiFi connection for the Vital Sync™ system:

1. **Setup** will already be selected on the Menu screen.
2. Navigate to and select **Service** on the Setup Menu Screen.
3. Input Service password and click **Done**.
4. Input the following settings: In **Institutional Defaults>Monitor>Interfaces**, the Comm Protocol Mode should be **Capnostream**. In **Institutional Defaults>Monitor>Interfaces>Server Settings**, input the **Server Port number** and the **Server IP Address**.
5. Click the **Back** button until you return to the main Service screen (reference [Figure 55. Service Mode Screen](#) on page 209). Navigate to and select the connectivity icon on the service screen to open the Service Connectivity screen.
6. Navigate to the WiFi Connection section of the screen and click the **On** button.
7. Navigate to and click the **Configure** button in the WiFi Connection section of the screen.
8. The Service WiFi Configuration screen will open. Navigate to and select the Vital Sync™ system network on the list of available networks. If the network does not require any security information, click **Connect** to connect to the network.
9. If the network requires security information, follow the instructions in [5.4.3.2 Configuring WiFi](#) on page 197, and then click **Connect** to connect to the network.

When the Vital Sync™ system network is successfully added, data will be transferred automatically from the Capnostream™35 monitor to the Vital Sync™ system.

The following measurement data is transferred:

- EtCO₂
- SpO₂
- Resp rate
- Pulse rate
- CO₂ waveform

For more information regarding the Covidien Vital Sync™ system, consult the user guide supplied with the Vital Sync™ system or your local representative.

Please note that the most recent configuration of the device will supersede previous configurations of the device, whether the configuration is done on the device itself or via Vital Sync™.



Warning: The Capnostream™35 monitor has been approved for use with the Vital Sync™ system. Use with hospital data systems other than the Vital Sync™ system is under the user's responsibility.

5.4.5. Printed Reports

Report printing is done via an external printer after storing a report using a USB device or micro SD card as described in [5.4 Data Transfer](#), above.

6. Preventive Maintenance

6.1. Introduction

Contact your local Service Center or refer to the Service Manual for service instructions and performance tests and checks. Maintenance should be performed by qualified service technicians only. Contact Capnographytechnicalsupport@medtronic.com for additional information.

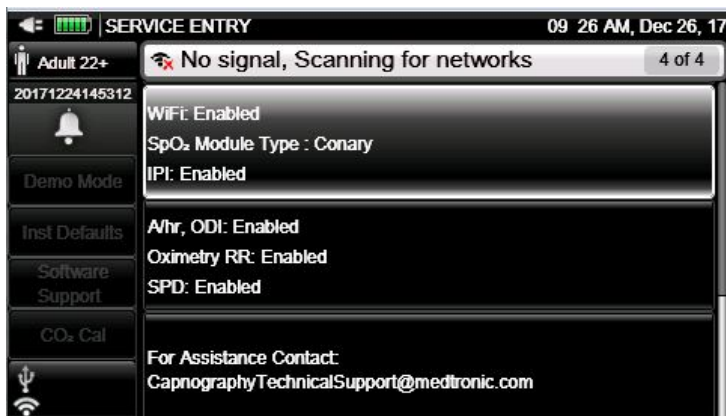
When the monitor is in Service Mode, no alarms, whether patient or equipment, will sound.

6.2. Service Entry and Service Mode Screens

Service-related information, including serial number, hardware and software revisions, and features provided by the device can be viewed on the Service Entry screen by clicking the **Menu** button on the front panel of the monitor and then **Setup>Service**.

To determine if A/hr, ODI, Oximetry RR, or SPD are available on your device, scroll down to the bottom of this screen.

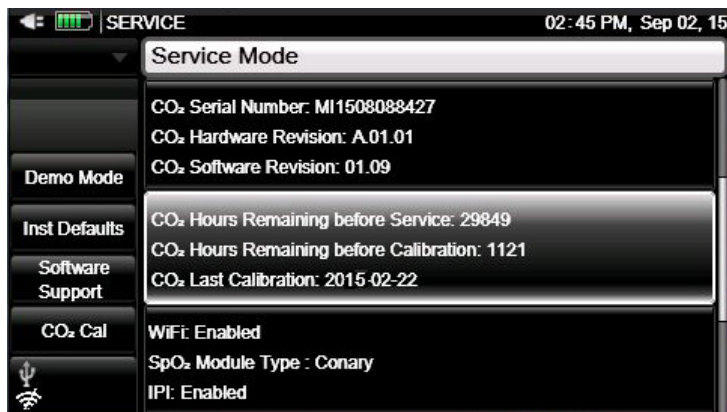
Figure 54. Service Entry Screen, bottom section



To open the Service Mode screen, input the service password (reference [13.1 Monitor Service Password](#) on page 287) on the Service Entry screen and click **Done**. Use the right arrow to move to the center of the Service Mode screen and use the down arrow to scroll down to the relevant section, in which the following information is provided:

- CO₂ Hours Remaining Before Service
- CO₂ Hours Remaining Before Calibration
- CO₂ Last Calibration

Figure 55. Service Mode Screen



6.3. Monitor Service Hours

The number of hours the monitor may continue working before service is required and before calibration is required appears in the main Service Mode screen; an example is seen in [Figure 55. Service Mode Screen](#) on page 209.

In addition to required service as described above, regular maintenance and safety checks should be scheduled with a qualified service technician every 24 months. Contact Capnographytechnicalsupport@medtronic.com for additional information.

Battery back-up time of the Li-ion battery may degrade over a period of time. To avoid degradation of battery capacity, it is recommended that the battery pack be replaced every two years. This should be part of the regular maintenance and safety checks scheduled every 24 months.

6.4. CO₂ Calibration

The unit is calibrated when it leaves the factory.

It is recommended that you calibrate the monitor within two weeks of the **Calibration Required** message appearing on the monitor.



Caution:

The calibration must be performed with a manufacturer authorized Calibration Kit containing a gas mixture of 5% CO₂ 21% O₂ and Bal N₂ and the authorized connecting means ("T" piece).

A manufacturer-approved Calibration Kit can be purchased from Airgas, Scott Medical Product Division (part number Z03NI748BDC002, legacy part number T4653ORF-2BD). Airgas may be contacted at 1-800-233-4334 (option "2" for customer Service). The kit includes:

- Calibration Gas containing 5% CO₂, 21% O₂ Bal N₂
- Tubing Adapter ("T" Piece)
- Calibration Line

If this process is performed while a battery powers the monitor, make sure that the battery is fully charged.

Prior to checking the calibration, verify that the Calibration Line supplied with the Calibration Kit is firmly attached.

Calibration shall be performed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.

4. Input Service password and click **Done**.
5. Select **CO₂ Cal**.
6. Select **Start** and follow the instructions on the screen.
7. The instructions on the screen will guide you regarding the rest of the process, including results, until you reach the screen which will provide you with calibration results.

6.5. CO₂ Calibration Check



Caution:

The calibration check must be performed with a manufacturer authorized Calibration Kit containing a gas mixture of 5% CO₂ 21% O₂ and Bal N₂ and the authorized connecting means ("T" piece).

A manufacturer-approved Calibration Kit can be purchased from Airgas, Scott Medical Product Division (part number Z03NI748BDC002, legacy part number T4653ORF-2BD). They may be contacted at 1-800-233-4334 (option "2" for customer Service). The kit includes:

- Calibration Gas containing 5% CO₂, 21% O₂ Bal N₂
- Tubing Adapter ("T" Piece)
- Calibration Line

If this process is performed while a battery powers the monitor, make sure that the battery is fully charged.

Prior to checking the calibration, verify that the Calibration Line supplied with the Calibration Kit is firmly attached.

6.5.1. Calibration Check Procedure



Note:

At any stage in the Calibration Check procedure, you can go back to the first screen of the calibration check process by clicking the **Back** button on the front panel.

The Calibration Check process includes the following steps:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input Service password and click **Done**.
5. Select **CO₂ Cal**.
6. Select **Cal Check** and follow the instructions on the screen.
7. Connect a FilterLine™ sampling line to the monitor and click **Start**.
8. Follow the instructions on the screen and click **Continue**, or click **Abort** to stop the Calibration Check.

The instructions on the screen will guide you regarding the rest of the process, including results.

6.6. Operations Log

The monitor provides the option of accessing a monitor operations log for troubleshooting, maintenance and other purposes.

The Operations log can be accessed as follows:

1. Click the **Menu** button on the front panel of the monitor.

2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input Service password and click **Done**.
5. Navigate to and select **Software Support** and then **Operations Log**.
6. Navigate to and select drive on which data shall be stored.
7. Select **Start** to begin downloading the data in txt format.

The following information is stored in the Operations Log (with date and time):

- Institutional defaults settings at turn-on
- Any change of settings
- Patient Admit/Discharge
- Patient IDs
- Alarms occurrences
- Times in pump-off mode
- Error code/messages
- User button presses
- Times in audio Alarm Paused/Alarm Off

The operations log includes a maximum of 4000 entries per category or 30 days of information.

6.7. Maintenance

Contact your local representative to order spare parts, calibration kits, or to get answers to any questions regarding service and periodic maintenance.

6.8. Cleaning

Germicidal disposable cloth wipes, or wipes moistened with Isopropyl alcohol or 96% Ethanol may be used to clean the monitor. Cleaning with a wipes moistened with these solutions should be followed by wiping the monitor with distilled water. Caustic or abrasive cleaners should not be used to clean the monitor.

Read the Instructions for Use enclosed with each reusable SpO₂ sensor before attempting to clean the sensor. Follow the sensor cleaning and disinfecting procedures in the particular sensor's Instructions for Use. SpO₂ sensors which are not reusable should not be cleaned. All Microstream™ CO₂ sampling lines are not reusable and should not be cleaned.



WARNING:

The device is not sterile. Do not autoclave or sterilize this device.



Caution:

Do not spray or pour any liquid directly on the monitor, accessories or consumables.



Caution:

Do not use caustic or abrasive cleaners, or harsh solvents, including petroleum-based or acetone solutions, to clean the device.



Caution:

Microstream™ etCO₂ consumables are designed for single patient use and are not to be reprocessed. Do not attempt to clean, disinfect or blow out the FilterLine™ sampling line as the monitor can be damaged.



Caution:

Do not expose SpO₂ sensor connector pins to cleaning solution as this may damage sensor.

7. Troubleshooting

7.1. Electrical Issues

Problem	Possible Cause	Action
Monitor does not turn on.	Both batteries are totally discharged and power cable is improperly attached or disconnected, or cable has faulty electrical connection.	Check power cable connection. Ensure that batteries are charging (the indicator on the front panel will indicate if the batteries are charging). If the power cable connection is working, a technician should be called to replace the battery. If there is a battery problem, the battery/power indicator on the front panel of the monitor will be red. Reference 3.2.5 Battery and AC Connections Indicators on page 55.
	AC wall outlet has no power and both batteries are not charged.	Check connections and correct problem.
AC mains power and monitor on indicator lights are on, but unit	Battery is not correctly placed into monitor.	Open the battery housing and check that the battery is firmly placed in the battery compartment.

Problem	Possible Cause	Action
will not operate on battery power when the AC mains power cable is disconnected.		Reference 3.2.2 Removable Battery Pack Installation on page 53.
Battery is not fully charged after being plugged in for 24 hours.	Battery has not recharged completely.	Unplug the monitor from AC power for 3-4 hours, and then plug in again. The battery will now recharge completely when connected to AC power. If the battery still does not recharge completely, replace the battery.

7.2. CO₂ Problems

Problem	Possible Cause	Action
No Breath message appears constantly and red alarm indicator flashes.	Physiological cause.	Check patient.
	Clogged or blocked FilterLine™ sampling line.	Check sampling line and replace if blocked.
	FilterLine™ sampling line caught in something or tube is kinked.	Check the sampling line from the monitor all the way to the patient to see if line is kinked, twisted closed or caught in bed or

Problem	Possible Cause	Action
		equipment.
FilterLine connected but pump is not working and no CO ₂ , etCO ₂ or RR readings are shown.	FilterLine™ sampling line not plugged in properly.	Check that the sampling line plug is fully screwed into the monitor.
	Gold ring worn or dirty.	Check that the gold ring on the end of the sampling line connector is present and not damaged or covered with dirt. Wipe off any dirt or replace sampling line as necessary.
EtCO ₂ values read erratically.	Mechanically ventilated patient who breathes spontaneously.	No action needed.
	A leak in the airway.	Check for connection and line leaks to patient and correct if necessary.
EtCO ₂ values are consistently higher or lower than expected.	Improper calibration.	Check calibration. Reference CO ₂ Calibration Check on page 211.
	BTPS setting turned off.	Check BTPS setting in the institutional settings. Reference 11.1.8 CO₂ Parameters on page 270 for details.

7.3. SpO₂ Sensor

Problem	Possible Cause	Action
No SpO ₂ signal: Zero display appears for oxygen saturation and pulse rate.	Sensor not properly connected to monitor or extension cable.	Check that the sensor and extension cable (if used) are properly connected to the monitor.
Replace SpO ₂ sensor message appears on screen	SpO ₂ board is not receiving information from the cable.	Detach and reattach the SpO ₂ sensor and try again. If the message still appears, replace the cable or sensor.
Loss of pulse or SpO ₂ signal: Zero display appears for oxygen saturation and pulse rate.	Sensor is improperly applied to patient.	Check sensor application.
	Patient's perfusion may be too poor.	Check the condition of the patient.
	Sensor or sensor extension cable may be damaged.	Replace sensor or sensor extension cable
	Sensor or sensor extension cable may be a non-Covidien-approved part.	Replace sensor or sensor extension cable with a part approved by Covidien

	Excessive patient motion or electro-surgical interference.	If possible, keep patient still. Check whether the sensor is secure and properly placed. Replace if necessary, move the sensor to a new site, or use a sensor that tolerates more motion.
Inaccurate SpO ₂ measurements appear.	Excessive illumination.	Check sensor placement or cover sensor with a dark or opaque material.
	Sensor placement on an extremity that has a blood pressure cuff, arterial catheter or intravascular line, or nail polish.	Check sensor placement.
	Patient's condition.	Check patient.
	Excessive patient movement.	If possible, keep patient still and use a sensor that tolerates more motion.

7.4. CO₂ Calibration

Problem	Possible Cause	Action
CALIBRATION REQUIRED message appears on the monitor, but service screen shows there are still hours remaining before next calibration.	It has been more than one year since the last CO ₂ calibration.	Perform a CO ₂ calibration.

7.5. Returning the Monitor

If it is necessary to return the monitor for repairs, contact your local representative for shipping instructions.

To repack the monitor, disconnect the accessories from the monitor. Pack the monitor in the original shipping carton. If the original carton is unavailable, use a suitable box filled with the appropriate amount of packing material. It is not necessary to return the sensors, Microstream™ etCO₂ consumables, or power cords.

If the monitor malfunctions, carefully package the monitor with an unused consumable from the same box or lot as the consumable used at the time of malfunction and return it with the monitor for inspection.

7.6. Technical Assistance

For technical information, contact your local Service Center or write to Capnographytechnicalsupport@medtronic.com.

The Service Manual includes information that is required by qualified personnel to service the monitor.

See also the Sales and Support Center>Technical Support area at <http://www.covidien.com/rms/brands/microstream>.

If it is necessary to return the monitor for repairs, contact your local representative for shipping instructions.

8. Accessories

8.1. Microstream™ EtCO₂ Consumables

H Products (for use in humidified environments) are marked with an asterisk (*) in the table below.

Table 25. Microstream™ Consumables

Microstream™ Consumables	
Intubated Consumables	
FilterLine™ Set Adult/Pediatric	XS04620
FilterLine™ Set Adult/Pediatric 100 unit	010579
FilterLine™ H Set Adult/Pediatric*	XS04624
FilterLine™ H Set Adult/Pediatric 100 unit*	010580
FilterLine™ H Set Infant/Neonatal*	006324
FilterLine™ Set Adult/Pediatric Long	007768
FilterLine™ H Set Adult/Pediatric Long*	007737
FilterLine™ H Set Infant/Neonatal Long*	007738
VitaLine™ H Set Adult/Pediatric*	010787
VitaLine™ H Set Infant/Neonatal*	010807

Non-intubated Consumables	
Smart CapnoLine™ Plus (O ₂ connector)	009818
Smart CapnoLine™ Plus (O ₂ connector) 100 unit	010209
Smart CapnoLine™ Plus Long (O ₂ connector)	010340
Smart CapnoLine™ Plus Long (O ₂ conn) 100 unit	010339
Smart CapnoLine™ Plus O ₂ (O ₂ tubing)	009822
Smart CapnoLine™ Plus O ₂ (O ₂ tubing) 100 unit	010210
Smart CapnoLine™ Plus O ₂ Long (O ₂ tubing)	009826
Smart CapnoLine™ Plus O ₂ Long (O ₂ tubing) 100 unit	010341
Smart CapnoLine™ Pediatric	007266
Smart CapnoLine™ O ₂ Pediatric (O ₂ tubing)	007269
Smart CapnoLine™ O ₂ Ped Long (O ₂ tubing)	007743
Smart CapnoLine™ H Plus O ₂ (O ₂ tubing)*	010433
Smart CapnoLine™ H Plus O ₂ (O ₂ tubing) 100 unit*	010625
Smart CapnoLine™ H Plus O ₂ Long (O ₂ tubing)*	012463
Smart CapnoLine™ H O ₂ Pediatric (O ₂ tubing)*	010582
Smart CapnoLine™ H O ₂ Pediatric Long (O ₂ tubing)*	012464
Smart CapnoLine™ Guardian (O ₂ connector)	012528
Smart CapnoLine™ Guardian (O ₂ connector) 100 unit	012537
Smart CapnoLine™ Guardian O ₂ (O ₂ tubing)	012529
Smart CapnoLine™ Guardian O ₂ (O ₂ tubing) 100 unit	012538

Smart CapnoLine™ Guardian O ₂ long (O ₂ tubing)	012530
Smart CapnoLine™ Guardian O ₂ long (O ₂ tubing) 100 unit	012539
Hook and Loop Strap	012542
O ₂ /CO ₂ Nasal FilterLine™ Adult (O ₂ tubing)	006912
O ₂ /CO ₂ Nasal FilterLine™ Adult (O ₂ tubing) 100 unit	010304
O ₂ /CO ₂ Nasal FilterLine™ Adult Long (O ₂ tubing)	007739
O ₂ /CO ₂ Nasal FilterLine™ Adult Long (O ₂ tubing) 100 unit	010344

8.2. Available Accessories and Spare Parts

See the list of available accessories and spare parts for the monitor below.

Table 26. Monitor Accessories and Spare Parts

Item	Medtronic Part Number	Use	Type
Extra battery	PM35BTY	Battery for monitor available for purchase separately	Accessory
External battery charger	PM35CHG	External charger for monitor battery	Accessory

AC adapter	PM35PSP1	AC adapter for monitor with power cord for use in the EU available for purchase separately	Spare part
AC adapter	PM35PSP2	AC adapter for monitor with power cord for use in the US available for purchase separately	Spare Part
Pole Clamp	PM35MNF	Connecting monitor with the Pole Clamp (VESA™* adapter assembly kit is also required for connection)	Accessory
VESA™* adapter assembly kit	PM35VAA	Connecting monitor with a VESA™* mount	Accessory
Capnostream35 UK Singapore Power Cord	PM35PCD3	Power cord for use in the UK and Singapore, available for purchase separately	Spare part
Capnostream35 Switzerland	PM35PCD4	Power cord for use in	Spare

Power Cord		Switzerland, available for purchase separately	part
Capnostream35 Australia Power Cord	PM35PCD5	Power cord for use in Australia, available for purchase separately	Spare part
Capnostream35 China Power Cord	PM35PCD6	Power cord for use in China, available for purchase separately	Spare part
Capnostream35 South Africa Power Cord	PM35PCD7	Power cord for use in South Africa. available for purchase separately	Spare part
Capnostream35 Argentina Power Cord	PM35PCD8	Power cord for use in Argentina, available for purchase separately	Spare part
Capnostream35 Brazil Power Cord	PM35PCD9	Power cord for use Brazil, available for purchase separately	Spare part
Capnostream35 Korea Power Cord	PM35PCD10	Power cord for use in South Korea,	Spare part

Available Accessories and Spare Parts

		available for purchase separately	
Capnostream35 Chile Power Cord	PM35PCD11	Power cord for use in Chile, available for purchase separately	Spare part

9. Theory of Operations

9.1. Introduction

The Capnostream™35 portable respiratory monitor provides accurate, continuous capnography and pulse oximetry monitoring for intubated and non-intubated patients from neonate to adult. Using Microstream™ capnography technology, patented FilterLine™ etCO₂ consumables, and Covidien Nellcor™ pulse oximetry technology, the monitor allows for simultaneous "hassle free" etCO₂ and SpO₂ monitoring.

9.2. Features

The monitor's features include:

- Dual parameter monitor that supports the current standard of care providing CO₂ and SpO₂ measurements
- Integrated Pulmonary Index™ (IPI) algorithm, which provides a simple, clear, and comprehensive indication of a patient's ventilatory status and trends
- Apneas per Hour and Oxygen Desaturation Index, indices used to help in the identification and quantification of apnea and oxygen desaturation events (if available)
- Simple user interface with color screen
- Routine functions are accessed with 2 clicks
- 48 hour trends to review patient history

- Alarm review
- SARA™ (Smart Alarm for Respiratory Analysis) algorithm, an embedded Smart Capnography alarm management technology, which reduces clinically insignificant alarms
- SatSeconds™ alarm management parameter, which monitors both degree and duration of desaturation as an index of desaturation severity, in order to help distinguish clinically significant events from minor and brief desaturations that may result in nuisance alarms.
- Event marking to compare events and medication administration to changes in patient status
- Case recording to help organize patient files
- Output to transfer patient data to USB flash memory devices or micro SD cards
- Wireless transfer of data via a WiFi network

9.3. Technology Overview

This section provides a basic overview of Capnography and Pulse Oximetry.

9.3.1. What is Capnography?

Capnography is a non-invasive method for monitoring the level of carbon dioxide in exhaled breath (etCO₂) to assess a patient's ventilatory status.

The monitor uses Microstream™ non-dispersive infrared (NDIR) spectroscopy to continuously measure the amount of CO₂ during every breath, the amount of CO₂ present at the end of exhalation (etCO₂), and the Respiratory Rate.

Infrared spectroscopy is used to measure the concentration of molecules that absorb infrared light. Because the absorption is proportional to the concentration of the absorbing molecule, the concentration can be determined by comparing its absorption to that of a known standard.

The Microstream™ etCO₂ consumables deliver a sample of the inhaled and exhaled gases from the ventilator consumable or directly from the patient (via an oral/nasal cannula) into the monitor for CO₂ measurement. Moisture and patient secretions are extracted from the sample, while maintaining the shape of the CO₂ waveform.

The 50 ml/min. sampling flow rate reduces liquid and secretion accumulation, decreasing the risk of obstruction in the sample pathway in humid ICU environments.

Once inside the Microstream™ CO₂ sensor, the gas sample goes through a micro-sample cell (15 microliters). This extremely small volume is quickly flushed, allowing for fast rise time and accurate CO₂ readings, even at high respiration rates.

The Micro Beam IR source illuminates the micro-sample cell and the reference cell. This proprietary IR light source generates only the specific wavelengths characteristic of the CO₂ absorption spectrum. Therefore, no compensations are required when different concentrations of N₂O, O₂, anesthetic agents and water vapor are present in the inhaled and exhaled breath. The IR that passes through the micro-sample cell and the IR that passes through the reference cell are measured by the IR detectors.

The microprocessor in the monitor calculates the CO₂ concentration by comparing the signals from both detectors.

9.3.2. What is Pulse Oximetry?

Pulse oximetry is based on: the difference in the absorption of red and infrared light (spectrophotometry) by oxyhemoglobin and deoxyhemoglobin; and changes in the volume of arterial blood in

tissue during the pulse cycle (plethysmography), and hence, light absorption by that blood.

A pulse oximeter determines Spot Oxygen Saturation (SpO₂) by passing red and infrared light into an arteriolar bed and measures changes in light absorption during the pulsatile cycle. Red and infrared low power light emitting diodes (LEDs) in the oximetry sensor serve as light sources; a photodiode serves as the photo detector.

Because oxyhemoglobin and deoxyhemoglobin differ in light absorption, the amount of red and infrared light absorbed by blood is related to hemoglobin oxygen saturation. To identify the oxygen saturation of arterial hemoglobin, the monitor uses the pulsatile nature of arterial flow. During systole, a new pulse of arterial blood enters the vascular bed and blood volume and light absorption increase. During diastole, blood volume and light absorption reach their lowest point. The monitor bases its SpO₂ measurements on the difference between maximum and minimum absorption (measurements at systole and diastole). The focus of light absorption by pulsatile arterial blood eliminates the effects of nonpulsatile absorbers such as tissue, bone, and venous blood.

Pulse oximetry provides arterial oxygen saturation (SpO₂) and pulse rate of adult, pediatric, and neonatal patients, at all perfusion levels. The pulse oximeter is intended for use during both no motion and motion conditions and for patients who are well or poorly perfused.

10. Product Specifications

10.1. Power Supply

Item	Value
Input	100-240 VAC, $\pm 10\%$ 50-60 Hz, 1.5A
Output Voltage and Power	12VDC, 2.5A

10.2. Battery

Item	Value
Battery Type	Lithium ion
Battery pack nominal voltage and energy	7.2V; 18.72Wh
Battery Operation	3 hours for removable battery pack; 20 minutes for internal battery pack
Battery Charging Time	Up to 5 hours when the monitor is off and up to 8 hours when the monitor is on, for both batteries
Battery Storage	There are differing requirements depending on how long you store the battery without use: Short-term storage (one month or less): The battery has an automatic discharge feature. You must periodically check the

	charge level of the battery. If the battery is not fully charged, charge before use. Long-term storage (3-12 months): The battery must be stored in a cold, dry area, not inside the monitor. Its charge decreases over time. To restore the battery to full power, recharge the battery before use. Reference 3.2.7 Storing the Battery on page 57 for more details.
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10.3. Controls

Item	Value
Front Panel	1 Switch for monitor On/Off control 4 specific function keys 1 Enter key with directional arrows

10.4. Display

Item	Value
Screen	109mm (4.3inch) Color TFT Display Pixel Pitch: 0.198 (horizontal) x 0.198(vertical) mm Active Display Area: 95.04 (horizontal) x 53.856 (vertical) mm Resolution 480 x 272 pixels Viewing angle (vertical) 125° Viewing angle (horizontal) 140°

Trace Speed	3.0, 6.3, 12.5 and 25 mm/sec
Waveform sampling rate	77.82 samples/sec for SpO ₂ (fixed) 20 samples/sec for Capnography (fixed)
Trend Storage	48 hours with 1 second resolution
Trend Display	Graphical Display: 1h, 2h, 4h, 8h, 12h views Tabular Display intervals: 1 sec, 5 sec, 15 sec, 30 sec, 1 min, 5 min, 15 min, 30 min, 1 hour

10.5. Sound Pressure Data

When measured at a distance of 1 m alarm volume is 45 dB(A) at minimum audio level and 65 dB(A) at maximum audio level. To adjust the audio volume level of device, reference [4.5.1 Alarm Volume](#) on page 69. To make a permanent change to alarm volume, reference [11.1.6.1 Main Monitor settings](#) on page 266.

10.6. Microstream™ Capnography Technology

Item	Value
CO ₂ Units	mmHg or kPa or Vol%
CO ₂ , etCO ₂ Range	0-150 mmHg
CO ₂ Waveform Resolution	0.1 mmHg
EtCO ₂ Resolution	1 mmHg

CO ₂ Accuracy	0-38 mmHg: ± 2 mmHg 39-150 mmHg: $\pm$ (5% of reading + 8% for every 1 mmHg above 38 mmHg) Accuracy applies for breath rates of up to 80 bpm. For breath rates above 80 bpm, accuracy is 4 mmHg or ± 12 % of reading whichever is greater, for EtCO₂ values exceeding 18 mmHg.
Respiration Rate Range	0-150 bpm
Respiration Rate Accuracy	0-70 bpm: ± 1 bpm 71-120 bpm: ± 2 bpm 121-150 bpm: ± 3 bpm
CO ₂ Alarms	No breath, EtCO ₂ high, EtCO ₂ low, RR high, RR low, IPI low (IPI also requires pulse oximetry information)
Flow Rate	50 ($42.5 \leq \text{flow} \leq 65$) ml/min, flow measured by volume
Waveform Sampling	20 samples/s
Rise Time	<190ms when used with sampling lines up to 4m length
Response Time	3.4s (typical); with use with 4m sampling lines, ~5.0 s
Initialization Time	40 s (typical)
Calibration Interval	Initially calibrate after 1,200 operating hours (or 12 months, whichever comes first), then once a year or after 4,000 operating hours, whichever comes first

10.7. Nellcor™ Pulse Oximetry

Item	Value
SpO ₂ Measurement Range	1-100%
SpO ₂ Accuracy ¹	
Adult and Pediatric Modes	
SpO ₂ range 70% - 100% ^{2,3,6,7}	±2 digits over the range of 70 to 100% (when using the accessories defined in this document), including under low perfusion; with motion, ±3 digits; with low saturation (60-80%) ±3 digits
Infant/Neonatal Mode	
SpO ₂ range 70% - 100% ^{4,5}	±2 digits over the range of 70 to 100% (when using the accessories defined in this document), with motion ±3; with low saturation (60-80%) ±3
Pulse Rate Range	20 to 250 beats per minute Pulse Rate values of < 20 beats per minute shall be displayed as 0 beats per minute Pulse Rate values of > 250 beats per minute shall be displayed as 250 beats per minute.
Pulse Rate Accuracy ^{2,3,4,6,7}	±3 digits over the range of 20 to 250 beats per minute inclusive, including under low perfusion; with motion, 48 to 127 bpm ±5 digits
Nellcor™ Respiration rate range and accuracy	Range: 4-40 bpm (breaths per minute) Accuracy: ±1 bpm

Alarms	SpO ₂ High, SpO ₂ Low, PR High, PR Low, Nellcor™ RR High, Nellcor™ Low
Sat Sec Range	10-100

1. Saturation accuracy varies by sensor type. Contact Covidien or a local Service Center for sensor accuracy information.

2. Accuracy specifications were validated using measurements of healthy non-smoking adult volunteers during controlled hypoxia studies spanning the specified saturation ranges. Subjects were recruited from the local population and comprised both men and women ranging in age from 18-50 years old, and spanned a range of skin pigmentations. Pulse oximeter SpO₂ readings were compared to SaO₂ values of drawn blood samples measured by hemoximetry. All accuracies are expressed as ± 1 SD. Because pulse oximeter equipment measurements are statistically distributed, about two-thirds of the measurements can be expected to fall in this accuracy (ARMS) range (refer to the Sensor Accuracy Grid for more details).

3. Adult specifications are shown for OXIMAX MAXA and MAXN sensors with the Nellcor™ bedside respiratory patient monitoring system.

4. Neonate specifications are shown for OXIMAX MAXN sensors with the Nellcor™ bedside respiratory patient monitoring system.

5. Clinical functionality of the MAXN sensor has been demonstrated on a population of hospitalized neonate patients. The observed SpO₂ accuracy was 2.5% in a study of 42 patients with ages of 1 to 23 days, weight from 750 to 4,100 grams, and 63 observations made spanning a range of 85% to 99% SaO₂.

6. Specification applies to Nellcor™ bedside respiratory patient monitoring system oximeter performance. Reading accuracy in the presence of low perfusion (detected IR pulse modulation amplitude 0.03% - 1.5%) was validated using signals supplied by a patient simulator. SpO₂ and pulse rate values were varied across the monitoring range over a range of weak signal conditions and compared to the known true saturation and pulse rate of the input signals.

7. Motion performance was validated during a controlled hypoxia blood study. Subjects performed rubbing and tapping movements 1-2 cm in amplitude with aperiodic intervals (randomly changing) with a random variation in frequency between 1-4 Hz. Applicability: OXIMAX MAXA, MAXAL, MAXP, MAXI, and MAXN sensors.

Contact Covidien or a local Service Center for a Nellcor™ oxygen saturation accuracy specification grid listing all of the Nellcor™ sensors used with the monitoring system.

10.8. Alarms

Item	Audible Alarm	Visual Alarm
High Priority Alarm	Beep pattern repeated every 12 seconds.	Flashing Red LED
Medium Priority Alarm	Triple beep repeated every 10 seconds	Flashing Yellow LED
Advisories Alarm	Single beep	No LED
Silent Advisories Alarm	No audible alarm	No LED

10.9. General Characteristics

Item	Value
Unit Dimensions	213 mm (w) x 137 mm (h) x 55 mm (d) [8.4 in (w) x 5.4 in (h) x 2.2 in (d)]
Unit Weight	1.0 kg (2.2 lb.)
Packaged Dimensions	370mm (w) x 200mm (h) x 125mm (d) (14.6in (w) x 7.9in (h) x 4.9in (d))
Packaged Weight	2.10 kg (4.6 lb.)
Operating Temperature	0° C to 40° C (32° F to 104° F) The monitor shall function for at least 20 minutes when placed in an environment with an ambient temperature of -20° C (-4° F) to 50° C (122° F).
Operating Pressure and Altitude	1250 feet (381m) below sea level to 15000 feet (4572m) above sea level (430 mmHg to 795 mmHg) without power

Item	Value
	supply 1250 feet (381m) below sea level to 9843 feet (3000m) above sea level (430 mmHg to 795 mmHg) with power supply
Operating Humidity	Ambient, non-condensing, relative humidity in the range of 10% to 95%
Storage and Transport Temperature	-20° C to +70° C (-4° to 158°F)
Storage and Transport Pressure and Altitude	Stored or transported at altitudes in the range of 1250 feet (381m) below sea level to 50000 feet (15,240m) above sea level (430 mmHg to 795 mmHg).
Storage and Transport Humidity	Ambient, non-condensing, relative humidity in the range of 10% to 90%.
Boot-up Time	Up to 60 seconds
Stabilization Time (from storage to operational conditions)	Up to 2 hours

10.10. Equipment Classification

Item	Value
Types of Protection against Electric Shock	Class 2
Degree of Protection against Electric Shock	Defibrillator-Protected Type BF
Mode of Operation	Continuous

Degree of protection against matter and water ingress	The monitor enclosure is IP54 protected (protected against dust and splashing water) when all port doors are closed. The level of protection against dust is reduced from level 5 to level 3 (protected against foreign solid objects above 2.5mm in diameter) when used with the SpO₂ sensors noted in Table 7. Nellcor™ SpO₂ Sensor Models and Patient Sizes on page 108 for EMS environment.
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10.11. Compliance

This product is designed to conform to the following standards:

- IEC 60601-1
- IEC 60601-1-2
- IEC 60601-2-49
- IEC 60601-1-12
- IEC 60601-1-6
- IEC 60601-1-8
- ISO 80601-2-61
- ISO 80601-2-55
- ISO 15223-1
- Waste Electrical and Electronic Equipment (WEEE) Directive 2002/96/EC
- Directive on the restriction of the use of certain hazardous substances (ROHS) in electrical and electronic equipment - 2011/65/EU

- IEC 62133
- Radio Equipment Directive: Directive 2014/53/EU

10.12. Electromagnetic Immunity

The monitor complies with IEC 60601-1-2.

The monitor is suitable for use in the specified electromagnetic environment. The user of the monitor should assure that it is used in an electromagnetic environment as described below.

Table 27. Guidance and Manufacturer's Declaration - Electromagnetic Emissions

Emissions Test	Compliance	Electromagnetic Environment – Guidance
RF emissions CISPR 11	Group 1	The monitor uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class B	
Harmonic emissions IEC 61000-3-2	Complies	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Complies	

Table 28. Guidance and Manufacturer's Declaration – Electromagnetic Immunity

Immunity Test	IEC 60601 test level	Compliance level	Electromagnetic Environment – Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	± 6 kV contact ± 8 kV air	± 6 kV contact ± 8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30 %.
Electrical fast transient/burst IEC 61000-4-4	± 2 kV for power supply lines ± 1 kV for input/output lines	± 2 kV for power supply lines ± 1 kV for input/output lines	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	± 1 kV line(s) to line(s) ± 2 kV line(s) to earth	± 1 kV line(s) to line(s) ± 2 kV line(s) to earth	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5 % UT ¹ (>95 % dip in UT) for 0,5 cycle 40 % UT (60 % dip in UT) for 5 cycles 70 % UT (30 % dip in UT)	<5 % UT (>95 % dip in UT) for 0,5 cycle 40 % UT (60 % dip in UT) for 5 cycles 70 % UT (30 % dip in UT)	Mains power quality should be that of a typical commercial or hospital environment. If the user of the monitor requires continued operation during power mains

	for 25 cycles <5 % UT (>95 % dip in UT) for 5 s	for 25 cycles <5 % UT (>95 % dip in UT) for 5 s	interruptions, it is recommended that the monitor be powered from an uninterruptible power supply or a battery.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 V	Portable and mobile RF communications equipment should be used no closer to any part of the monitor, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distances: 150 kHz to 80 MHz: $d = 1.2\sqrt{P}$ 80 MHz to 800 MHz: $d = 1.2\sqrt{P}$ 800 MHz to 2.5 GHz: $d = 2.3\sqrt{P}$
61000-4-3	20 V/m, 1 kHz modulation 80 MHz to 2.5 GHz (per ISO 80601-2- 55, 1st ed.)	20 V/m	

			where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m).² Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey, should be less than the compliance level in each frequency range.^{a,b} Interference may occur in the vicinity of equipment marked with the following symbol: 
¹ UT is the a.c. mains voltage prior to application of the test level.			
² At 80 MHz and 800 MHz, the higher frequency range applies. NOTE: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people			
^a Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the monitor is used exceeds the applicable RF			

compliance level above, the monitor should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the monitor.

^bOver the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

The monitor is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the monitor can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the monitor as recommended below, according to the maximum output power of the communications equipment.

Table 29. Recommended Separation Distances between Portable and Mobile RF Communications Equipment and the Monitor

Rated maximum output power of transmitter W	Separation distance according to frequency of transmitter in m1		
	150 kHz to 80 MHz $d = 1.2\sqrt{P}$	80 MHz to 800 MHz $d = 1.2\sqrt{P}$	800 MHz to 2,5 GHz $d = 2.3\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.27
100	12	12	23

For transmitters rated at a maximum output power not listed above, the recommended separation distance d in metres (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

¹ At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.

NOTE: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

11. Institutional Settings

11.1. Institutional Defaults

Institutional Defaults provide a way for users at a particular institution to maintain settings as determined by their own institution even when the monitor is turned off. For example, if a monitor is in use in a department that requires specific alarm limits, specific event names, or any other setting, on a regular basis, this can be set up in institutional settings. This avoids the need to constantly change settings.

11.1.1. Changing Institutional Defaults



WARNING:

Changing the settings might adversely affect the monitoring of patients. Changes to the Institutional settings must only be made by authorized personnel.

Patients will not be monitored while in service mode, even if the monitor is connected to a patient. (The monitor is in service mode once the service password is entered.) Therefore, you may want to remove the sampling line from the patient or disconnect the sampling line from the monitor while putting the monitor into service mode. No data will be recorded while the monitor is in service mode, and thus trying to monitor while in service mode will lead to missing data issues.

Institutional Defaults may be changed using the following steps:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Setup Menu screen.

3. Navigate to and select **Service** on the Setup Menu screen.
4. Enter the Service password and select **Done**.
5. Navigate to and select **Institutional Defaults**.
6. Navigate to and select the desired section to update defaults in that section. The sections are: Alarms, Trend, Monitor, Parameters.
7. After any change to the device settings done in service mode, the device will request that you shut down. Click **Enter** to shut down and the restart the device.

For specific data regarding these types of Institutional Defaults, reference below.

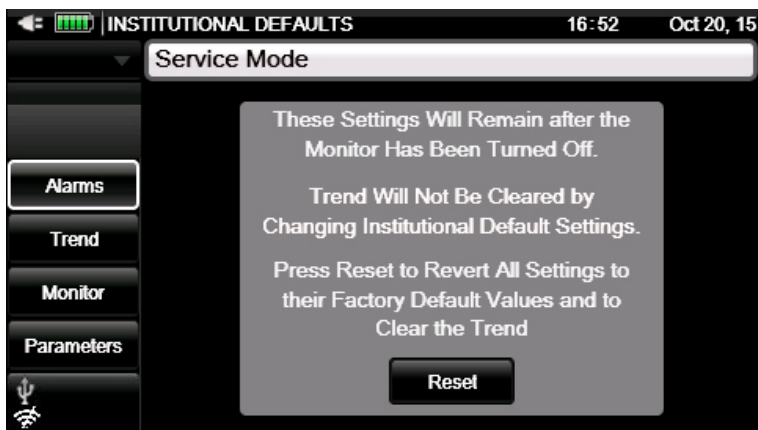
11.1.2. Reset to Factory Defaults

Reset the monitor to use factory defaults as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup will already be selected** on the Setup Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Enter the Service password and select **Done**.
5. Navigate to and select the desired section to update defaults in that section. The sections are: Alarms, Trend, Monitor, Parameters.

The following screen will appear:

Figure 56. Revert To Factory Defaults Screen



6. Select **Reset** to reset all institutional settings to factory defaults.

If you want to reset only one or more sections of the defaults to factory defaults, Navigate to and select the desired section and reset that section only to factory defaults by clicking the **Factory Defaults** button in the relevant section.

In all Institutional Default screens, clicking **Reset** will also clear the trend memory, including the entire patient information database.

11.1.3. Institutional Defaults Export/Import

Institutional Defaults may be exported or imported as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Setup Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Enter the Service password and select **Done**.
5. Navigate to and select **Software Support**.

6. Navigate to and select **Software Download>Default Transfer**.
7. Select **Export Defaults** to export current defaults to a USB flash memory drive or SD card, or **Import Defaults** to import defaults from a USB flash memory drive or micro SD card.
8. Select a drive and click **Start**.
9. When the export/import is complete, the monitor will inform you, with the following message: Export Completed, or Import Completed, Shutdown and Restart the Monitor to Resume Normal Operation.

11.1.4. Monitor Alarms Settings

Select **Factory Defaults** in the Inst. Def.: Alarms screen to restore all Alarms Factory Defaults.

11.1.4.1. Alarm Limits

The factory default settings for Adult/Pediatric and Infant/Neonatal alarm limits are given below. Units appear in range column.

Table 30. Factory Default Alarm Limits

Parameter	Adult 22+	Adult 12-22	Pediatric 6-12	Pediatric 3-6	Pediatric 1-3	Infant /Neonatal	Alarm Range
EtCO ₂ High	60	60	55	55	55	60	5-150 mmHg
EtCO ₂ Low	15	15	15	15	15	20	0-145 mmHg
RR High	35	35	40	40	50	70	5-150 bpm

Parameter	Adult 22+	Adult 12-22	Pediatric 6-12	Pediatric 3-6	Pediatric 1-3	Infant /Neonatal	Alarm Range
RR Low	6	6	10	10	10	12	0-145 bpm
Oximetry RR High	35	NA	NA	NA	NA	NA	9-40bpm
Oximetry RR Low	6	NA	NA	NA	NA	NA	4-35 bpm
No Breathing	30	30	20	20	20	15	10-60 sec
SpO ₂ High	100	100	100	100	100	100	25-100% saturation
SpO ₂ Low	85	85	85	85	85	90	20-95% saturation
Pulse Rate High	140	140	140	140	150	180	25-250 bpm
Pulse Rate Low	50	50	50	60	60	100	20-245 bpm
IPI Low	3	3	3	3	3	N/A	1-9 or OFF
SatSe conds	100	100	100	100	100	OFF	10, 25,

Parameter	Adult 22+	Adult 12-22	Pediatric 6-12	Pediatric 3-6	Pediatric 1-3	Infant /Neonatal	Alarm Range
TM level*							50, 100 or Off
SPD sensitivity**	1	NA	NA	NA	NA	NA	1-3

*SatSecondsTM is not an alarm, but can trigger an SpO2 alarm

** For patient type Adult 22+ only

Under Alarm Limits, any alarms can be enabled or disabled as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**.
5. Navigate to and select **Institutional Defaults>Alarms**.
6. Navigate to and select an Alarm Limits set which includes the relevant alarm.
7. Navigate to and select the relevant alarm.
8. Navigate to and select the **Enable** button to enable the alarm if it is currently disabled. If the relevant alarm is currently enabled, the button will show **Disable**. Select the Disable button to disable the alarm. If the relevant alarm is currently disabled, the button will show **Enable**. Select the **Enable** button to enable the alarm.

This change will be implemented only for the particular set in which it was made, but not in other sets. For example, if you disable the RR High

alarm for pediatric patients, Age 1-3 years, it will still be available for other pediatric patients and for adult patients.

If the alarm limit to be changed is IPI, note that only the following Alarm Limits sets include IPI: Pediatric 1-3 Years Alarm Limits, Pediatric 3-6 Years Alarm Limits, Pediatric 6-12 Years Alarm Limits, and Adult Alarm Limits. IPI is not available for Infant/Neonatal Patients.

11.1.4.2. Alarm Priority settings

Parameter	Choices	Factory Default
EtCO ₂ High	Medium, High	High
EtCO ₂ Low	Medium, High	High
RR High	Medium, High	High
RR Low	Medium, High	High
SpO ₂ High	Medium, High	High
SpO ₂ Low	Medium, High	High
Pulse Rate High	Medium, High	High
Pulse Rate Low	Medium, High	High
IPI Low	Medium, High	High
CO ₂ Error	Medium, High	Medium
Pulse Not Found	Medium, High	Medium
FilterLine Blockage	Medium, High	Medium
Battery Low	Medium, High	Medium
Communications Stopped	Medium, High	Medium

System Reset – Check Settings	Medium, High	Medium
SpO ₂ Error	Medium, High	Medium
SpO ₂ Sensor Not On Patient	Medium, High	Medium
Replace SpO ₂ Cable	Medium, High	Medium
Check SpO ₂ Sensor Connection	Medium, High	Medium

The No Breath alarm is always a High Priority alarm.

CO₂ FilterLine Disconnected, SpO₂ Sensor Disconnected, and Temperature High are always Medium Priority alarms.

The behavior of the WiFi Communication Stopped alarm differs from other alarms in this category. When set at Medium Priority, the Communications Stopped alarm will provide a visual alert only, with no audio alarm. When that alarm is set at High Priority, it will provide both a visual and an audio alert as usual for High Priority alarms.

Alarm Priority settings may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**.
5. Navigate to and select **Institutional Defaults>Alarms>Alarm Priority**.
6. Navigate to the alarm for which you want to change Alarm Priority and select the desired Alarm Priority level.

7. Repeat for other alarms as required.
8. Return to the Home Screen by clicking the **Home** button.
9. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

11.1.4.3. Display Config settings

The order of the parameters below as they appear on the Alarm Review screen may be changed if desired in Institutional Defaults, to any desired order. The Alarm Review screen displays nineteen parameters.

Table 31. Parameters Available on the Alarm Review Screen

Parameter
No Breath
EtCO ₂
RR
SpO ₂
PR
IPI
SatSec

Display Config settings (the order of the parameters as they appear on the Alarm Review screen) may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.

3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**.
5. Navigate to and select **Institutional Defaults>Alarms>Display Config**.
6. The list of parameters that appear on the Alarm Review screen with the factory default order is given in the center of the screen. To change the order, select the parameter you want to show in the first place on the screen for the first line of the list, the parameter you want to show in the second place on the screen for the second line of the list, and so on.
7. This is done by navigating to and clicking on the first line, and then navigating to the desired parameter for the first line, and clicking **Enter**. This will replace only the selected parameter with the parameter you have chosen as your first line; it will not move any parameters down the list.
8. To change the order of additional parameters, repeat as many times as required.

Blank (a dash) may be chosen for any line to view less than nine parameters.

9. Return to the Home Screen by selecting the Home button.
10. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

11.1.4.4. Alarm Delay settings

Alarm delay options for the following alarms are listed below. For all alarms, the alarm delay default is 0 sec; that is, no alarm delay.

Table 32. Alarm Delay Settings

Parameter	Choices	Factory Default
EtCO ₂ High	0, 10, 15, 20, 30	0 sec
EtCO ₂ Low	0, 10, 15, 20, 30	0 sec
RR High	0, 10, 15, 20, 30	0 sec
RR Low	0, 10, 15, 20, 30	0 sec
SpO ₂ High	0, 10, 15, 20, 30	0 sec
SpO ₂ Low	0, 10, 15, 20, 30	0 sec
Pulse Rate High	0, 10, 15, 20, 30	0 sec
Pulse Rate Low	0, 10, 15, 20, 30	0 sec
IPI Low	0, 10, 15, 20, 30	0 sec

Alarm Delay settings may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**.
5. Navigate to and select Institutional **Defaults>Alarms>Alarm Delay**.
6. Navigate to the alarm for which you want to change Alarm Delay and select the desired Alarm Delay level.
7. Repeat for other alarms as required.
8. Return to the Home Screen by clicking the **Home** button.

9. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

11.1.5. Monitor Trend Settings

Selecting Factory Defaults in the Inst. Def.: Trend screen will restore all Trends Factory Defaults.

Trend Defaults settings may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**.
5. Navigate to and select **Institutional Defaults>Trend**.
6. Navigate to the trend issue which you want to change and select the desired level.
7. Repeat for other issues as required.
8. Return to the Home Screen by selecting the **Home** button.

Changing the Trend resolution will not clear the Trend memory, thus maintaining any patient data that was in the monitor.

11.1.5.1. Main Trend settings

Parameter	Choices	Factory Default
Default Tabular Increment Trend View	1 Sec, 5 Sec, 15 Sec, 30 Sec, 1 Min, 5 Min, 15 Min, 30 Min, 1 Hour	30 Sec

Default Graphical Trend View	1, 2, 4, 8, 12 Hours	4 Hours
Home Screen Trend Display View	1, 2, 4 Hours	1 Hour
Default Event Marking Mode	Quick, Detailed	Detailed
Calendar/Recorded Display Mode	Calendar, Recorded	Calendar

11.1.5.2. Events

11.1.5.2.1. Medication events

The following medication events are currently available in the monitor: Fentanyl, Versed, Midazolam, Morphine, Demerol, Propofol, Surfactant, Medication-Other.

If your institution requires different or additional events, events may be added or changed, as seen below.

11.1.5.2.2. Patient events

The following patient events are currently available in the monitor: Eating, Drinking, Coughing, Ambulating, Chest Pt, Turned, Snoring, Patient-Other.

If your institution requires different or additional events, events may be added or changed, as seen below.

11.1.5.2.3. Intervention events

The following intervention events are currently available in the monitor: Oxygen, Suction, Adj Airway, Narcan, Romazicon, Neb tx, Stimulated, CO₂ Insufflation), ABG, Intervention-Other.

If your institution requires different or additional events, events may be added or changed, as seen below.

Event defaults may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**.
5. Navigate to and select **Institutional Defaults>Trend>Edit Event**.
6. Select the event type in which you want to make changes.
7. A list of current event names in this event type category will appear.
8. Navigate to and select the event name you want to change.
9. Input the new name and select **Done**.
10. Select **Back** to retain the change. Select **Home** to return to the Home screen.
11. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

11.1.5.3. Trend Display Config settings

The parameters shown or the order of the parameters shown may be changed in Institutional Defaults.

In both the Graphical Trend screen and the Tabular Trend screen, six parameters may be chosen to view; only three parameters can be seen at one time. Blank (a dash) may be chosen for any line to view less than five parameters.

In order to view trend parameters on the screen on a larger proportion of the screen, the following can be done:

When choosing parameters to view, if you select the same parameter for multiple lines in a row, that parameter will appear in a larger portion of the Graphical Trend screen. That parameter will simply be repeated in multiple rows in the Tabular Trend screen.

This process may be done for up to five parameters, so that all parameters may be viewed in this larger format.

The Tabular Trend stored report shows all parameters, with dashes indicating unavailable parameters. The Graphical Trend stored reports will show up to three parameters at a time.

Table 33. Parameters Available on the Trend Screens

Parameter
EtCO ₂
RR
SpO ₂
PR
IPI

Display Config settings (the order of the parameters on trend screens) may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**.
5. Navigate to and select **Institutional Defaults>Trend>Display Config.>Standard**.
6. The list of parameters with the factory default order as they appear in trend screens is given in the center of the screen. To

change the order, select the parameter you want to show in the first place on the screen for the first line of the list, the parameter you want to show in the second place on the screen for the second line of the list, and so on. This is done by navigating to and clicking on the first line, and then navigating to the desired parameter for the first line, and clicking **Enter**.

7. To change the order of additional parameters, repeat as many times as required.
8. Return to the Home Screen by selecting the **Home** button.
9. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

11.1.6. Monitor Settings

Selecting **Factory Defaults** in the Inst. Def.: Monitor screen will restore all monitor Factory Defaults.

The following monitor settings may be changed in the Institutional Defaults: Monitor screen.

11.1.6.1. Main Monitor settings

Parameter	Choices	Factory Default
Date Format	Dd mmm yy, mmm dd, yy	Mmm dd, yy
Time Format	12 Hour, 24 Hour	12 Hour
Language	English, French, German, Spanish, Italian, Dutch, Russian, Swedish, Norwegian, Japanese, Chinese, Korean, Finnish, Polish, Portuguese	English
Alarm Volume	Maximum, Last Setting,	Last Setting

	Audio Off	
Reminder Signal	Enabled, Disabled	Disabled
Patient Type	Adult 22+, Adult 12-22 yrs, Pediatric 1-3 yrs. Pediatric 3-6 yrs, Pediatric 6-12 yrs, Infant/Neonatal	Adult 22+
Patient Weight Unit	Kg, lbs	lbs
RS232 Function	Standard, VueLink: IPI; VueLink: IPI, A/hr, ODI	Standard
Brightness Index	1,2,3,4,5,6,7,8,9,10, Auto	Auto
Parameter Standby Mode	Enabled, Disabled	Disabled
EMS Mode	Enabled, Disabled	Enabled
Clinical Report	Enabled, Disabled	Disabled

Main monitor settings may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**. Navigate to and select **Institutional Defaults>Monitor**.
5. Navigate to the monitor issue which you want to change and select the desired level.
6. Click the **Enter** button to return to the list of issues. For IP Settings, use the **Back** button to return to the list of issues.
7. Repeat for other issues as required.
8. Return to the Home Screen by selecting the **Home** button.

9. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

11.1.6.2. Interface settings

Parameter	Choices	Factory Default
RS232 Baud rate	Automatic, 9.6K, 19.2K, 38.4K, 57.6K, 115.2K	Automatic
Comm Protocol Mode	Microcap, Capnostream	Capnostream
English Text Report Format	Unicode	ASCII, Unicode
WiFi Settings	DHCP Disable, Enable IP address, Subnet Mask, Gateway, DNS Server	DHCP Enabled

Monitor Interface settings may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and click **Done**. Navigate to and select **Institutional Defaults>Monitor>Interfaces**.
5. Navigate to the monitor interface issue which you want to change and select the desired level.
6. Click the **Enter** button to return to the list of issues. For A/hr Visual Alert Level, use the **Back** button to return to the list of issues.

7. Repeat for other issues as required.
8. Return to the Home Screen by selecting the **Home** button.
9. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.



Note:

WiFi Settings will be available only once WiFi is turned on.

11.1.7. Home Screen Configuration

The device provides the option of choosing a Home Screen configuration that best fits your needs as your default Home Screen configuration. Reference [2.3.5 Monitoring Display Screen Options](#) on page 26 for a list of configurations.

The default Home screen may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and click **Done**. Navigate to and select **Institutional Defaults>Monitor>Home Screen Config**.
5. The first Home Screen option will appear. Using the right directional arrow, scroll between the Home Screen options until you reach the one that you want to use as your default Home Screen.
6. When the chosen Home Screen appears on the screen, click **Enter** to select the chosen screen. If you want to choose more than one default home screen, navigate to another screen and click **Enter** to select that screen as well. When you have chosen the final default home screen, click the down directional button

and then the **Enter** button to select **Apply**. The **Institutional Defaults: Monitor** screen will appear.

7. Click **Home** to return to the Home Screen. You will be asked to Shut down the device. When the monitor is turned on again after being turned off, this chosen Home Screen will be the default home screen that appears on the screen. If you choose more than one default screen, you may toggle between the chosen home screens using the directional buttons.

11.1.8. CO₂ Parameters

Institutional Defaults can be set for all CO₂ parameters that are settable in the monitor. To change the parameters, select CO₂ in the Institutional Defaults: Parameters screen.



Note:

Selecting Factory Defaults in the Inst. Def.: Parameters screen will restore all Parameters Factory Defaults.

Parameter	Choices	Factory Default
CO ₂ Unit	mmHg, kPa, Vol%	mmHg
BTPS	On, Off	On
Pump Off Timeout (min)	5, 10, 15, 30	15
CO ₂ Waveform Scale (mmHg)	20, 60, 120, 150, Auto	60
EtCO ₂ Trend Display Scale (mmHg)	50, 100, 150	50
RR Trend Display Scale (bpm)	50, 100, 150	50
A/hr Visual Alert Level	1-99	10

* BTPS denotes the standard correction used during measurement for body temperature, pressure, and saturation. BTPS should be set to ON during all measurement procedures. The monitor automatically turns off the BTPS correction during calibration procedures and turns it on again following these procedures. There is no need for the user to make any changes to the BTPS setting.

CO₂ Parameters settings may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**. Navigate to and **select Institutional Defaults>Parameters>CO₂**.
5. Navigate to the CO₂ parameter for which you want to make changes and select the desired value.
6. Click the **Enter** button to return to the list of issues. Repeat for other parameters as required.
7. Return to the Home Screen by selecting the **Home** button.
8. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

If the operator changes the capnography-based respiration rate trend scale settings in the Institutional Defaults: Parameters: CO₂ screen described above, the RR trend scale settings in the Institutional Defaults: Parameters: SpO₂ screen, which refers to Nellcor™ RR, will automatically be changed to match those settings.

11.1.9. SpO₂ Parameters

Institutional Defaults can be set for all SpO₂ parameters that are settable in the monitor. To change the parameters, select SpO₂ in the Institutional Defaults screen.

**Note:**

Selecting Factory Defaults in the Inst. Def.: Parameters screen will restore all Parameters Factory Defaults.

Parameter	Choices	Factory Default
Pulse Tone	On, Off	Off
SpO ₂ Scale for Trend Display (%)	0-100, 50-100	50-100
PR Scale for Trend Display (bpm)	150, 300	150
ODI Visual Alert Level	1-99	10
Oximetry RR	Enabled, disabled	Enabled
RR Trend Display Scale (bpm)	50, 100, 150	50

SpO₂ Parameters settings may be changed as follows:

1. Click the **Menu** button on the front panel of the monitor.
2. **Setup** will already be selected on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and select **Done**. Navigate to and select **Institutional Defaults>Parameters>SpO₂**.
5. Navigate to SpO₂ parameter for which you want to make changes and select the desired value.
6. Click the **Enter** button to return to the list of issues. Repeat for other parameters as required.
7. Return to the Home Screen by selecting the **Home** button.

If the operator changes the Nellcor™ RR trend scale settings in the Institutional Defaults: Parameters: SpO2 screen described above, the RR trend scale settings in Institutional Defaults:Parameters:CO2 screen, which refers to capnography-based respiration rate, will automatically be changed to match those settings.

11.1.10. Parameters Settings – Set Colors

The device provides the option to set colors for the parameters seen on the visual display.

1. Click the **Menu** button on the front panel of the monitor.
2. Navigate to and select **Setup** on the Menu screen.
3. Navigate to and select **Service** on the Setup Menu screen.
4. Input the Service password and click **Done**. Navigate to and select **Institutional Defaults>Parameters>Set Colors**.
5. Navigate to the parameter for which you want to change color and click the right directional button to select another color. Navigate to the color, click Enter and then navigate to and click **Apply**.
6. Click the **Back** button to return to the parameters list. Repeat for other parameters as required. Note that the same color can not be chosen for different parameters, except for capnography-based RR and Nellcor™ RR, which are currently both white and can be defined with the same color.
7. Return to the Home Screen by selecting the **Home** button.
8. The device will ask you to save settings and shut down. Click **Shutdown** to shut down, and then restart to retain the settings change you have just made.

12. Appendix A: Pulse Oximetry Clinical Studies

12.1. Clinical Studies Introduction

This section presents data from clinical studies conducted with the Nellcor™ bedside respiratory patient monitoring system equipped with the pulse oximetry module used in Capnostream™35 monitor.

Prospective clinical studies were conducted in accordance to EN ISO80601-2-61:2011 to demonstrate accuracy of pulse oximetry for Nellcor™ sensors with OxiMax™ technology used with this pulse oximetry module during both motion and non-motion conditions.

12.2. Clinical Studies Methodology

12.2.1. Hypoxia Methodology (Accuracy, Low Saturation, and Motion Studies)

The general purpose of invasive controlled desaturation study is to validate the SpO₂ and pulse rate accuracy in comparison to reference-standard measurements of blood SaO₂ by a CO-oximeter and ECG heart rate. This is achieved through paired observations of SpO₂ and SaO₂ values over the SaO₂ accuracy range of 70% to 100% on a group of healthy adult volunteers. The fraction of inspired oxygen (FiO₂) delivered to test subjects is varied to achieve a series of targeted steady-state

saturation periods. Arterial blood samples are periodically taken from an indwelling arterial catheter for use in the comparison.

In accordance to EN ISO80601-2-61:2011, desaturation to 70% is conducted in a gradual continuous process targeting multiple saturation plateaus (e.g. 98, 90, 80 and 72%). In these studies, six arterial samples were taken, 20 seconds apart at each plateau, resulting in a total of approximately 24 samples per subject. Each sample was drawn while SpO₂ data were simultaneously collected and marked for direct comparison to CO-oximetry.

End tidal CO₂, respiratory rate, respiratory pattern and electrocardiogram were continuously monitored throughout the study.

12.2.2. Low Saturation Methodology (Low Saturation Study Only)

The methodology and purpose of the Low Saturation study is the same as the hypoxia methodology, 12.2.1 Hypoxia Methodology (Accuracy, Low Saturation, and Motion Studies) on page 275. However, the desaturation is to 60% instead of 70%. This lower saturation is obtained by the addition of a new plateau at 60% SaO₂, increasing the range from 70 to 100% to 60 to 100%. This results in approximately 30 arterial samples instead of 24 for this study.

12.2.2.1. Motion Methodology (Motion Study Only)

Standard motions include tapping and/or rubbing at aperiodic intervals with amplitudes of 1-2 cm and 1-4 Hz with a random variation in frequency to simulate physiological motion. In this study, the subject was instructed to tap with finger tips to maintain consistency of area of effect on the pressure pad and to prevent resting hand on pressure pad between motions so that only qualified taps are recorded by the pressure pad system.

Each plateau (70 to 100%) has both an interval of tapping and rubbing. In this study, the order of tapping and rubbing was alternated between subjects.

Two video cameras were used to capture motion of the subjects. These videos were then reviewed to determine if any data points should be removed if the appropriate amplitudes were not being reached during the blood samples.

12.3. Clinical Studies Results

12.3.1. Accuracy Results (No Motion)

The following summary describes the demographic information of the subjects enrolled into the MAXA, MAXN, and MAXFAST Accuracy and Low Saturation study: A total of 11 subjects were analyzed, 5 (45%) males and 6 (55%) females. The mean age of the subjects was 31.8 ± 5.2 years, with a range of 25 to 42 years of age. Two subjects had dark pigmentation (dark olive to extremely dark). Weight ranged from 49 kg to 103.6 kg, and height ranged from 143.5 cm to 192 cm.

The following summary describes the demographic information of the subjects enrolled into the SC-A Sensor study: A total of 16 subjects were analyzed. There were 6 (37.5%) males and 10 (62.5%) female subjects enrolled into the study. The mean age of study participants was 31.44 ± 6.7 years, with a range of 24 to 42 years of age. Three subjects had dark pigmentation (dark olive to extremely dark). Weight ranged from 48.7 kg to 96.9 kg, and height ranged from 143.5 cm to 188 cm.

The following summary describes the demographic information of the subjects enrolled into the study for all other sensors (listed in Table 34. SpO₂ Accuracy Results (No Motion)): A total of 11 subjects were analyzed. There were 4 (36.4%) males and 7 (63.6%) female subjects enrolled into the study. The mean age of study participants was 30.36 ± 7.85 years, with a range of 22 to 46 years of age. Three subjects had dark pigmentation (dark olive to extremely dark). Weight ranged from 58.4 kg to 114.4 kg, and height ranged from 159 cm to 187 cm.

Accuracy results for SpO₂ can be found in [Table 34. SpO₂ Accuracy Results \(No Motion\)](#). ARMS (Accuracy root mean square) is used to describe the accuracy of pulse oximetry, which is affected by both bias and precision. As shown in the table, SpO₂ meets the acceptance criteria for all of the listed sensors during non-motion conditions.

Table 34. SpO₂ Accuracy Results (No Motion)

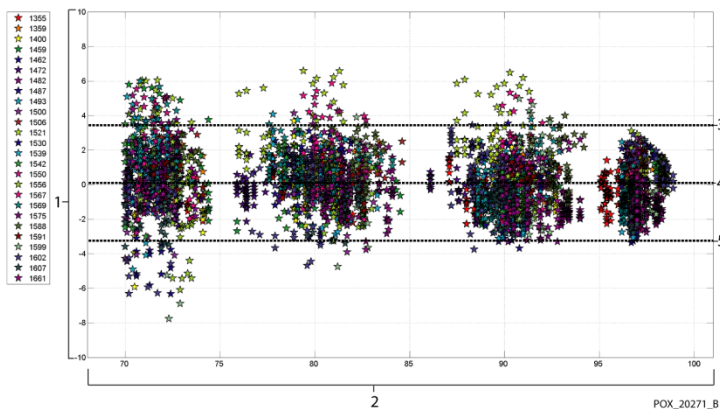
Sensor	Motion	# of Data Points	ARMS (%)	SpO ₂ Acceptance Criteria 70%-100% (%)	Pass/Fail
MAXA	No	976	1.54	≤2.0	Pass
MAXN	No	723	1.41	≤2.0	Pass
MAXFAST	No	235	1.42	≤2.0	Pass
SC-A	No	659	1.86	≤2.0	Pass
DS100A	No	411	2.16	≤3.0	Pass
OxiCliq-A	No	480	1.58	≤2.5	Pass
D-YSE	No	458	1.96	≤3.5	Pass



Note:

Each individual subject is represented by a unique color on the plots. Subject identification numbers are indicated in the legend to the left of each plot.

Figure 57. Modified Bland-Altman for SpO₂ (All Data - No Motion): SaO₂ vs. (SpO₂ - SaO₂)



Label	Description
1	SpO ₂ - SaO ₂ (%)
2	SaO ₂ (%)
3	Upper 95% LoA
4	Mean Bias
5	Lower 95% LoA

Pulse oximeters have been known to perform better at the higher saturation levels compared to the lower end. Though, when presenting the ARMS, the common methodology is to provide the data across the whole range (70% to 100%). The data below is presented to show each decade, which includes the RMSD (root mean square difference) and N values. RMSD and ARMS are the same. ARMS is used for pooled data across the whole study to represent accuracy of the system, whereas RMSD is used as the general term. There is no acceptance criteria associated with decade levels of hypoxia, thus represented as RMSD. In [Table 35. RMSD of SpO2 per Decade \(No Motion\)](#), SpO₂ RMSD is presented per decade.

Table 35. RMSD of SpO₂ per Decade (No Motion)

SpO ₂ Range	100%-90%	89%-80%	79%-70%
N	1693	1037	1212
RMSD (%)	1.46	1.66	2.01

The plateaus that were used during the study were 70 - 76, 76.01 - 85, 85.01 - 94 and >94%, as presented in [Table 36. RMSD of SpO₂ per Plateau \(No Motion\)](#), below.

Table 36. RMSD of SpO₂ per Plateau (No Motion)

SpO ₂ Range	Room Air Plateau	90% Plateau	80% Plateau	70% Plateau
N	978	1102	1034	828
RMSD (%)	1.27	1.65	1.69	2.15

12.3.2. Clinical Studies Accuracy Results (Low Saturation)

The accuracy results for SpO₂ can be found in [Table 37](#) across a SaO₂ range of 60 to 80%. As shown in the table, SpO₂ meets the acceptance criteria for the MAXA, MAXN, and MAXFAST sensors during low saturation conditions.

Table 37. SpO₂ Accuracy Results (60 to 80% SaO₂, low saturation)

Sensor	Motion	# of Data Points	ARMS (%)	SpO ₂ Acceptance Criteria 60%-80%	Pass/Fail

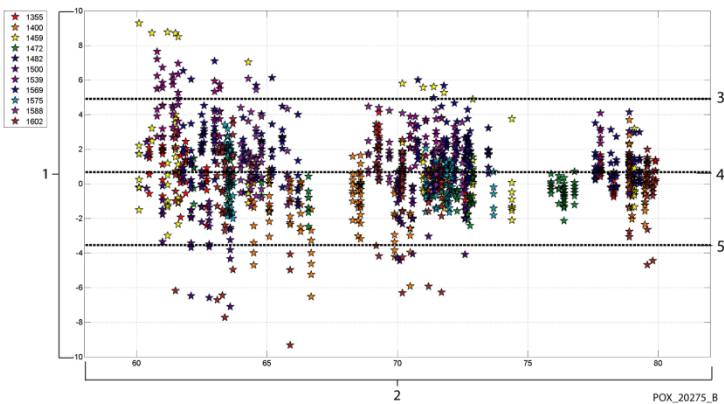
				(%)	
MAXA	No	610	2.40	≤ 3.0	Pass
MAXN	No	453	1.92	≤ 3.0	Pass
MAXFAST	No	143	2.41	≤ 3.0	Pass
ALL	No	1206	2.24	≤ 3.0	Pass

Modified Bland-Altman plots for all the data are presented in [Figure 58](#).
Modified Bland-Altman for SpO₂ (All Data - Low Saturation): SaO₂ vs. (SpO₂ - SaO₂).



Note:
 Each individual subject is represented by a unique color on the plots. Subject identification numbers are indicated in the legend to the left of each plot.

Figure 58. Modified Bland-Altman for SpO₂ (All Data - Low Saturation): SaO₂ vs. (SpO₂ - SaO₂)



Label	Description
1	SpO ₂ - SaO ₂ (%)
2	SaO ₂ (%)
3	Upper 95% LoA
4	Mean Bias
5	Lower 95% LoA

In [Table 38](#), SpO₂ RMSD is presented per decade. In the following table, [Table 39](#), RMSD is presented per plateau.

Table 38. RMSD of SpO₂ per Decade (Low Saturation)

SpO ₂ Range	80%-70%	69%-60%
N	637	569
RMSD (%)	1.73	2.69

Table 39. RMSD of SpO₂ per Plateau (Low Saturation)

SpO ₂ Range	70% Plateau	60% Plateau
N	506	483
RMSD (%)	1.93	2.79

12.3.3. Clinical Studies Accuracy Results (Motion)

The following describes the demographic information of the subjects enrolled into the study: A total of 14 subjects were analyzed, 5 (35.7%) males and 9 (64.3%) female subjects. The mean age was 31.57 ± 6.8 years, with a range of 24 to 42 years of age. Three subjects had dark pigmentation (dark olive to extremely dark). Weight ranged from 48.7 kg to 88.6 kg, and height ranged from 143.5 cm to 185 cm.

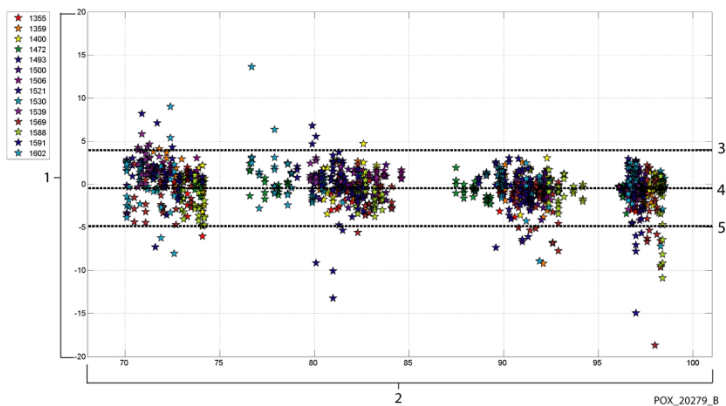
The accuracy results for both SpO₂ and pulse rate during motion are presented in [Table 40](#). As shown in the table, SpO₂ meets the acceptance criteria for both the MAXA and MAXN sensor during motion.

Table 40. SpO₂ Accuracy Results during Motion

Sensor	Motion	# of Data Points	ARMS (%)	SpO ₂ Acceptance Criteria 70%-100% (%)	Pass/Fail
MAXA	Yes	637	1.70	≤3.0	Pass
MAXN	Yes	618	2.76	≤3.0	Pass
ALL	Yes	1255	2.28	≤3.0	Pass

A modified Bland-Altman plot for the data is presented in [Figure 59](#). *Modified Bland-Altman for SpO₂ (All Data - Motion): SaO₂ vs. (SpO₂ - SaO₂).*

Figure 59. Modified Bland-Altman for SpO₂ (All Data - Motion): SaO₂ vs. (SpO₂ - SaO₂)



Label	Description
1	SpO ₂ - SaO ₂ (%)
2	SaO ₂ (%)
3	Upper 95% LoA
4	Mean Bias
5	Lower 95% LoA

Pulse oximeters have been known to perform better at the higher saturation levels compared to the lower end. Though, when presenting the ARMS, the common methodology is to provide the data across the whole range (70% to 100%). The data below is presented to show each decade, which includes the RMSD and N values. There is no acceptance criteria associated with decade levels of hypoxia. In, SpO₂ RMSD is presented per decade.

Table 41. RMSD of SpO₂ per Decade (Motion)

SpO ₂ Range	100%-90%	89%-80%	79%-70%
N	589	322	344
RMSD (%)	2.36	1.97	2.41

RMSD for each decade are well within the acceptance criteria of 3%. The plateaus that were used during the study were 70 - 76, 76.01 - 85, 85.01 - 94 and >94%, as presented in Table 42.

Table 42. RMSD of SpO₂ per Plateau (Motion)

SpO ₂ Range	Room Air Plateau	90% Plateau	80% Plateau	70% Plateau
N	318	330	318	289
RMSD (%)	2.61	1.96	2.19	2.33

12.4. Clinical Studies Conclusion

12.4.1. No Motion

The pooled results indicate that the observed SpO₂ ARMS values met the listed specification (dependent on the sensor used) with pulse oximetry module used in this device for SpO₂ during non-motion conditions across the SaO₂ saturation range of 70 to 100%.

The pooled results indicate that for a saturation range of 60-80% for SpO₂, the acceptance criterion was met for the pulse oximetry module used in this device when tested with MAXA, MAXN, and MAXFAST sensors.

12.4.2. Motion

The pooled results indicate that the observed SpO₂ ARMS values met the listed specification of 3% when tested with MAXA, MAXN sensors, with the pulse oximetry module used in this device, for SpO₂ during motion conditions across the SaO₂ saturation range of 70 to 100%.



13. Appendix B: Monitor Service Password

13.1. Monitor Service Password

The service password is: SERV.



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